

SPATIALAB: CAN VISION–LANGUAGE MODELS PERFORM SPATIAL REASONING IN THE WILD?

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ABSTRACT

Spatial reasoning is a fundamental aspect of human cognition, yet it remains a major challenge for contemporary vision–language models (VLMs). Prior work largely relied on synthetic or LLM-generated environments with limited task designs and puzzle-like setups, failing to capture the real-world complexity, visual noise, and diverse spatial relationships that VLMs encounter. To address this, we introduce **SPATIALAB**, a comprehensive benchmark for evaluating VLMs’ spatial reasoning in realistic, unconstrained contexts. SPATIALAB comprises 1,400 visual question–answer pairs across six major categories: *Relative Positioning*, *Depth & Occlusion*, *Orientation*, *Size & Scale*, *Spatial Navigation*, and *3D Geometry*, each with five subcategories, yielding 30 distinct task types. Each subcategory contains at least 25 questions, and each main category includes at least 200 questions, supporting both multiple-choice and open-ended evaluation. Experiments across diverse state-of-the-art VLMs, including open- and closed-source models, reasoning-focused, and specialized spatial reasoning models, reveal a substantial gap in spatial reasoning capabilities compared with humans. In the multiple-choice setup, InternVL3.5-72B achieves 54.93% accuracy versus 87.57% for humans. In the open-ended setting, all models show a performance drop of around 10–25%, with GPT-5-mini scoring highest at 40.93% versus 64.93% for humans. These results highlight key limitations in handling complex spatial relationships, depth perception, navigation, and 3D geometry. By providing a diverse, real-world evaluation framework, SPATIALAB exposes critical challenges and opportunities for advancing VLMs’ spatial reasoning, offering a benchmark to guide future research toward robust, human-aligned spatial understanding. SPATIALAB will be released upon acceptance.

1 INTRODUCTION

Spatial reasoning is a fundamental aspect of human cognition that involves understanding spatial layouts, imagining and manipulating objects, and navigating environments. It supports applications ranging from robotics, autonomous driving and augmented/virtual reality to geospatial analytics, computer graphics, human–computer interaction, and education (Huang et al., 2024b; Qi et al., 2024; 2025). For artificial agents, autonomous systems that perceive and act in dynamic environments, reliable spatial competence requires interpreting geometric relations, depth cues, occlusion patterns, and agent-centric perspectives in scenes that are visually noisy and structurally complex (Bar-Anan et al., 2006; Cai et al., 2025; Trope & Liberman, 2010; Ramalho et al., 2018). Recent vision–language models have advanced multimodal representation and language grounding, but their spatial judgments remain fragile in realistic environments (Chen et al., 2025; Kosoy et al., 2025; Pothiraj et al., 2025).

Existing spatial benchmarks, however, are narrow and simplified. Most focus on elementary tasks such as binary spatial relations, low-resolution depth categories, or schematic navigation cues, often in synthetic or simplified settings that reduce visual clutter and homogenize objects (Du et al., 2024; Jia et al., 2025; Szymanska et al., 2024; Shiri et al., 2024; Kamath et al., 2023; Fu et al., 2024; Song et al., 2024; Yang et al., 2024; Qi et al., 2025; Cai et al., 2024). These controlled setups lessen perceptual and reasoning demands, leading to apparent saturation while masking failures under distribution shift. Critical challenges such as robust occlusion inference, scale-consistent judgments across viewpoints, and planning under partial observability remain under-sampled (Kosoy et al., 2025; Pothiraj et al., 2025; Chen et al., 2025; Cai et al., 2025). As a result, models that perform well on

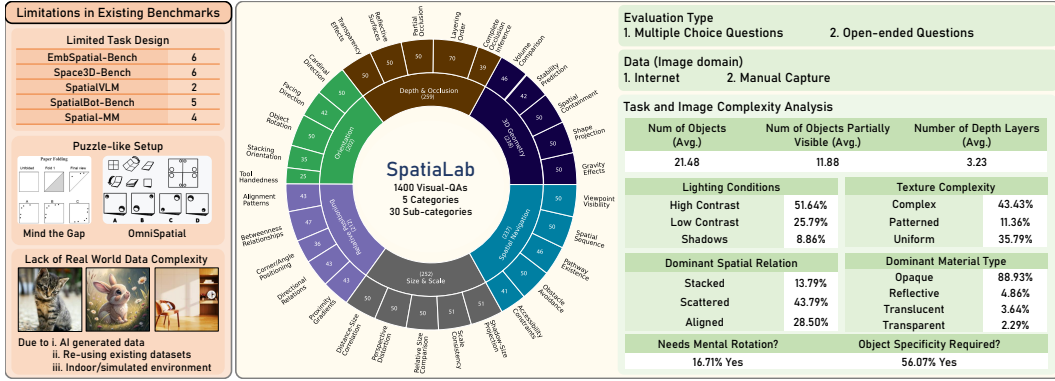


Figure 1: **Overview of SPATIALLAB**. The benchmark addresses limitations of prior datasets (left), introduces 1,400 visual QA pairs spanning 5 categories and 30 subcategories (center), and enables systematic evaluation through multiple-choice and open-ended tasks. It features diverse task and image complexity, with varied object counts, layers, lighting, textures, relations, and materials (right).

existing benchmarks often fail on integrated, multi-step inference tasks, limiting safe deployment in real-world environments where spatial errors carry practical consequences.

A meaningful evaluation must therefore span the all core axes of spatial reasoning: relative position, depth and occlusion, orientation, size and scale, navigation, and 3D geometry. Unlike current task-isolated setups, human cognition integrates these dimensions seamlessly under noise and ambiguity (Driess et al., 2023; Brohan et al., 2023; Zitkovich et al., 2023; Collaboration et al., 2023; Yuan et al., 2024; Qi et al., 2025). Robust benchmarks should therefore test both choice and generation, benchmark against human performance, and draw from real-world scenes that reveal failures masked by simulator datasets-principles that motivate our benchmark.

To address the above needs, we introduce **SPATIALLAB**, a benchmark designed to assess spatial reasoning of vision-language models in realistic, unconstrained visual contexts, as outlined in Figure 1. **SPATIALLAB** provides a balanced suite of visual question-answer items covering a broad taxonomy of spatial tasks and supports both multiple-choice and open-ended evaluation formats. The benchmark emphasizes photographic diversity, complex spatial relational queries, and tasks that require integrating depth, perspective, and 3D structure. We evaluate a wide collection of state-of-the-art VLMs, including open- and closed-source models and reasoning-focused systems, compare them to human performance, and perform extensive error analysis. Our key contributions are as follows:

1. We introduce **SPATIALLAB**, a large-scale benchmark of 1,400 visual question-answer pairs spanning six principal categories and thirty task types in realistic, unconstrained contexts. With at least 200 questions per category and 25 per subcategory, it ensures balanced coverage and supports both multiple-choice (**SPATIALLAB-MCQ**) and open-ended (**SPATIALLAB-OPEN**) evaluation.
2. We perform a **comprehensive evaluation** of 25+ state-of-the-art vision-language models: open- and closed-source, reasoning-oriented, and spatially specialized reasoning models against human baselines, exposing a consistent and significant gap in spatial reasoning.
3. We deliver **in-depth error analyses and diagnostic studies** that uncover systematic failure modes in depth perception, navigation, occlusion, and 3D geometry, yielding concrete insights and actionable directions for advancing robust, human-aligned spatial understanding.
4. We explore multiple strategies to enhance spatial reasoning in VLMs, including **supervised fine-tuning (SFT)**, **chain-of-thought (CoT) prompting**, **CoT with self-reflection**, and **multi-agent architectures (SPATIOXOLVER) with multi-step reasoning**, analyzing their benefits and limitations across multiple-choice and open-ended tasks.

In the multiple-choice setting, leading models achieve roughly 50–55% accuracy, while human annotators exceed 85%. In open-ended generation, model performance drops by another 10–25 percentage points, with the best systems near 40% versus about 65% for humans. Error analysis reveals concentrated failures in occlusion inference, scale-consistency, and multi-step navigational

Table 1: Comparison of spatial reasoning benchmarks.

Dataset	E.	T.C.	Data				QAs			Venue/	Eval		A.L.	Md.	Best Baseline	B.B.P.
			Domain	Source	A.	Size	Count	Df.	Cmpl.	Year	Type	Depth				
ScanQA	Y	7	Indoor	E.D.	M	800	41K	H	E	CVPR'22C	Close-ended	L	L	I	-	-
Visual Spatial	Y	7	R.W.	E.D.	T	10K	10K	E	E	TACL'23	MCQ,Bin.	L	M	I	LXMERT	70.10
What's up	N	6	Indoor	E.D.	T	5K	5K	E	E	EMNLP'23C	MCQ	S	Hi	I	XVLM-COCO	60.40
EmbSpatial-Bench	Y	6	Indoor	E.D.	T	2.2K	3.6K	H	E	ACL'24C	MCQ,%	M	S	I	Qwen-VL-Max	49.11
Space3D-Bench	N	6	Indoor	E.D.	M	211	1K	Mm	E	ECCV'24W	Mix	L	S	I	RAG3D-Chat	66.80
SpatialRGPT-Bench	N	12	Mix	E.D.	T	1.5K	1.5K	Mm	E	NeurIPS'24C	MCQ,Bin.	M	L	I	GPT-4o	57.83
BLINK-Spatial	Y	14	Mix	E.D.	M	7.3K	3.8K	H	E	ECCV'24C	MCQ,Bin.	M	M	I	GPT-4o	59.03
Spatial-MM	N	4	Internet	Internet	T	2.3K	2.3K	E	Mm	EMNLP'24C	Open	L	Hi	I	GPT-4 Vision	63.82
RoboSpatial	Y	4	Indoor	E.D.	T	1M	3M	H	E	CVPR'25C	Open,Bin.	L	S	I	LEO	71.90
SpatialVLM	N	2	Web	E.D.	T	10M	2B	H	E	CVPR'24C	Open	L	S	I	GPT-4V	68.00
VSI-Bench	Y	8	Indoor	M.E.D.	T	288	5K	Mm	Mm	CVPR'25C	MCQ,Nm.	M	M	V	Gemini-1.5 Pro	45.40
OmniSpatial	Y	50	Internet	Internet	M,P	1387	1.5K	H	Mm	2025	MCQ,Bin.	M	S	I	o3-2025-04-16	56.33
Mind the Gap	N	6	Mix	AI-Gen	M,P	1.8K	1.8K	E	Mm	2025	Mix	M	S	I	InternVL2.5-26B	48.83
VLM4D	N	4	Mix	Mix	M	1K	1.8K	E	H	ICCV'25	MCQ	Hi	M	Mix	Gemini-2.5 Pro	62.00
SPATIALAB-MCQ	Y	30	Mix	Mix,M	M	1.2K+	1.5K	H	H	2025	MCQ	Hi	H	Mix	InternVL3.5-72B	54.93
SPATIALAB-OPEN	Y	30	Mix	Mix,M	M	1.2K+	1.5K	H	VH	2025	Open	Hi	H	Mix	GPT-5-mini	40.93

planning, indicating limits in compositional spatial representations and reasoning. Our comprehensive study highlights which interventions provide robust gains, which amplify existing biases, and the critical role of perceptual grounding for generalizable spatial reasoning. By providing tasks, baseline results, and diagnostic analyses, SPATIALAB establishes a foundation for progress and reproducible comparison in spatial reasoning evaluation.

2 PRELIMINARIES: SPATIAL REASONING IN VLMs

Visual-spatial reasoning, the ability to perceive, represent, and manipulate spatial relations among objects, plays a central role in human cognition and in enabling embodied agents to navigate and act in the world (Palmer, 1999; Chabris et al., 2006). Formally, we can describe a scene as a set of objects $O = \{o_1, \dots, o_n\}$ with attributes (e.g., position, orientation, size) in a world coordinate system W . Visual-spatial reasoning then seeks to learn a mapping $f : (I, Q) \mapsto R$, where I is an image (or image sequence), Q is a linguistic query, and R is a structured representation of spatial relations $R = \{r(o_i, o_j) \mid o_i, o_j \in O\}$ capturing predicates such as *left-of*, *occludes*, or *supports*. For vision-language models (VLMs), learning f is highly challenging because it requires integrating noisy visual perception with compositional linguistic abstraction under embodied constraints such as perspective, gravity, and physical interaction (Dai et al., 2017; Alayrac et al., 2022).

Early approaches to spatial reasoning in VLMs have largely relied on basic spatial relations or narrow task designs (Cheng et al., 2024; Du et al., 2024; Szymanska et al., 2024), often situated in synthetic datasets like CLEVR (Johnson et al., 2016) or structured question-answering benchmarks such as GQA (Hudson & Manning, 2019) or puzzle-like simulated setups such as MGT (Stogiannidis et al., 2025) and OmniSpatial (Stogiannidis et al., 2025), as we can notice in Table 1¹. While useful for controlled evaluation, these settings oversimplify spatial structure, omit environmental noise, and neglect dynamic factors such as temporal changes or viewpoint shifts. To overcome these limitations, we argue for the need to benchmark spatial reasoning in rich, real-world contexts that capture scene complexity, temporal variation, and multiple perspectives. Such settings expose VLMs to challenges they would encounter in embodied or interactive applications, including robotics, AR/VR, and autonomous navigation (Mu et al., 2023; Collaboration et al., 2023).

In SPATIALAB, we organize spatial reasoning into six principal categories, *Relative Positioning*, *Depth & Occlusion*, *Orientation*, *Size & Scale*, *Spatial Navigation*, and *3D Geometry*, each linked to a distinct cognitive faculty studied in psychology and spatial cognition (Baddeley, 1998; Previc, 1998). As shown in Figure 1 and Appendix B, each category is further decomposed into five subcategories, producing a taxonomy of thirty task types that isolate fine-grained reasoning skills, such as directional alignment, occlusion inference, or stability prediction. This classification scheme is motivated by the need to capture the full spectrum of visual-spatial reasoning challenges. By systematically breaking down complex reasoning into measurable and testable units, we can evaluate not only whether models “solve” spatial tasks but also which specific dimensions remain fragile or underdeveloped (Gardner, 2011; Huang et al., 2022; Gupta & Kembhavi, 2023; Yang et al., 2023).

¹ Here, E. → Embodied, R.W. → Real World, T.C. → Task Categories, E.D. → Existing Dataset, A. → Annotation, T → Template, M → Manual, P → Puzzles, Md. → Modality (I → Image, V → Video), Bin. → Bin., Nm. → Nm., Df. → Difficulty, Cmpl. → Complexity (E → Easy, Mm → Medium, H → Hard, S → Shallow, L → Low, Hi → Hi, VH/VHi → Very Hard/Very High), A.L. → Analysis Level, B.B.P. → Best Baseline Performance.

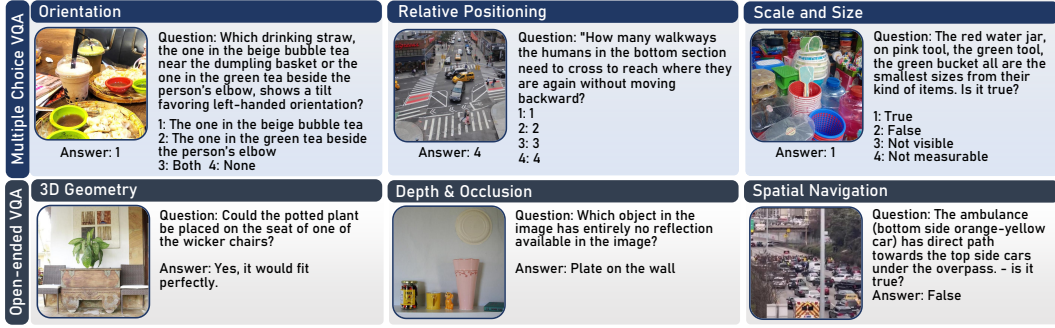


Figure 2: Representative examples from six categories in open-ended and MCQ Tasks.

Moreover, we adopt two complementary evaluation formats: multiple-choice and open-ended question answering. The multiple-choice format probes recognition and fine discrimination among distractors, while the open-ended format measures generative reasoning and compositional description (Driess et al., 2023; Liu et al., 2025; Zheng et al., 2022). Together, these formats reflect real-world reasoning demands and expose different strengths and weaknesses in model capabilities. By combining a principled taxonomy, diverse real-world imagery, and dual evaluation formats, SPATIALAB provides a structured framework for systematic benchmarking. This setup enables fine-grained model comparison, alignment with human-level performance, and deeper insights into the development of robust spatial cognition in artificial agents.

3 SPATIALAB BENCHMARK

Spatial reasoning remains a persistent challenge for vision-language models (VLMs), especially in naturalistic conditions where occlusion, perspective distortion, and noisy backgrounds co-occur. To address this, we present **SPATIALAB**, a benchmark designed to systematically evaluate VLMs' capacity for complex spatial cognition under realistic visual conditions. The benchmark comprises 1,400 visual question-answer pairs organized into six major categories: *Relative Positioning*, *Depth & Occlusion*, *Orientation*, *Size & Scale*, *Spatial Navigation*, and *3D Geometry*, each further divided into five subcategories, yielding 30 distinct task types. Every subcategory includes at least 25 QA pairs, while each top-level category contributes a minimum of 200, ensuring balanced coverage across the taxonomy. **SPATIALAB** supports both multiple-choice and open-ended evaluation, enabling assessment of discriminative accuracy as well as generative reasoning and explanatory competence, thus aligning with broader multimodal evaluation practices while emphasizing spatial-specific reasoning. High visual diversity, fine-grained annotation, and rigorous quality control make **SPATIALAB** a reliable and scalable testbed, and Figure 2 illustrates representative examples from the six categories, showcasing both open-ended and multiple-choice formats that capture the richness and difficulty of real-world spatial reasoning tasks.

3.1 BENCHMARK CONSTRUCTION

Image Collection. The image corpus was curated using three complementary strategies: (i) automated large-scale web crawling, (ii) targeted online retrieval of scene-specific images, and (iii) manual snapshots collected in diverse indoor and outdoor environments (Figure 3). This multi-source pipeline ensures diversity across object categories, environmental conditions, and spatial configurations. To explicitly capture natural variability, images were profiled along six meta-dimensions: *lighting condition* (high contrast, low contrast, shadows, and reflective settings), *texture complexity* (uniform, patterned, and complex), *edge complexity* (sharp and smooth boundaries), *dominant spatial relation* (stacked, scattered, aligned), *material type* (transparent, translucent, opaque, reflective), and *gravity constraints* (normal, floating, and unconstrained). Each dimension was systematically represented in the final corpus, ensuring that the benchmark reflects a broad spectrum of real-world, in-the-wild scenarios rather than simplified or synthetic setups. Complexity analysis of the corpus reveals that on average images contain 21.48 objects, of which 11.88 are partially visible, distributed across 3.23 distinct depth layers. Spatial reference chains (e.g., "object A is left of B, which is behind C") contain 2.07 links, indicating the presence of multi-step relational reasoning. Also, 16.71% of the questions

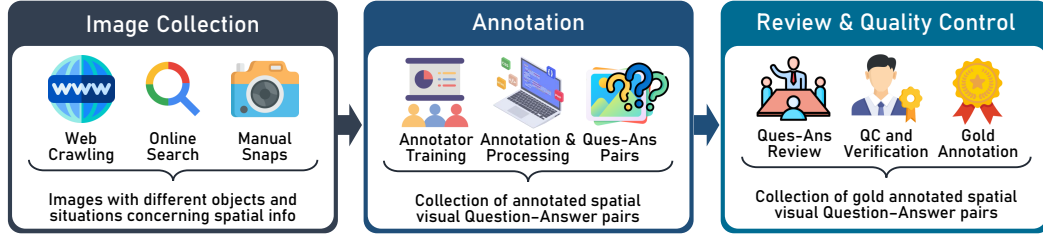


Figure 3: **Data creation pipeline for SPATIALAB.** Images are collected via web crawling, targeted search, and manual snapshots, followed by structured annotation of spatial question-answer pairs.

require mental rotation; 56.07% require object specificity (specific color, pattern, or other details). Such statistics posits that **SPATIALAB** captures cluttered, layered, and relationally rich scenes, which are characteristic of real-world spatial reasoning, underrepresented in synthetic datasets.

Annotation. Annotation proceeded in three phases. First, annotators underwent targeted training (details are available in Appendix C.2) to ensure consistency in interpreting spatial categories and subcategories. Second, each image was paired with one or more spatial QA tasks covering a range of reasoning types such as proximity gradients, alignment patterns, occlusion inference, and stability prediction. Questions were designed to balance perceptual grounding with higher-order inference (e.g., predicting stability under gravity constraints). Third, QA pairs were encoded in both multiple-choice (04 options) and open-ended formats, ensuring comparability across evaluation paradigms. In total, it produces around 1,500 spatial visual QA pairs, across six categories and 30 subcategories.

Review and Quality Control. To ensure reliability, annotation outputs underwent a three-tier review and validation pipeline. In the first stage, QA items were checked for semantic validity and alignment with the designated subcategory. In the second stage, independent annotators verified correctness of both the question phrasing and the designated answer, focusing on eliminating ambiguity and enforcing task clarity. In the third stage, a gold-standard annotation round was conducted, establishing a set of high-confidence QA pairs for evaluation. All metadata was removed from the images, and the dataset was carefully curated to avoid copyright concerns, though minor oversights may still occur. This procedure yielded a final benchmark of 1,400 validated QA items.

4 EXPERIMENTS AND EVALUATION

Models. We categorize the evaluated models into four groups. **Proprietary models** include GPT-4o-mini (OpenAI, 2024), GPT-5-nano, Gemini-2-Flash and Gemini-2.5-Flash (Team, 2025a), and Claude 3.5 Haiku (Anthropic, 2024). **Open-source models** comprise Intern-VL3 (1B, 2B, 4B) (Zhu et al., 2025), Qwen-VL2.5 (3B, 7B, 32B, 72B Instruct) (Bai et al., 2025), GLM 4.5V-3.06B and GLM-4.1V-9B-Thinking (Team et al., 2025), Gemma-3 (4B, 27B)-it (Gemma Team, Google DeepMind, 2025), Llama-3.2 (11B, 90B) Vision-Instruct (Grattafiori et al., 2024), and Step-3-321B-MoE (StepFun, 2025). **Reasoning models** include o4-mini (OpenAI, 2025), Gemini-2-Flash and Gemini-2.5-Flash thinking variants (Team, 2025a), and Kimi-VL-A3B-Thinking-2506 (Team, 2025b). **Spatial reasoning specialists** comprise SpaceOm, SpaceThinker-Qwen2.5VL-3B, and SpaceQwen2.5-VL-3B-Instruct (Chen et al., 2024a). For GLM-4.5V-106B-MoE, GLM-4.1V-9B-Thinking, and Step-3-321B-MoE we only performed open-ended evaluation, as these models are not instruction-tuned and were trained primarily on open-ended tasks, resulting in excessive refusals during MCQ evaluation. For ease of comparison in our results tables, we use color coding: proprietary, open-source up to 7B, larger than 7B, proprietary reasoning, open-source reasoning, spatial specialists, and human baseline.

Evaluation Process and Metrics. We measure accuracy as the primary metric for both MCQ and open-ended tasks. For **MCQ evaluation**, we use direct prompting, where models return the option number corresponding to their selected answer. Responses are automatically checked against the correct answer. Detailed prompts and details are provided in Appendix D.1. For **open-ended evaluation**, models are prompted to generate free-form answers, which are then assessed using a large language model judge. Prompts and details are available Appendix D.2. To evaluate the performance of LLM Judge with humans, we performed LLM-human agreement analysis too, as

Table 2: Per-category multiple-choice accuracy (% , $\uparrow$ higher is better) on SPATIALAB-MCQ, reported for each spatial question category alongside the overall score. Within each model group, the strongest entry is rendered in **bold** and underlined.

Model	3D Geom. (#238)	Dep. & Occu. (#259)	Orientation (#202)	Relat. Posit. (#212)	Size & Scale (#252)	Spati. Navig. (#237)	Overall (#1400)
<i>Random Choice</i>	25.00	25.00	25.00	25.00	25.00	25.00	25.00
<i>Proprietary Models</i>							
GPT-4o-mini	47.06	39.00	47.03	47.17	49.60	49.79	46.50
GPT-5-mini	48.74	54.83	60.40	62.74	44.84	56.54	54.29
Gemini-2.0-Flash	47.06	55.21	53.96	58.02	54.37	46.84	52.50
Gemini-2.5-Flash	44.96	48.26	48.02	56.13	42.46	51.05	48.29
Gemini-2.5-Pro	47.48	50.19	49.50	58.49	43.65	52.32	50.07
Claude 3.5 Haiku	42.44	42.08	46.53	46.23	35.71	45.99	42.93
Mistral Medium 3.1	46.64	49.81	47.52	61.79	41.67	41.77	47.93
<i>Open-Source Models</i>							
InternVL3.5-1B	33.61	32.43	23.27	37.26	31.75	30.80	31.64
InternVL3.5-2B	34.03	31.66	31.68	40.57	32.54	32.49	33.71
Qwen-VL2.5-3B-Instruct	41.18	35.52	46.04	40.09	47.22	39.24	41.43
InternVL3.5-4B	42.86	42.86	42.08	54.72	36.51	42.19	43.29
Gemma-3-4B-it	43.70	34.36	46.53	45.75	37.30	37.97	40.57
Qwen-VL2.5-7B-Instruct	42.86	37.84	42.57	46.23	42.06	35.44	41.00
Llama-3.2-11B-Vision-Instruct	26.47	30.50	20.30	42.92	30.56	32.07	30.50
Gemma-3-27B-it	43.28	40.15	48.02	54.25	48.02	47.26	46.57
Qwen-VL2.5-32B-Instruct	41.18	40.15	46.53	45.28	45.24	41.77	43.21
InternVL3.5-72B	50.00	57.14	53.47	66.04	49.21	54.85	54.93
Qwen-VL2.5-72B-Instruct	47.06	48.65	51.98	54.25	43.65	48.95	48.86
Llama-3.2-90B-Vision-Instruct	46.22	52.12	50.50	58.96	46.83	48.52	50.36
<i>Reasoning Models</i>							
o3-mini	50.00	54.83	51.98	61.32	38.89	50.21	50.93
o4-mini-medium	51.26	58.30	54.95	64.15	40.87	51.48	53.21
Gemini-2-Flash-Thinking	37.82	41.31	41.58	45.75	50.40	43.04	43.36
Gemini-2.5-Flash-Thinking	45.80	53.67	52.97	56.60	55.16	53.59	52.93
Kimi-VL-A3B-Thinking-2506	42.86	41.31	40.59	51.42	39.68	41.35	42.71
<i>Spatial Reasoning Models</i>							
SpaceOm	42.44	38.61	48.02	37.74	42.86	39.24	41.36
SpaceThinker-Qwen2.5VL-3B	40.34	37.84	47.03	38.21	43.25	37.97	40.64
SpaceQwen2.5-VL-3B-Instruct	31.51	35.14	37.62	37.74	50.79	47.26	40.14
<i>Human Baseline</i>	93.70	74.13	91.58	91.51	88.89	87.76	87.57

detailed in Appendix D.2. Table 5 demonstrates that the LLM judge (Gemini-2.5-Flash) achieves substantial agreement with human annotators, with a Cohen’s kappa of 0.738 against the majority vote and 0.681–0.795 against individual annotators. Raw accuracy against the majority vote is 0.880, while Fleiss’ kappa among human annotators is 0.774, indicating strong reliability. For additional details on experiment setup, prompts, and evaluation procedures, see Appendix D.

Improving Visual Reasoning Capabilities. To enhance spatial reasoning on our diverse benchmark, we explore reasoning strategies including inherent reasoning in the VLMs (Appendix H.1), Chain-of-Thoughts (CoT) (Appendix H.2), CoT with self-reflection (Appendix H.3), supervised fine-tuning (SFT) (Appendix H.4), and multi-agent systems (Appendix H.5). For SFT, specifically, Qwen-VL2.5-3B-Instruct was fine-tuned on 40% of the dataset with stratified sampling, improving generalization across relational and geometric tasks. Evaluation on the remaining 60% showed reduced errors in navigation and orientation, with additive gains when integrating CoT and SFT. However, agentic reasoning revealed that while orientation benefits substantially, other categories stagnate or degrade, underscoring that deeper perceptual and spatial grounding is required beyond multi-step reasoning.

5 FINDINGS AND ANALYSIS

5.1 OVERALL RESULTS

SPATIALAB-MCQ. Table 2 demonstrates model performances in multiple choice evaluation on SPATIALAB-MCQ. We observe substantial variation in performance across models, tasks, and families: overall model accuracies span roughly 30–55% (random choice = 25%), while the human baseline is 87.57%, indicating large headroom. InternVL3.5-72B, an open source model, is the best performing model, with 54.93% accuracy. Reasoning- and instruction-tuned models (e.g., o4-mini: 53.21%, Gemini-2.5-Flash-Thinking: 52.93%, GPT-5-mini: 54.29%) and some very large open-source models (e.g., InternVL3.5-72B: 54.93%, Llama-3.2-90B: 50.36%) rank near the top, but model scale alone is not determinative: for example, Llama-3.2-11B attains only 30.50% despite its substantial size. Category-wise, Orientation and 3D Geometry are where many strong models excel

Table 3: Per-category open-ended (free-form generation) accuracy (% , $\uparrow$ higher is better) on SPATIALAB-OPEN, reported for each spatial question category alongside the overall score. Within each model group, the top result is rendered in **bold** and underlined.

Model	3D Geom. (#238)	Dep. & Occu. (#259)	Orientation (#202)	Relat. Posit. (#212)	Size & Scale (#252)	Spati. Navig. (#237)	Overall (#1400)
Proprietary Models							
GPT-4o-mini	23.53	16.60	23.27	30.66	17.86	21.94	26.00
GPT-5-mini	45.38	34.75	37.13	49.53	42.46	37.13	40.93
Gemini-2.0-Flash	31.93	24.32	27.23	31.13	26.19	24.47	27.43
Gemini-2.5-Flash	34.03	26.64	31.68	38.68	26.59	29.54	30.93
Gemini-2.5-Pro	37.14	45.45	36.36	37.14	23.91	24.44	33.61
Claude 3.5 Haiku	26.05	18.92	24.75	25.94	20.24	21.10	22.64
Mistral Medium 3.1	25.21	19.31	21.78	29.25	15.08	16.88	21.00
Open-Source Models							
InternVL3.5-1B	05.88	09.65	9.90	13.68	09.13	10.13	09.64
InternVL3.5-2B	12.18	11.20	10.89	23.58	11.90	18.14	14.50
Qwen-VL2.5-3B-Instruct	15.55	8.49	15.35	10.85	18.25	9.28	12.93
InternVL3.5-4B	19.33	17.76	15.84	19.81	16.27	18.99	18.00
Gemma-3-4B-it	20.17	13.13	14.85	23.58	15.08	19.83	17.64
Qwen-VL2.5-7B-Instruct	15.13	15.83	20.30	27.83	15.87	19.83	18.86
Llama-3.2-11B-Vision-Instruct	16.81	16.99	22.28	25.00	13.49	18.57	18.57
Gemma-3-27B-it	22.69	16.22	24.75	34.43	22.62	21.94	23.43
Qwen-VL2.5-32B-Instruct	16.39	09.65	14.85	16.98	09.92	13.50	13.36
InternVL3.5-72B	22.69	20.46	20.30	31.60	19.84	26.16	23.36
Qwen-VL2.5-72B-Instruct	26.89	20.85	25.25	30.66	24.60	20.68	24.64
Llama-3.2-90B-Vision-Instruct	22.69	23.17	21.29	28.30	21.83	27.00	24.00
GLM-4.5V-106B-MoE	31.09	20.46	25.25	26.42	24.21	24.47	25.21
Reasoning Models							
o3-mini	39.08	30.12	29.70	39.15	41.27	30.38	35.00
o4-mini-medium	40.76	32.82	32.18	42.92	44.05	34.18	37.86
Gemini-2-Flash-Thinking	31.09	27.41	31.19	34.43	29.37	29.54	30.36
Gemini-2.5-Flash-Thinking	37.14	45.45	36.36	37.14	21.74	22.22	32.77
Kimi-VL-A3B-Thinking-2506	13.45	14.29	11.88	17.92	12.70	18.57	14.79
GLM-4.1V-9B-Thinking	21.43	19.31	26.73	24.53	19.84	22.78	22.21
Step-3-321B-MoE	32.35	18.53	27.72	27.36	29.76	27.85	27.14
Spatial Reasoning Models							
SpaceOm	12.61	06.95	15.84	11.79	18.65	12.24	12.93
SpaceThinker-Qwen2.5VL-3B	13.45	09.27	17.82	10.38	19.44	10.13	13.36
SpaceQwen2.5-VL-3B-Instruct	12.61	03.86	13.86	09.43	11.90	11.39	10.36
Human Baseline	73.53	50.19	70.30	69.81	65.48	62.87	64.93

(several models exceed 60% on orientation or 3D geometry), whereas Spatial Navigation, Depth & Occlusion, and Size & Scale are more variable and generally harder across the board. Notably, the spatially specialized models (SpaceOm, SpaceThinker, SpaceQwen) achieve only mid-40s to low-40s overall, so specialization does not automatically translate to higher MCQ accuracy across our diverse question set. Depth & Occlusion and Size & Scale show mixed results, some architectures handle occlusion relatively well while others fail, pointing to differing inductive biases or training data. These patterns imply that instruction-tuning and reasoning capability particularly help geometric and orientation reasoning, while multi-step grounding tasks like navigation remain an open weakness.

SPATIALAB-OPEN. Open-ended evaluation scores presented in Table 3 are substantially lower than MCQ results and show a wide spread: overall accuracy ranges from about 9.6% (InternVL3.5-1B) up to 40.9% (GPT-5-mini), while the human baseline sits at 64.9%, indicating a large gap to close. Proprietary models generally lead the leaderboard on open-ended tasks (e.g., GPT-5-mini 40.93%, GPT-4o-mini 26.00%, Gemini-2.5-Flash 30.93%), but reasoning-tuned variants stand out in particular, the o4-mini (37.86%) and Gemini-2.5-Flash-Thinking (32.77%) perform noticeably better than many baseline open-source counterparts, suggesting instruction/CoT-style tuning substantially improves generative outputs. Most open-source small models cluster at the bottom (many in the teens; InternVL3.5-1B = 9.64%, Qwen-VL2.5-3B = 12.93%), whereas some very large open models improve to the low-20s (e.g., InternVL3.5-72B = 23.36%, Qwen-VL2.5-72B = 24.64%), showing that scale alone yields only modest gains. Spatial-specialist models (SpaceOm, SpaceThinker, SpaceQwen) attain only 10–13% overall, implying that architecture or task specialization does not automatically translate to robust open-ended generation across our diverse question set. Across subtasks we observe relatively better performance on orientation and certain relative-position items for top models (e.g., GPT-5-mini Rel.Pos = 49.53%, Ori = 37.13%), while depth & occlusion, size & scale, and especially spatial navigation remain consistently hard (many models score <30% on these subtasks). This pattern reinforces the earlier finding that multi-step grounding and perceptual reasoning are the bottlenecks for open generative answers.

5.2 PERFORMANCE DROP IN OPEN-ENDED EVALUATION COMPARED TO MCQ

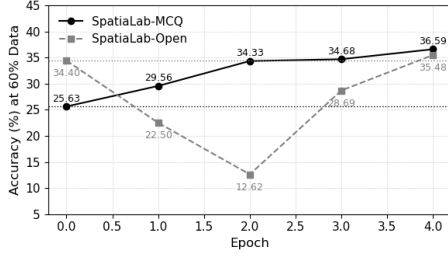
Across 25 models, the average MCQ→open-ended performance gap in SPATIALAB is 23.0% ($\sigma = 5.5\%$), with subtask means ranging from 22.89% (spatial navigation) to 24.57% (3D geometry). Specialist spatial reasoning models exhibit the largest gaps (about 27%), particularly in spatial navigation (up to 36.68%) and orientation (34.44%), while reasoning-oriented models achieve smaller gaps (around 19%) and lower subtask variance, reflecting the stabilizing effect of instruction-tuning and CoT decoding. Correlations confirm that spatial navigation dominates overall disparity (Pearson $r = 0.99$), with orientation ($r = 0.83$) and 3D geometry ($r = 0.79$) also contributing. Negative or near-zero gaps (e.g., Llama-3.2-11B: -1.98 in depth & occlusion; o4-mini: -3.18 in relative position) indicate that poorly designed MCQ distractors can misrepresent true competence. We hypothesize that format sensitivity arises from MCQ’s structural advantage, specialization bias toward categorical tasks, sequential reasoning challenges in spatio-navigation, and calibration differences in generative output. These findings suggest that MCQ alone can overestimate practical spatial reasoning ability. We recommend complementing MCQ with open-ended evaluations, auditing distractors, employing stepwise generation or instruction-tuning, and reporting per-subtask diagnostics to capture true model performance and reduce format-dependent artifacts. More details are in Appendix G and Table 18.

5.3 ERROR ANALYSIS

SPATIALAB-MCQ. Our analysis (Appendix E) shows that VLMs achieve selective peaks but lack holistic spatial competence. Closed-source models reach the highest scores (e.g., 85.71% in *Stacking Orientation*), yet collapse to near-chance in *Relative Size Comparison*. Open-source scaling improves ceilings (80.56% in *Corner/Angle Positioning*) but does not prevent catastrophic failures (2.0% in *Object Rotation*). Reasoning-augmented and spatially specialized models add localized gains, such as in occlusion inference or navigation, but remain capped below 55% in many physical abstraction tasks (*Gravity Effects*, *Stability Prediction*). Error distributions further reveal systematic weaknesses in embodied reasoning (*Tool Handedness*), recursive relational chaining (*Pathway Existence*), and non-local cues (*Reflective Surfaces*), while stronger results in *Obstacle Avoidance* suggest shortcut exploitation rather than robust planning. Taken together, these patterns confirm that current VLMs rely heavily on surface-level correlations and lack stable encodings of orientation, physics, and compositional logic. Progress will likely require both richer training distributions (e.g., embodied and physics-driven data) and architectural mechanisms for geometric and reference-frame grounding.

SPATIALAB-OPEN. Error analysis of SPATIALAB-OPEN (Appendix F) reveals that even top-performing closed-source models (e.g., GPT-5-mini: 58.14% in *Directional Relations*) collapse on tasks like *Proximity Gradients*, indicating that dominance remains task-specific. Large open-source models achieve strong results in *Depth & Occlusion* (e.g., 60.0%) but still fall to 9.3% on *Proximity*, showing that scale alone cannot ensure robustness. Smaller models exhibit isolated strengths (e.g., 26.0% in *Relative Size Comparison*) but often break down completely (e.g., 0.0% in *Betweenness*), reflecting the limitations of compact architectures. Reasoning-tuned systems show promise, Gemini-2.5-Flash-Thinking reaches 75.0% in *Tool Handedness*, yet similar models fall to single digits, revealing fragile integration. Specialized spatial models perform worst overall, rarely exceeding 20% and frequently failing on core relational tasks. Failure patterns highlight systemic weaknesses in occlusion handling, orientation, and multi-step relational chaining, while partial success in size-scale cues suggests reliance on superficial correlations. These results point to three root causes: missing representations for spatial analogies, architectural limits tied to scale and training diversity, and brittle reasoning pipelines underscoring the requirement for a unified geometric encodings, physics-aware reasoning, and embodied data to move from isolated peaks toward consistent spatial competence.

Qualitative Error Analysis. To better understand model weaknesses, we conducted a qualitative error analysis (see Appendix I) and found that failures cluster into a small number of recurring classes rather than being random noise. Common errors include spatial mislocalization, where models confuse referents in crowded scenes; perspective and scale mistakes, where reliance on object-size priors overrides image-based cues; and occlusion or ordering failures, especially with thin or partially hidden structures. We also observe attribute confusion, such as mixing perceptual properties with functional ones, and open-ended rationalizations that generate fluent but visually ungrounded narratives. These errors are consistent across architectures and prompting styles, suggesting structural biases in current VLMs rather than dataset-specific noise. Importantly, failure rates spike when multiple cues must be fused, such as combining depth ordering with relative size, and confidence



Category	MCQ			Open-ended		
	Before	After	Gain	Before	After	Gain
3D Geometry	30.07	37.76	7.69	20.28	23.78	3.50
Depth & Occlusion	23.87	32.90	9.03	43.23	45.81	2.58
Orientation	23.33	36.67	13.33	39.67	42.15	2.48
Relative Positioning	26.19	27.78	1.59	51.18	50.39	-0.79
Size & Scale	27.63	44.74	17.11	18.42	14.47	-3.95
Spatial Navigation	22.38	38.46	16.08	36.62	39.44	2.82
Grand Total	25.63	36.59	10.97	34.40	35.48	1.07

(a) SFT performance trends over epochs.

(b) Performance comparison (Before vs After) across categories for MCQ and Open-ended formats.

Figure 4: SFT training results: (a) learning trends over epochs, and (b) final accuracy values.

calibration is especially poor in open-ended settings. Our diagnostic protocols confirm that models often ignore minimal but decisive visual features, rely on brittle heuristics, and fail to update when counterfactual edits are applied. The root causes point to insufficient object-centric binding, lack of geometric supervision, and training objectives that reward plausibility over grounding. Together, these findings highlight that current VLMs achieve strong coarse perception but struggle with multi-cue integration and grounded reasoning, underscoring the need for geometry-aware supervision, multi-scale feature retention, and verification pipelines.

5.4 PERFORMANCE IMPROVEMENT APPROACHES

Inherent Reasoning Mechanisms. We notice that reasoning-enabled models consistently outperform their baselines in both MCQ and open-ended formats, with the largest gains in relational and orientation tasks (e.g., +13.1% in Relative Positioning for MCQ). While reasoning helps stabilize open-ended performance, improvements remain uneven across categories (notably declining in Size & Scale), underscoring that reasoning modules boost logical consistency but do not fully solve grounding or scale sensitivity. Detailed per-category comparisons are provided in Appendix H.1.

Chain-of-Thought (CoT) Prompting. Across models, CoT prompting provides little benefit and often reduces accuracy (Table 19), with orientation being the only category showing consistent gains. This indicates that while step-by-step reasoning helps with directional alignment, it fails in tasks requiring robust perceptual grounding such as depth, scale, and spatial navigation. Unlike in textual reasoning, CoT here tends to amplify flawed priors rather than correct them, underscoring that solving SPATIALAB requires perceptual understanding beyond logical chaining (details in Appendix H.2).

Chain-of-Thought (CoT) with Self-Reflection. Adding self-reflection to CoT prompting, where models are explicitly prompted to review their previous reasoning step-by-step and correct mistakes (see Figure 8 in Appendix for prompt), yields modest benefits in multiple-choice settings, especially for geometry and depth. However, this fails to generalize to open-ended tasks (Appendix H.3). This confirms that reflective reasoning cannot compensate for missing perceptual grounding, highlighting a core limitation of current VLMs.

Supervised Fine-Tuning (SFT). As shown in Figure 4, supervised fine-tuning (SFT), conducted on Qwen2.5-VL-3B-Instruct using a stratified 40% split of SPATIALAB (~560 samples), consistently improves MCQ accuracy across all spatial reasoning categories. However, it offers little, and sometimes negative, transfer to open-ended tasks. This divergence suggests overfitting to task-specific answer distributions, reinforcing concerns that SFT often *teaches to the test* rather than fostering generalizable reasoning (Wang et al., 2025). The sharp drop in open-ended performance points to biased internal representations, tuned for categorical discrimination but unstable in generative settings. A likely cause is catastrophic forgetting of linguistic priors during fine-tuning (Huang et al., 2024a), compounded by objectives that favor discrete-choice alignment over spatial reasoning in language. Overall, while SFT enhances performance in structured formats, it risks creating an algorithmic straightjacket, boosting accuracy in constrained tasks while stalling progress in naturalistic spatial reasoning. More details are available in Appendix H.4.

AI Agents for Spatial Reasoning (SPATIOXOLVER). To explore agentic capabilities, we employ SPATIOXOLVER, a multi-agent system adapted and extended from Xolver (Hosain et al., 2025) designed to perform structured spatial reasoning on images. This framework decomposes complex visual reasoning into specialized sub-tasks managed by dedicated agents, including modules for object segmentation, attribute extraction, spatial relation mapping, and transformation tracking. By

consolidating these outputs into a unified structured representation, the system attempts to propagate low-level perceptual cues through higher-order relational inferences. We implemented the pipeline using Gemini-2.5-Flash with low temperature to ensure deterministic outputs during the multi-stage refinement process. Experimental results demonstrate substantial gains in the Orientation category, improving by +8.00% in MCQ and an impressive +36.00% in open-ended evaluation. However, this success is not universal; categories such as Depth & Occlusion and Spatial Navigation suffered severe declines of -24.00% and -12.00% in open evaluation, respectively. These drops indicate that when inherent perceptual priors—such as depth or occlusion cues—are weak, multi-step reasoning tends to reinforce misconceptions rather than correct them. Ultimately, these findings suggest that while agentic workflows enhance tasks reducible to sequential alignment, they cannot compensate for a lack of fundamental perceptual grounding in complex 3D environments. More details are available in Appendix H.5.

6 DISCUSSION

Spatial reasoning is central to embodied intelligence, yet most benchmarks rely on synthetic scenes, templated questions, or narrow task scopes that overlook real-world visual variability. We introduce SPATIALAB, a benchmark of 1,400 visual question-answer pairs covering six major spatial categories and 30 subcategories, including 3D geometry, occlusion, and navigation. The benchmark uses both Multiple-Choice and Open-Ended formats to separate option-based pattern use from genuine generative reasoning. Its quality is supported by a multi-stage review process and complexity checks covering lighting, texture, and spatial diversity. Evaluations of more than 25 contemporary models and human baselines show that SPATIALAB functions as a precise diagnostic tool for identifying weaknesses in current vision-language systems.

Detailed error analysis uncovers systematic failures in handling transparency, multi-step navigation, and occlusion, indicating that models often rely on surface-level texture cues rather than internalized 3D geometric representations. While interventions such as Supervised Fine-Tuning (SFT) yielded clear gains in structured MCQ tasks, they simultaneously triggered instability in open-ended reasoning, suggesting a “catastrophic forgetting” of linguistic priors when optimizing for discrete answers. Furthermore, reasoning-based approaches like Chain-of-Thought proved effective for sequential tasks like Orientation but failed to correct fundamental perceptual errors in Depth and Scale, often amplifying hallucinations instead of resolving them. Agentic decompositions showed similar trade-offs, improving specific alignment tasks while degrading performance in holistic scene understanding contexts. These divergent behaviors confirm that scaling laws and reasoning scaffolds alone are insufficient; achieving human-level spatial intelligence will require architectural innovations that explicitly integrate geometric grounding and physics-aware representations.

7 CONCLUDING REMARKS

Spatial reasoning remains a core bottleneck for VLMs, particularly in visually complex, real-world settings. SPATIALAB introduces 1,400 diverse QA pairs across six spatial categories and thirty subcategories, enabling fine-grained evaluation via both MCQ and open-ended formats. Despite applying several performance-improving methods, our experiments reveal substantial gaps between state-of-the-art models and human performance, with open-ended generation showing an additional 10–25% drop relative to multiple-choice accuracy. Error analysis reveals persistent failures in depth, occlusion, 3D geometry, and multi-step navigation, reflecting reliance on superficial cues over grounded spatial representations. SPATIALAB exposes these limitations and provides a structured testbed for diagnosing spatial competence. Our benchmark enables holistic evaluation and provides actionable insights for developing VLMs with human-aligned spatial understanding. Future progress will require richer training signals (e.g., embodied interaction, action-conditioned feedback) and architectures capable of geometric grounding, relational chaining, and consistent multi-scale reasoning.

ETHICS STATEMENT

We have taken multiple steps to ensure ethical compliance throughout the development of this work. All metadata associated with the images was removed to protect privacy, and any personally identi-

able features (such as human faces or license plates) were minimized or excluded wherever possible. The dataset was carefully curated to avoid copyright concerns, with diligent attention paid to sourcing and licensing, though minor oversights may occur despite best efforts. The benchmark construction process was designed with transparency and fairness in mind, ensuring that no demographic groups were singled out or negatively represented. We adhered to the ICLR Code of Ethics and confirm that no human subjects or sensitive personal data were involved.

REPRODUCIBILITY STATEMENT

To facilitate reproducibility, we provide a detailed description of the dataset construction process, annotation pipeline, and evaluation protocol in the main text and appendix. All filtering, quality-control, and validation steps are explicitly documented, with clear definitions of task categories and annotation criteria. Hyperparameters, model configurations, and training procedures are reported in the appendix to support consistent replication of our results.

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A RELATED WORK

Spatial reasoning in multimodal models. Spatial reasoning, the capacity to apprehend geometric structure and relational layout within visual scenes, constitutes a cornerstone of embodied intelligence. Early multimodal benchmarks such as CLEVR (Johnson et al., 2016) and GQA (Hudson & Manning, 2019) probed compositional reasoning through synthetic renderings or curated templates; while such controlled conditions enabled precise analysis, they afforded limited ecological validity. The emergence of large vision–language models (VLMs) (Li et al., 2023; Alayrac et al., 2022; OpenAI, 2023) has prompted a reappraisal of spatial reasoning within more naturalistic datasets, yet systematic weaknesses persist. Despite commanding performance on captioning and visual question answering, these models fail consistently on tasks that demand depth ordering, occlusion inference, or coherent reference-frame alignment. This dissociation exposes a fundamental limitation: scale alone does not confer progress in structured spatial understanding.

Spatial reasoning benchmarks. A succession of benchmarks has recently been introduced to interrogate these gaps (Table 1). SCANQA (Azuma et al., 2022) numbered among the earliest embodied QA datasets, requiring reasoning over 3D indoor environments; however, its modest scale (800 scenes) and comparatively shallow complexity curtail broader generalization. Datasets such as VISUAL SPATIAL (Liu et al., 2023) and WHAT’S UP (Kamath et al., 2023) broadened the scope to real-world and household contexts, yet their reliance on template-driven questions attenuated linguistic variability and overrepresented elementary binary or multiple-choice relations. More recent efforts, including EMBSPATIAL-BENCH (Du et al., 2024) and SPACE3D-BENCH (Szymanska et al., 2024), enriched task variety but remained circumscribed by scale (2.2K and 211 samples, respectively) and annotation style, whether template-heavy or manual yet narrow. SPATIALRGPT-BENCH and BLINK-SPATIAL contributed mixed domains and up to 14 categories, advancing diversity, but continued to privilege either static snapshots or limited multi-view simulation.

Embodiment, difficulty, and diversity. The movement toward embodiment is evident in datasets such as ROBOSPATIAL (Song et al., 2024), which scales to 1M scenes with robot-centric questions. Its template-based annotation, however, renders questions less natural and susceptible to heuristic shortcuts. SPATIALVLM (Chen et al., 2024b) likewise offers unprecedented scale (billions of QA pairs), yet its two-category coverage precludes fine-grained diagnostic evaluation. At the opposite extreme, high-quality manual datasets such as VSI-BENCH (Yang et al., 2024) attain superior annotation fidelity but, with only 288 scenes, remain inadequate for training or robust cross-model comparison. With respect to complexity, most existing benchmarks plateau at “medium” (Mm in Table 1), seldom reaching the “hard” or “extremely hard” regimes that necessitate multi-step causal inference. A substantial evaluation gap thus persists for next-generation models.

Evaluation protocols. Evaluation protocol and analytic depth are of comparable consequence. The preponderance of prior work adopts close-ended QA or binary judgments (e.g., VISUAL SPATIAL, EMBSPATIAL-BENCH), which streamline scoring at the expense of output expressivity. Although a handful of datasets, including ROBOSPATIAL, SPATIAL-MM, and SPATIALAB-OPEN, admit open-ended evaluation, only SPATIALAB-OPEN pairs it with extreme complexity, manual puzzle-style annotation, and a comprehensive treatment of error patterns. This combination surfaces systematic weaknesses that lie beyond raw accuracy, spanning calibration and cross-model divergence, which existing datasets seldom capture. Absent such granularity, reported gains can prove misleading, as models may overfit dataset-specific biases rather than acquire robust, transferable spatial competence.

Positioning of SPATIALAB. SPATIALAB resolves these limitations directly. Whereas prior datasets span between 2 and 14 task categories, SPATIALAB encompasses 30 distinct categories across geometry, occlusion, orientation, relative positioning, size and scale, and navigation, delivering unmatched task diversity. In contrast to template-based benchmarks, every question is manually authored and puzzle-inspired, securing linguistic richness while foreclosing shortcut exploitation. SPATIALAB is moreover fully embodied and multi-modal, incorporating both static and dynamic visual contexts, and thereby bridges the divide between robotics-style datasets (ROBOSPATIAL) and web-scale corpora (SPATIALVLM). As Table 1 attests, even the strongest baselines, InternVL3.5-72B and GPT-5-mini, remain well below 60%, and the open-ended variant yields only 40.93% accuracy.

This pronounced performance gap establishes both the difficulty of the benchmark and its value for driving future research.

Toward next-generation evaluation. By uniting high difficulty, broad coverage, manual annotation, and open-ended tasks, SPATIALAB transforms the spatial reasoning landscape from template-driven, narrow evaluations into comprehensive, embodied testing. This design permits not merely a verdict on whether a model answers correctly, but the identification of systematic error patterns, the interrogation of causal and counterfactual reasoning, and the measurement of calibration reliability. SPATIALAB accordingly furnishes a rigorous foundation for the next phase of multimodal research, complementing datasets such as MIND THE GAP and VLM4D that probe temporal or generative facets yet remain narrower in coverage. We anticipate that future work will adopt SPATIALAB as a high-bar diagnostic instrument, propelling models beyond surface-level recognition toward genuine grounded spatial intelligence.

A.1 DETAILED ANALYSIS OF LIMITATIONS IN PRIOR BENCHMARKS

To situate SPATIALAB within the broader landscape, we undertake a systematic examination of existing spatial reasoning benchmarks. Prior efforts have laid important foundations, yet each carries structural and methodological constraints that curtail its value for evaluating contemporary VLMs. We organize these constraints along three axes: synthetic-domain bias, restricted task scope, and constrained evaluation formats. Table 4 consolidates the principal limitations identified.

Synthetic and Template-Based Constraints. Benchmarks such as CLEVR and GQA depend on synthetic environments or template-generated linguistic structures. While these settings afford controlled conditions for isolating compositional reasoning, their visual domain lacks real-world complexity and their linguistic space remains narrow, so models may exploit template regularities in lieu of grounded spatial interpretation. Related simulated benchmarks such as ROBOSPATIAL exhibit the same failure mode, incurring a pronounced sim-to-real gap.

Domain and Scale Limitations. Datasets grounded in 3D reconstruction, including ScanQA and Space3D-Bench, supply depth supervision but confine themselves to indoor scenes, precluding assessment in the outdoor and mixed environments that embodied agents must ultimately navigate. High-quality manual datasets such as VSI-Bench deliver richer annotation yet remain small in scale, which constrains their utility for high-resolution subcategory analysis.

Taxonomic and Evaluation Narrowness. Large-scale datasets such as SpatialVLM span broad tasks but lack fine-grained taxonomies, rendering them ill-suited to diagnosing specific weaknesses in orientation, occlusion, or layout reasoning. Many recent benchmarks, including EMBSPATIAL-BENCH and WHAT’S UP, lean heavily on Binary or MCQ formats, which can inflate measured performance by rewarding option elimination rather than genuine generative spatial reasoning. SPATIALAB redresses these deficiencies by uniting MCQ and Open-Ended formats across a detailed taxonomy of 30 spatial subcategories.

Table 4: Limitations identified in representative prior spatial benchmarks set against SPATIALAB, organized by domain constraints, evaluation format, and task taxonomy. SPATIALAB alone combines in-the-wild imagery, a 30-subcategory taxonomy, and paired MCQ and open-ended evaluation.

Benchmark	Domain	Eval. Format	Taxonomy	Primary Limitations Identified
CLEVR and GQA	Synthetic	Closed-Ended	Relational	Artificial visual domain and template-driven language invite shortcut learning.
ScanQA	Indoor 3D	Open-Ended	Navigation	Confined to indoor point clouds at limited scale ($< 1K$ scenes).
SpatialVLM	Web Data	Open-Ended	General	Spans only a handful of high-level spatial categories.
EmbSpatial-Bench	Indoor	MCQ / Binary	6 Categories	Depends on closed-ended formats with shallow reasoning chains.
RoboSpatial	Simulator	Hybrid	Embodied	Template-generated questions in simulated scenes yield a sim-to-real gap.
Space3D-Bench	Indoor	MCQ	6 Categories	Very small scale (~ 200 samples) precludes robust statistical evaluation.
OmniSpatial	Mix	MCQ / Binary	Metric	Prioritizes metric outputs with scant support for explanatory reasoning.
SPATIALAB (Ours)	In-the-Wild	MCQ + Open	30 Sub-cats	Resolves the identified limitations through real-world imagery, a fine-grained taxonomy, and paired evaluation formats.

B TASK DESIGN AND DESCRIPTIONS

Spatial reasoning demands far more than the recognition of individual objects; it hinges on how those objects relate, align, and interact within physical space. To capture this complexity, we introduce *SpatialLab*, a benchmark expressly designed to probe fine-grained facets of spatial understanding across

a diverse taxonomy of categories. Its tasks mirror the ways humans naturally describe and reason about their surroundings, spanning relative positioning (e.g., *What's to the left of the microwave?*) through to geometric inference (e.g., *Would this stack fall if another book is added?*). Every category decomposes into subcategories that isolate subtle yet essential dimensions of reasoning, including occlusion, orientation, navigation, and physical effects such as gravity. By articulating each subtask in accessible, everyday terms, SpatiaLab compels models to move beyond mere object detection and engage in structured spatial interpretation. This design lets us rigorously assess how well systems generalize across the contexts that matter most for robotics, AR/VR, autonomous driving, and human-computer interaction. The sections that follow provide concise, task-level descriptions for every category and subcategory in SpatiaLab.

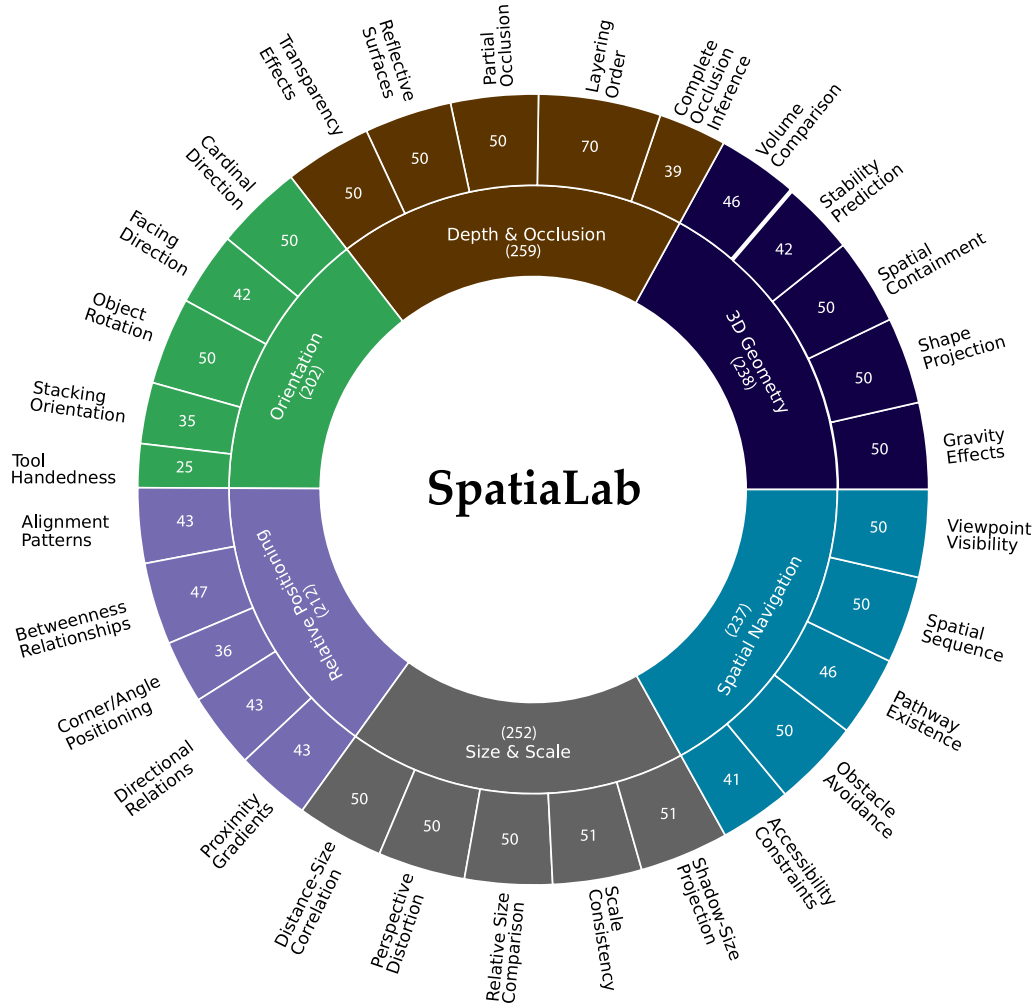


Figure 5: **Categories and subcategories of spatial reasoning in SPATIALAB.** Each category decomposes into five subcategories, yielding thirty task types in total.

We now articulate the rationale and real-world utility that motivate each task design, organized by category:

B.1 RELATIVE POSITIONING

Directional Relations. This subtask targets the directional predicates that situate objects relative to one another, such as left, right, above, and below. Mastering these predicates allows a model to parse spatial layouts as fluently as a human narrator would, grounding descriptions in a shared frame of

reference. Resolving what lies immediately to the left of an appliance, for instance, is foundational to scene understanding and to robot navigation.

Proximity Gradients. Proximity reasoning requires ordering objects by their closeness to a designated reference point, isolating the relative-distance dimension of spatial cognition. It underlies queries such as “Which object is nearest to the doorway?” Models capable of estimating such gradients of closeness directly support navigation tasks and assistive technologies.

Alignment Patterns. Alignment concerns whether a set of objects conforms to a common orientation or ordering, for instance being stacked vertically or arranged along a straight line. Detecting such regularities reveals how environments are organized, from shelves and tables to warehouses. This capacity is equally central to robotic sorting and automated inspection.

Betweenness Relationships. Betweenness isolates the relation of interposition, identifying the object that occupies the intermediate region between two references. Determining what sits between the lamp and the armchair, for example, demands reasoning about spatial continuity along an axis. Such judgments feed directly into navigation, retrieval, and situational-awareness tasks.

Corner/Angle Positioning. Corners and angled locations serve as salient reference anchors in spatial reasoning. Questions such as “What’s positioned in the back-right corner?” require the model to localize an object within a tightly constrained region of the scene. This form of reasoning is indispensable in robotics, where it underpins mapping and placement.

B.2 DEPTH & OCCLUSION

Layering Order. Layering demands ordering objects along the depth axis, distinguishing what stands in front from what recedes behind. Identifying the frontmost item on a table, for example, is a direct expression of depth reasoning. This skill is especially consequential in AR/VR applications, where visual stacking shapes the user experience.

Partial Occlusion. Partial occlusion arises when one object conceals a portion of another from view, forcing the model to infer what remains unseen. Recovering the identity of an item partly hidden behind a coffee cup, for instance, requires reconstructing missing evidence. This inferential robustness is vital for perception systems operating in cluttered environments.

Complete Occlusion Inference. When an object is fully occluded, its presence may still be betrayed by indirect cues such as cast shadows or telltale bulges. Inferring what lies beneath a blanket compels the model to reason from such indirect evidence rather than direct observation. This mirrors the way humans draw confident inferences under uncertain visual conditions.

Transparency Effects. Transparent media, such as glass or water, permit partial visibility of whatever lies behind them. Determining what is discernible through a fish tank requires fusing occlusion reasoning with the perception of transparency. This interplay is pivotal for simulation, graphics, and embodied AI.

Reflective Surfaces. Reflective reasoning asks models to interpret mirrors, polished metals, and glass reflections as sources of spatial information. Recognizing what is reflected in a mirror entails an indirect mapping between a virtual image and its physical origin. This competence is essential to safety in autonomous vehicles and to situational awareness in robotics.

B.3 ORIENTATION

Cardinal Direction. Cardinal reasoning grounds objects in compass directions such as north, south, east, and west. Establishing which way a front door faces requires integrating global orientation cues into a consistent frame. This capability is critical for navigation and geospatial alignment.

Object Rotation. Rotation reasoning quantifies how far an object has been turned about an axis. Estimating that a chair has been rotated by a given angle is essential to accurate 3D modeling. Such estimates enable robots to align objects correctly prior to manipulation.

Facing Direction. Facing direction pins down the orientation of an object’s front-facing surface. Deciding whether a monitor faces north, for example, requires mapping that surface orientation onto global references. This reasoning is important for surveillance and for robotics.

1296 **Stacking Orientation.** Stacking orientation captures whether objects are piled vertically, horizontally,
1297 or in mixed configurations. Correctly reading these arrangements is what ensures stability in settings
1298 from warehouses to homes. It is likewise critical to safe robotic stacking.

1299 **Tool Handedness.** Many tools, such as scissors or gloves, are engineered for either left- or right-
1300 handed use. Discerning their handedness calls for fine-grained perception of shape and orientation.
1301 This precision advances assistive robotics in human-centered environments.

1302

1303 **B.4 SIZE & SCALE**

1304

1305 **Relative Size Comparison.** Relative comparison determines whether one object is larger, taller,
1306 or smaller than another. Judging whether a vase stands taller than a statue exercises proportional
1307 reasoning. This ability is crucial to packing, fitting, and recognition tasks.

1308 **Perspective Distortion.** Perspective can render distant objects apparently smaller than nearer ones
1309 even when their true dimensions say otherwise. Recognizing illusions such as a faraway car appearing
1310 smaller than a nearby bicycle demands compensating for depth cues. Mastering this makes models
1311 robust to the vagaries of camera perspective.

1312 **Distance-Size Correlation.** Here size judgments are coupled to distance estimation, so that apparent
1313 scale must be disentangled from true scale. A nearby bicycle looming larger than a distant car should
1314 not mislead reasoning about actual size. Getting this right benefits navigation, AR/VR, and visual
1315 grounding.

1316 **Scale Consistency.** Scale consistency tests whether object sizes are plausible given their context.
1317 A mouse rendered larger than a chair signals a violation that models must detect. This vigilance
1318 strengthens realism in simulated environments.

1319 **Shadow-Size Projection.** Shadows encode indirect cues about both object size and the position of
1320 the light source. Inferring how high a light must be from the length of a cast shadow requires explicit
1321 geometric reasoning. This is valuable in graphics, architecture, and physics-based modeling.

1322

1323 **B.5 SPATIAL NAVIGATION**

1324

1325 **Pathway Existence.** Pathway reasoning determines whether an unobstructed route connects two
1326 points in a scene. Judging whether one can reach the desk without moving the chairs directly probes
1327 navigability. This capacity underpins indoor robot planning.

1328 **Obstacle Avoidance.** Avoidance requires selecting routes that minimize the risk of collision. Re-
1329 questing the clearest path from door to window maps directly onto both robotic and human navigation.
1330 It is essential in autonomous driving and indoor mobility aids.

1331 **Viewpoint Visibility.** From any given viewpoint, some objects fall into view while others are blocked
1332 from it. Determining what is obscured when seated on the couch requires simulating that specific
1333 perspective. This strengthens AR/VR realism and robotic vision.

1334 **Spatial Sequence.** Sequence reasoning establishes the order in which objects are encountered along
1335 a trajectory. Walking from the kitchen to the balcony, one would pass the dining table first. Such
1336 ordering mirrors the navigational memory that humans rely on.

1337 **Accessibility Constraints.** Constraint reasoning identifies which objects are rendered inaccessible
1338 by intervening obstacles. A cabinet, for instance, may be unreachable because a table stands in front
1339 of it. This is crucial for assistive AI and ergonomic design.

1340

1341 **B.6 3D GEOMETRY**

1342

1343 **Volume Comparison.** Volume comparison asks which object occupies the greater share of three-
1344 dimensional space. Judging whether a box exceeds a sphere in volume supports packing and
1345 containment tasks. This is important for logistics and simulation.

1346 **Stability Prediction.** Stability reasoning forecasts whether adding or displacing objects will topple
1347 a stack. Anticipating whether placing a book will collapse a pile mirrors human physical intuition.
1348 This supports safety in robotics and construction.

1349

Shape Projection. Projection tasks ask what silhouette an object would cast under a given rotation or illumination. A sphere, for instance, always projects a circle, whereas a cube may project a square or a hexagon. Reasoning of this kind strengthens genuine geometric understanding.

Spatial Containment. Containment evaluates whether one object can be accommodated inside another. Reasoning about whether a ball fits within a box supports packing and design. This task aligns closely with intuitive geometry.

Gravity Effects. Gravity reasoning predicts how objects move once physical forces act upon them. A rolling ball, for example, may come to rest beneath a table owing to friction and intervening barriers. This strengthens real-world simulation capabilities.

C MORE DETAILS ON BENCHMARK CONSTRUCTION

C.1 ANNOTATOR RECRUITMENT

Our annotation cohort was composed exclusively of engineering graduates between the ages of 24 and 30, a deliberately curated pool whose technical background equipped them to reason precisely about spatial structure. All annotators participated on a strictly voluntary basis and worked independently, without external incentives, a design choice that safeguarded the integrity and impartiality of their contributions. Prior to commencing annotation, each participant completed a structured training regimen calibrated to establish an unambiguous, shared understanding of the task requirements and to enforce consistency across the cohort. Throughout the entire process, we adhered rigorously to established ethical and quality-control standards, ensuring that every stage of data collection satisfied a uniformly high bar. Taken together, these measures yielded annotations that are demonstrably reliable and tightly aligned with the design principles underpinning the dataset.

C.2 ANNOTATOR TRAINING AND PROTOCOL

In advance of any annotation activity, we architected a comprehensive, structured training program engineered to guarantee consistency and reliability across the entirety of the task taxonomy. Annotators were first systematically oriented to the six core categories (*Relative Positioning, Depth & Occlusion, Orientation, Size & Scale, Spatial Navigation, and 3D Geometry*) and the full complement of thirty subcategories that they subsume. For each subcategory, we furnished carefully curated exemplars that made the governing spatial relations explicit (e.g., directional relations, occlusion inference, stability prediction), and we paired these with deliberately constructed counter-examples that foregrounded the most common failure modes and interpretive pitfalls. Extending beyond the taxonomy proper, we trained annotators to reason systematically across six meta-dimensions of image variability, namely lighting, texture complexity, edge density, dominant relation, material type, and gravity constraints, so that the questions they authored would deliberately exploit the full breadth of visual diversity. We further illustrated how these dimensions interact in authentic real-world settings, for instance, adjudicating occlusion under low-contrast lighting or preserving scale consistency in the presence of perspective distortion, thereby anchoring abstract principles in concrete perceptual challenges.

The training additionally encompassed the two complementary evaluation modes that define the benchmark: multiple-choice and open-ended. Within the multiple-choice regime, annotators were instructed in the craft of engineering precise, targeted distractors capable of probing fine-grained perceptual discrimination; within the open-ended regime, they practiced formulating prompts calibrated to elicit compositional, multi-step reasoning. To operationalize and consolidate this training, we conducted dedicated calibration sessions in which annotators produced both question types for a common set of sample images and iteratively refined their designs in response to structured feedback, progressively converging toward a shared standard of quality. To situate task design within the broader research landscape, we jointly reviewed representative prior benchmarks (e.g., SpatialBot-Bench, EmbSpatial, SpatialMM) with the annotators and conducted a critical analysis of their limitations. Such resources frequently privilege simplified relations, synthetic or staged scenes, and a narrow repertoire of reasoning types, while systematically neglecting the natural complexity introduced by phenomena such as shadows, reflections, and gravity-sensitive stability. This deliberate gap analysis crystallized the imperative for spatial contexts that are richer, more genuinely ambiguous, and dynamically layered, and it directly motivated the design decisions embodied in our benchmark.

Finally, we delivered exhaustive demonstrations of the annotation protocol itself. These demonstrations codified precise criteria for selecting images of sufficient spatial complexity, for mapping each question unambiguously to its governing subcategory, and for scrupulously avoiding trivial or under-constrained formulations. Annotators were explicitly directed to privilege tasks demanding genuine multi-step reasoning, such as depth layering, relational chains, and cross-view scale inference, in preference to reductive binary labels. Through this systematic training coupled with sustained iterative feedback, we ensured that the resulting dataset achieves a principled equilibrium between comprehensive taxonomic coverage and authentic real-world relevance.

D MORE DETAILS ON EXPERIMENTS AND EVALUATION

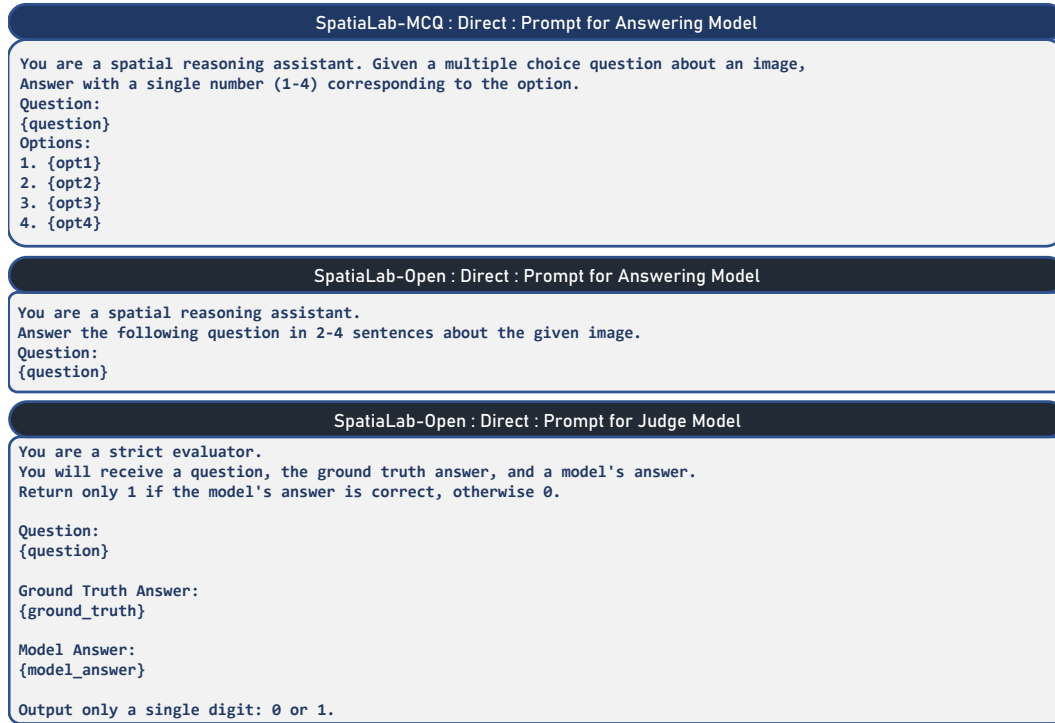


Figure 6: Prompts used for evaluation.

D.1 MULTIPLE CHOICE EVALUATION

The prompts supplied to the answer model are reproduced in Figure 6. To render decoding fully deterministic and thereby guarantee reproducible predictions, we fix `temperature` to 0 and cap `max_tokens` at 10, retaining the default settings for all remaining generation parameters.

D.2 OPEN-ENDED EVALUATION

The prompts governing both the answer model and the judge model are documented in Figure 6. To accommodate the free-form nature of open-ended responses, we set `temperature` to 0.7 and raise `max_tokens` to 1200, leaving every other generation parameter at its default value.

D.2.1 EVALUATION OF LLM-AS-A-JUDGE FOR OPEN-ENDED EVALUATION

To establish the trustworthiness of *Gemini-2.5-Flash* as an automatic judge, we conducted a controlled validation study on a balanced set of 240 samples that spans every category. Each sample comprised a complete evaluation instance, namely the question, its ground truth, and the model output, and was scored independently by three human annotators. To remove ambiguity from the annotation

protocol, judgments were recorded as binary labels, with 1 denoting a correct response and 0 an incorrect one. We then benchmarked the judge’s decisions against these human annotations through a suite of complementary agreement metrics, computed both in aggregate and at the level of individual categories. Concretely, we adopt **Cohen’s Kappa**, **Accuracy with Majority Vote**, and **Fleiss’ Kappa**, each defined precisely below.

Cohen’s Kappa. Cohen’s kappa (κ) quantifies the pairwise agreement between two raters after discounting the agreement expected from chance alone. For a binary task it is defined as $\kappa = \frac{p_o - p_e}{1 - p_e}$, in which p_o denotes the observed agreement and p_e the agreement anticipated by chance, $p_e = \sum_{c \in \{0,1\}} p_c^{(1)} p_c^{(2)}$. Here $p_c^{(1)}$ and $p_c^{(2)}$ are the marginal probabilities with which each rater assigns class c . The coefficient admits an intuitive scale: $\kappa = 1$ signals perfect agreement, $\kappa = 0$ indicates agreement no better than chance, and $\kappa < 0$ reflects systematic disagreement. We evaluated Cohen’s kappa between the judge and each annotator individually (κ_{Human1} , κ_{Human2} , κ_{Human3}), and likewise between the judge and the annotators’ majority vote (κ_{majority}).

Accuracy with Majority Vote. As a complementary, uncorrected measure, we also quantify the raw agreement between the judge and the annotator majority. Accuracy is defined as $\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}(y_i^{\text{LLM}} = y_i^{\text{maj}})$, where y_i^{LLM} is the judge’s decision for sample i and y_i^{maj} is the majority vote among the annotators. In contrast to Cohen’s kappa, accuracy applies no correction for chance; it reports directly the proportion of labels on which the judge and the majority coincide.

Fleiss’ Kappa. To gauge the reliability of the human reference itself, we computed Fleiss’ kappa (κ_F), which extends Cohen’s kappa from a pair of raters to an arbitrary number of them. Let n_{ij} denote the number of annotators assigning category j to item i , and define $P_i = \frac{1}{n(n-1)} \sum_j n_{ij}(n_{ij} - 1)$, in which n is the number of annotators. The observed agreement is the mean of these per-item quantities, $\bar{P} = \frac{1}{N} \sum_{i=1}^N P_i$, while the chance-expected agreement is $\bar{P}_e = \sum_j p_j^2$, with $p_j = \frac{1}{Nn} \sum_i n_{ij}$. Fleiss’ kappa follows as $\kappa_F = \frac{\bar{P} - \bar{P}_e}{1 - \bar{P}_e}$. Larger values attest to stronger inter-annotator reliability beyond what chance would produce.

Table 5: Agreement between Gemini-2.5-Flash as an automatic judge and three human annotators, reported both overall and for every category. For each row we list Cohen’s kappa against the majority vote and against each annotator individually, raw accuracy relative to the majority, and Fleiss’ kappa quantifying inter-annotator reliability. Bold values mark the strongest agreement attained within each category.

Category	κ_{Majority}	κ_{Human1}	κ_{Human2}	κ_{Human3}	Acc. (majority)	Fleiss κ (all)
Overall	0.738	0.738	0.681	0.795	0.880	0.774
3D Geometry	1.000	1.000	0.875	1.000	1.000	0.916
Depth & Occlusion	0.805	0.805	0.899	0.697	0.909	0.871
Orientation	0.500	0.500	0.426	0.588	0.810	0.571
Relative Positioning	0.833	0.657	0.676	0.833	0.917	0.778
Size & Scale	0.417	0.571	0.417	0.696	0.714	0.704
Spatial Navigation	0.815	0.815	0.667	1.000	0.933	0.773

Interpretation Table 5 establishes that the LLM judge (Gemini-2.5-Flash) is in strong agreement with the human annotators, attaining a Cohen’s kappa of 0.738 against the majority vote and a range of 0.681–0.795 against the annotators taken individually. Overall raw accuracy relative to the majority reaches 0.880, and the Fleiss’ kappa of 0.774 among the human annotators evidences substantial consistency in the reference itself. Several categories are especially compelling: 3D Geometry achieves perfect agreement, with $\kappa = 1.000$ and an accuracy of 1.000, while Depth & Occlusion likewise records high agreement ($\kappa_{\text{majority}} = 0.805$, accuracy = 0.909). By contrast, Orientation ($\kappa = 0.500$, accuracy = 0.810) and Size & Scale ($\kappa = 0.417$, accuracy = 0.714) exhibit only moderate kappa agreement, though both remain within acceptable reliability ranges.

Taken together, these results demonstrate that Gemini-2.5-Flash scores correctness on a par with human evaluators: its Cohen’s kappa values fall predominantly within the substantial range (0.61–0.80),

Table 6: Multiple-choice Evaluation Accuracy on SPATIALAB-MCQ by Question Sub-Categories for Closed-source Models. We bold and underline the best score within each model category.

Category	Sub-Category	GPT-4o-mini	GPT-5-mini	Gemini-2.0-Flash	Gemini-2.5-Flash	Claude 3.5 Haiku	Mistral Medium 3.1
3D Geometry	Gravity Effects	52.0	50.0	46.0	52.0	52.0	46.0
	Shape Projection	54.0	48.0	50.0	50.0	46.0	60.0
	Spatial Containment	36.0	46.0	44.0	38.0	34.0	38.0
	Stability Prediction	52.38	52.38	59.52	50.0	45.24	52.38
	Volume Comparison	41.3	47.83	39.13	34.78	34.78	36.96
Depth and Occlusion	Complete Occlusion Inference	61.54	69.23	61.54	61.54	61.54	46.15
	Layering Order	32.86	55.71	50.0	42.86	32.86	45.71
	Partial Occlusion	42.0	54.0	64.0	50.0	44.0	52.0
	Reflective Surfaces	34.0	54.0	46.0	48.0	44.0	44.0
	Transparency Effects	32.0	44.0	60.0	44.0	36.0	62.0
Orientation	Cardinal Direction	26.0	48.0	44.0	30.0	44.0	38.0
	Facing Direction	45.24	57.14	57.14	47.62	35.71	45.24
	Object Rotation	40.0	60.0	50.0	46.0	38.0	44.0
	Stacking Orientation	85.71	77.14	62.86	65.71	68.57	62.86
	Tool Handedness	52.0	68.0	64.0	64.0	56.0	56.0
Relative Positioning	Alignment Patterns	44.19	58.14	55.81	53.49	46.51	58.14
	Betweenness Relationships	51.06	63.83	70.21	63.83	44.68	61.7
	Corner/Angle Positioning	47.22	63.89	50.0	58.33	50.0	77.78
	Directional Relations	48.84	79.07	67.44	69.77	53.49	69.77
	Proximity Gradients	44.19	48.84	44.19	34.88	37.21	44.19
Size and Scale	Distance-Size Correlation	56.0	56.0	56.0	42.0	34.0	42.0
	Perspective Distortion	50.0	40.0	46.0	36.0	38.0	34.0
	Relative Size Comparison	42.0	44.0	58.0	46.0	32.0	38.0
	Scale Consistency	47.06	45.1	47.06	39.22	33.33	41.18
	Shadow-Size Projection	52.94	39.22	64.71	49.02	41.18	52.94
Spatial Navigation	Accessibility Constraints	56.1	60.98	65.85	65.85	48.78	56.1
	Obstacle Avoidance	38.0	48.0	34.0	34.0	46.0	40.0
	Pathway Existence	54.35	41.3	43.48	45.65	39.13	36.96
	Spatial Sequence	66.0	78.0	48.0	62.0	56.0	44.0
	Viewpoint Visibility	36.0	54.0	48.0	50.0	40.0	34.0

and several categories rise into the almost-perfect range (above 0.81). The Fleiss’ kappa of 0.774 further confirms that the annotators are highly consistent among themselves, which in turn validates the reliability of the reference scores against which the judge is measured. The judge secures perfect or near-perfect agreement in well-specified categories such as 3D Geometry ($\kappa = 1.000$, accuracy = 1.000) and Spatial Navigation ($\kappa = 0.815$, accuracy = 0.933), evidencing dependable recognition of the intended patterns. Where agreement is merely moderate, as in Orientation and Size & Scale, the residual gap is most plausibly attributable to the intrinsic difficulty and subjectivity of those tasks rather than to any deficiency of the judge. On balance, the validation confirms that the LLM-as-judge is a robust and reliable scoring instrument, one that closely mirrors human evaluation and can be deployed for automated scoring across all categories of the benchmark.

E ERROR ANALYSIS OF SPATIALAB-MCQ

E.1 SUB-CATEGORY-WISE QUANTITATIVE ERROR ANALYSIS

Closed-source Models. Closed-source models (Table 6) anchor the upper envelope of the benchmark, establishing the strongest ceilings observed anywhere in SPATIALAB-MCQ. The apex is set by **GPT-4o-mini**, whose 85.71% in *Stacking Orientation* constitutes the single highest score across every model and subcategory. This peak is not isolated: **Gemini-2.0-Flash** demonstrates command of occlusion and projective inference, recording 64.0% in *Partial Occlusion* and 64.71% in *Shadow-Size Projection*, while **GPT-5-mini** extends the trend into embodied and layering tasks with 68.0% in *Tool Handedness* and 55.71% in *Layering Order*. These selective triumphs, however, coexist with pronounced fragility, epitomized by **Claude 3.5 Haiku**, which attains merely 32.0% in *Relative Size Comparison*. Aggregated across categories, closed-source systems lead their open-source counterparts by a substantial 15–25 percentage points, yet the within-model dispersion frequently exceeds 50 percentage points. The diagnostic verdict is therefore unambiguous: proprietary training pipelines elevate peak attainment without conferring the uniform spatial competence that robust reasoning demands.

Open-source Large Models. Scaling to large open-source models (Table 7) delivers demonstrable gains in peak attainment while simultaneously exposing severe representational fragilities. At the high end, **InternVL3.5-72B** secures 80.56% in *Corner/Angle Positioning*, one of the strongest results any open-source system achieves, and both **InternVL3.5-72B** and **llama-3.2-90b-vision-instruct** converge at 66.67% in *Complete Occlusion Inference*. Against these advances stands a striking

Table 7: Multiple-choice Evaluation Accuracy on SPATIALAB-MCQ by Question Sub-Categories for Open-source Large Models. We bold and underline the best score within each model category.

Category	Sub-Category	llama-3.2-11b-vision-instruct	Gemma-3-27B-it	qwen-2.5-vl-72b-instruct	InternVL3-5-72B	qwen-2.5-vl-72b-instruct	llama-3.2-90b-vision-instruct
3D Geometry	Gravity Effects	22.0	48.0	54.0	52.0	52.0	56.0
	Shape Projection	32.0	42.0	44.0	50.0	46.0	50.0
	Spatial Containment	32.0	34.0	36.0	44.0	46.0	36.0
	Stability Prediction	33.33	54.76	38.1	52.38	47.62	52.38
	Volume Comparison	13.04	39.13	32.61	52.17	43.48	36.96
Depth and Occlusion	Complete Occlusion Inference	30.77	61.54	58.97	66.67	48.72	66.67
	Layering Order	34.29	34.29	37.14	50.0	50.0	38.57
	Partial Occlusion	32.0	34.0	44.0	60.0	48.0	58.0
	Reflective Surfaces	28.0	38.0	26.0	54.0	50.0	44.0
	Transparency Effects	26.0	40.0	40.0	60.0	46.0	62.0
Orientation	Cardinal Direction	22.0	48.0	34.0	46.0	48.0	40.0
	Facing Direction	21.43	42.86	52.38	54.76	50.0	50.0
	Object Rotation	2.0	40.0	38.0	48.0	48.0	44.0
	Stacking Orientation	28.57	65.71	65.71	68.57	62.86	71.43
	Tool Handedness	40.0	48.0	52.0	56.0	56.0	56.0
Relative Positioning	Alignment Patterns	27.91	55.81	48.84	53.49	55.81	53.49
	Betweenness Relationships	42.55	55.32	55.32	65.96	61.7	61.7
	Corner/Angle Positioning	58.33	55.56	36.11	80.56	50.0	75.0
	Directional Relations	60.47	62.79	51.16	83.72	62.79	65.12
	Proximity Gradients	27.91	41.86	32.56	48.84	39.53	41.86
Size and Scale	Distance-Size Correlation	36.0	50.0	54.0	52.0	44.0	46.0
	Perspective Distortion	22.0	42.0	36.0	46.0	34.0	52.0
	Relative Size Comparison	28.0	48.0	40.0	48.0	52.0	46.0
	Scale Consistency	29.41	47.06	41.18	43.14	37.25	37.25
	Shadow-Size Projection	37.25	52.94	54.9	56.86	50.98	52.94
Spatial Navigation	Accessibility Constraints	29.27	58.54	43.9	75.61	53.66	43.9
	Obstacle Avoidance	24.0	38.0	38.0	46.0	42.0	40.0
	Pathway Existence	28.26	54.35	36.96	39.15	54.35	47.83
	Spatial Sequence	34.0	52.0	56.0	66.0	54.0	62.0
	Viewpoint Visibility	44.0	36.0	34.0	50.0	42.0	48.0

collapse: **llama-3.2-11b-vision-instruct** manages a mere 2.0% in *Object Rotation*. The pattern of localized strength recurs in *Stacking Orientation*, where **llama-3.2-90b-vision-instruct** reaches 71.43%, though such peaks are purchased at the expense of other categories, and gravity reasoning holds at only a modest 56.0%. With within-model spread routinely exceeding 70 percentage points, the evidence points to brittle specialization rather than broad mastery. We conclude that parameter scaling amplifies selective strengths and raises ceilings, yet leaves the foundational representation gaps structurally intact.

Open-source Small Models. Open-source small models (Table 8) exhibit the sharpest performance contrasts observed within any single family, spanning the full dynamic range of the taxonomy. The divergence is stark: **qwen-2.5-vl-7b-instruct** attains 74.29% in *Stacking Orientation*, whereas **InternVL3.5-1B** collapses to 12.0% in *Cardinal Direction*. Between these poles, competence remains thin where abstraction is required, with the best models tied at only 34.78% in *Pathway Existence* and gravity-related reasoning capped at 50.0% for **InternVL3.5-4B** and **Gemma-3-4B-it**. That within-model spread frequently exceeds 60 percentage points is diagnostic of brittle generalization: these systems are acutely cue-dependent, excelling where strong visual templates are available but faltering wherever tasks demand deeper spatial abstraction. We read this inconsistency as direct evidence that representation learning proceeds unevenly across the subcategory space. The composite picture is one of localized competence that never consolidates into robustness across the full spectrum of spatial reasoning.

Reasoning Models. Reasoning-enhanced models (Table 9) confer tangible advantages on select tasks yet fail to translate this depth into consistent breadth. **o4-mini** surpasses most open baselines with 65.12% in *Alignment Patterns* and 56.0% in *Gravity Effects*, while **Gemini-2.0-Flash-Thinking** signals sharper geometric reasoning at 62.75% in *Shadow-Size Projection* and 52.17% in *Volume Comparison*. The family’s strongest showing belongs to **Gemini-2.5-Flash-Thinking**, which records 71.79% in *Complete Occlusion Inference* and 70.21% in *Betweenness Relationships*. Tellingly, that same model concedes ground elsewhere, dropping to 34.0% in *Cardinal Direction* and 34.0% in *Shape Projection*. This coexistence of deep peaks and shallow troughs, with within-model spreads frequently exceeding 35 percentage points, demonstrates that explicit reasoning mechanisms deepen performance within particular subcategories without broadening coverage across them. We conclude that reasoning augmentation supplies localized capacity while leaving the structural gaps in spatial understanding unaddressed: these systems evidence genuine promise, but not yet generalizable competence.

Spatial Reasoning Models. Spatially specialized models (Table 10) deliver precisely targeted improvements while conspicuously lacking generality. **SpaceQwen2.5-VL-3B-Instruct** distinguishes itself in size-scale reasoning, clearing many open-source baselines with 48.0% in *Distance-Size*

Table 8: Multiple-choice Evaluation Accuracy on SPATIALAB-MCQ by Question Sub-Categories for Open-source Small Models. We bold and underline the best score within each model category.

Category	Sub-Category	InternVL3.5-1B	InternVL3.5-2B	qwen-2.5-vl-3b-instruct	InternVL3.5-4B	Gemma-3-4B-it	qwen-2.5-vl-7b-instruct
3D Geometry	Gravity Effects	46.0	38.0	48.0	50.0	50.0	50.0
	Shape Projection	30.0	42.0	38.0	42.0	40.0	44.0
	Spatial Containment	30.0	28.0	34.0	34.0	36.0	34.0
	Stability Prediction	30.95	35.71	50.0	52.38	45.24	52.38
	Volume Comparison	30.43	26.09	36.96	36.96	47.83	34.78
Depth and Occlusion	Complete Occlusion Inference	30.77	30.77	41.03	64.1	33.33	41.03
	Layering Order	30.0	24.29	30.0	32.86	22.86	27.14
	Partial Occlusion	38.0	30.0	38.0	54.0	46.0	42.0
	Reflective Surfaces	38.0	34.0	36.0	32.0	42.0	40.0
	Transparency Effects	26.0	42.0	36.0	40.0	32.0	44.0
Orientation	Cardinal Direction	12.0	18.0	36.0	24.0	38.0	30.0
	Facing Direction	21.43	28.57	52.38	40.48	42.86	38.1
	Object Rotation	12.0	28.0	42.0	34.0	42.0	34.0
	Stacking Orientation	45.71	57.14	62.86	68.57	68.57	74.29
	Tool Handedness	40.0	36.0	40.0	60.0	48.0	48.0
Relative Positioning	Alignment Patterns	27.91	30.23	39.53	44.19	41.86	41.86
	Betweenness Relationships	38.3	38.3	46.81	65.96	46.81	53.19
	Corner/Angle Positioning	50.0	55.56	47.22	69.44	55.56	50.0
	Directional Relations	51.16	58.14	27.91	60.47	46.51	48.84
	Proximity Gradients	20.93	23.26	39.53	34.88	39.53	37.21
Size and Scale	Distance-Size Correlation	32.0	20.0	40.0	36.0	38.0	36.0
	Perspective Distortion	26.0	40.0	42.0	38.0	44.0	44.0
	Relative Size Comparison	34.0	32.0	46.0	38.0	30.0	42.0
	Scale Consistency	27.45	35.29	41.18	31.37	27.45	41.18
	Shadow-Size Projection	39.22	35.29	66.67	39.22	47.06	47.06
Spatial Navigation	Accessibility Constraints	36.59	29.27	51.22	41.46	39.02	43.9
	Obstacle Avoidance	24.0	26.0	38.0	38.0	38.0	36.0
	Pathway Existence	34.78	30.43	32.61	34.78	34.78	28.26
	Spatial Sequence	24.0	42.0	40.0	56.0	42.0	44.0
	Viewpoint Visibility	36.0	34.0	36.0	40.0	36.0	26.0

Table 9: Multiple-choice Evaluation Accuracy on SPATIALAB-MCQ by Question Sub-Categories for Reasoning Models. We bold and underline the best score within each model category.

Category	Sub-Category	o4-mini	Gemini-2.0-Flash-Thinking	Gemini-2.5-Flash-Thinking	Kimi-VL-A3B-Thinking-2506
3D Geometry	Gravity Effects	56.0	38.0	46.0	50.0
	Shape Projection	50.0	36.0	40.0	34.0
	Spatial Containment	56.0	28.0	48.0	42.0
	Stability Prediction	50.0	35.71	50.0	50.0
	Volume Comparison	43.48	52.17	45.65	39.13
Depth and Occlusion	Complete Occlusion Inference	69.23	51.28	71.79	51.28
	Layering Order	57.14	30.0	41.43	35.71
	Partial Occlusion	50.0	42.0	58.0	40.0
	Reflective Surfaces	54.0	40.0	52.0	42.0
	Transparency Effects	64.0	50.0	54.0	42.0
Orientation	Cardinal Direction	46.0	34.0	50.0	28.0
	Facing Direction	50.0	42.86	54.76	45.24
	Object Rotation	56.0	36.0	48.0	30.0
	Stacking Orientation	71.43	57.14	57.14	65.71
	Tool Handedness	56.0	44.0	60.0	44.0
Relative Positioning	Alignment Patterns	65.12	41.86	44.19	34.88
	Betweenness Relationships	65.96	55.32	70.21	63.83
	Corner/Angle Positioning	75.0	55.56	61.11	47.22
	Directional Relations	65.12	44.19	65.12	60.47
	Proximity Gradients	51.16	32.56	41.86	48.84
Size and Scale	Distance-Size Correlation	44.0	42.0	56.0	44.0
	Perspective Distortion	40.0	44.0	50.0	36.0
	Relative Size Comparison	46.0	46.0	58.0	32.0
	Scale Consistency	39.22	56.86	52.94	43.14
	Shadow-Size Projection	35.29	62.75	58.82	43.14
Spatial Navigation	Accessibility Constraints	56.1	41.46	53.66	31.71
	Obstacle Avoidance	32.0	44.0	44.0	36.0
	Pathway Existence	45.65	41.3	47.83	34.78
	Spatial Sequence	64.0	48.0	68.0	54.0
	Viewpoint Visibility	60.0	40.0	54.0	48.0

Correlation and 49.02% in *Scale Consistency*, and it further leads the navigation subcategories at 53.66% in *Accessibility Constraints* and 44.0% in *Obstacle Avoidance*. Consistent with their domain-specific mandate, **SpaceOm** and **SpaceThinker-Qwen2.5VL-3B** both reach 50.0% in *Stability Prediction*. These gains, however, do not generalize: performance falls sharply on adjacent tasks such as *Spatial Containment*, where accuracy is only 24.0%, and even the strongest areas rarely exceed 55.0%, with the subcategory spread routinely surpassing 30 percentage points. The evidence indicates that domain targeting sharpens narrow peaks but stops short of holistic competence; specialization enhances select abilities without integrating them into the robust, general framework that spatial reasoning in the wild requires.

Table 10: Multiple-choice Evaluation Accuracy on SPATIALAB-MCQ by Question Sub-Categories for Spatial Reasoning Models. We bold and underline the best score within each model category.

Category	Sub-Category	SpaceOm	SpaceThinker-Qwen2.5-VL-3B	SpaceQwen2.5-VL-3B-Instruct
3D Geometry	Gravity Effects	48.0	46.0	32.0
	Shape Projection	38.0	36.0	36.0
	Spatial Containment	40.0	36.0	24.0
	Stability Prediction	50.0	50.0	28.57
	Volume Comparison	36.96	34.78	36.96
Depth and Occlusion	Complete Occlusion Inference	38.46	35.9	38.46
	Layering Order	30.0	32.86	24.29
	Partial Occlusion	44.0	46.0	38.0
	Reflective Surfaces	44.0	44.0	38.0
	Transparency Effects	40.0	32.0	42.0
Orientation	Cardinal Direction	40.0	38.0	30.0
	Facing Direction	59.52	54.76	50.0
	Object Rotation	40.0	42.0	28.0
	Stacking Orientation	62.86	62.86	40.0
	Tool Handedness	40.0	40.0	48.0
Relative Positioning	Alignment Patterns	39.53	44.19	39.53
	Betweenness Relationships	44.68	42.55	53.19
	Corner/Angle Positioning	41.67	41.67	38.89
	Directional Relations	25.58	25.58	20.93
	Proximity Gradients	37.21	37.21	34.88
Size and Scale	Distance-Size Correlation	40.0	44.0	48.0
	Perspective Distortion	40.0	38.0	42.0
	Relative Size Comparison	42.0	46.0	48.0
	Scale Consistency	33.33	33.33	49.02
	Shadow-Size Projection	58.82	54.9	66.67
Spatial Navigation	Accessibility Constraints	48.78	46.34	53.66
	Obstacle Avoidance	36.0	32.0	44.0
	Pathway Existence	32.61	34.78	41.3
	Spatial Sequence	48.0	44.0	50.0
	Viewpoint Visibility	32.0	34.0	48.0

Overall Discussion. Viewed across the full taxonomy and every model family (Table 8, Table 7, Table 6, Table 10, Table 9), the benchmark diagnoses a single, consistent signature: fragmentation. Small open-source models peak at 74.29% in *Stacking Orientation* yet collapse to 12.0% in *Cardinal Direction*; larger models lift the ceiling to 80.56% in *Corner/Angle Positioning* but still fail catastrophically at 2.0% in *Object Rotation*. Closed-source systems set the benchmark high-water mark, with **GPT-4o-mini** reaching 85.71% in *Stacking Orientation*, and nonetheless descend to 32.0% in *Relative Size Comparison*. Specialized models convert their domain focus into navigation (53.66%) and stability (50.0%) strengths while remaining capped below 55% overall, and reasoning models post gains such as 71.79% in *Complete Occlusion Inference* even as other subcategories languish at 34.0%. No approach escapes the pattern: within-model spreads routinely exceed 50 percentage points, and the subcategories demanding physical abstraction, *Gravity Effects*, *Stability Prediction*, and *Spatial Containment*, score below 35–40% irrespective of family. We trace this fragility to two root causes: the absence of stable internal encodings of orientation and physics, and an over-reliance on surface-level correlations. The sheer variance, from 85.71% highs to 2.0% lows, betrays pattern-matching in lieu of robust spatial reasoning. Scaling, proprietary training, and reasoning augmentation each confer real but partial benefits, yielding isolated peaks that coexist with glaring blind spots rather than any general coverage. The precision of this diagnosis is itself the contribution: SPATIALAB-MCQ localizes competence to specific axes and exposes exactly where it fails. Progress toward holistic spatial intelligence will accordingly require architectural change aimed at embedding consistent geometric grounding and physics-aware mechanisms; our results document both remarkable selective gains and the persistence of fundamental, systematically identifiable gaps.

E.2 MODEL FAILURE ANALYSIS

The per-item distribution of correct answers across categories and subcategories in Table 11 sharpens the diagnostic picture and isolates where competence concentrates and where it dissolves. 3D Geometry and Size & Scale present relatively balanced distributions, yet both retain a sizeable fraction of items that no model answered correctly, indicating that even elementary geometric invariances are not reliably internalized. Depth & Occlusion, by contrast, carries a heavier mid-range

Table 11: Item-difficulty distribution on SPATIALAB-MCQ, tabulating how many questions were solved by a given number of models (0–5), broken down by category and sub-category. Here “0” indicates that no model answered the item correctly, whereas “5” indicates that only five of the models tested succeeded on it; lower bins thus isolate the questions that are hardest across the model pool.

Category	Sub-Category	Number of Models Successfully Answered						Total
		0	1	2	3	4	5	
3D Geometry	Gravity Effects	9	7	3	2	3	1	25
	Shape Projection	4	6	0	9	5	7	31
	Spatial Containment	6	9	6	3	6	1	31
	Stability Prediction	5	6	4	2	2	3	22
	Volume Comparison	6	5	6	5	4	2	28
3D Geometry Total		30	33	19	21	20	14	137
Depth & Occlusion	Complete Occlusion Inference	0	4	0	6	3	4	17
	Layering Order	9	8	7	8	6	5	43
	Partial Occlusion	2	2	5	7	5	2	23
	Reflective Surfaces	6	5	5	7	4	1	28
	Transparency Effects	4	6	3	2	5	7	27
Depth & Occlusion Total		21	25	20	30	23	19	138
Orientation	Cardinal Direction	1	7	10	7	3	4	32
	Facing Direction	2	3	8	5	3	1	22
	Object Rotation	4	6	7	3	5	4	29
	Stacking Orientation	2	3	0	2	1	1	9
	Tool Handedness	2	1	4	3	1	1	12
Orientation Total		11	20	29	20	13	11	104
Relative Positioning	Alignment Patterns	3	4	4	4	4	1	20
	Betweenness Relationships	3	1	2	2	5	7	20
	Corner/Angle Positioning	0	1	3	2	3	3	12
	Directional Relations	1	4	1	2	3	2	13
	Proximity Gradients	4	6	6	5	2	3	26
Relative Positioning Total		11	16	16	15	17	16	91
Size & Scale	Distance-Size Correlation	6	5	6	1	1	7	26
	Perspective Distortion	8	7	3	3	6	4	31
	Relative Size Comparison	5	5	7	4	4	4	29
	Scale Consistency	7	6	7	4	5	2	31
	Shadow-Size Projection	5	7	2	1	1	4	20
Size & Scale Total		31	30	25	13	17	21	137
Spatial Navigation	Accessibility Constraints	3	2	1	5	4	3	18
	Obstacle Avoidance	2	6	9	6	10	2	35
	Pathway Existence	2	10	4	1	4	3	24
	Spatial Sequence	3	2	3	2	7	8	25
	Viewpoint Visibility	2	2	5	9	8	5	31
Spatial Navigation Total		12	22	22	23	33	21	133
Grand Total		116	146	131	122	123	102	740

tail (2–4 correct), the signature of partial but inconsistent reasoning about layered scenes. Orientation stands out as the weakest category overall, its especially low totals in Stacking Orientation and Tool Handedness revealing that subtle directional and embodied affordances are poorly captured. Relative Positioning is more heterogeneous still: tasks such as Proximity Gradients attract moderate success, whereas Corner/Angle Positioning accumulates near-zero correct counts, exposing brittleness in compositional reasoning over reference frames. Spatial Navigation registers comparatively stronger performance, notably in Obstacle Avoidance and Viewpoint Visibility, yet its persistent errors in Pathway Existence show that relational chaining remains a bottleneck. In aggregate, no category attains uniformly high accuracy, and the Grand Total spreads evenly across the 0–5 bins, confirming that success is acutely task-dependent rather than the expression of general spatial competence. The

consolidated evidence establishes that current models command isolated strengths while lacking systematic reliability across the spatial dimensions the benchmark probes.

These distributions resolve into a small number of identifiable root causes, each tied to a specific representational or reasoning bottleneck. The elevated error rates in Stacking Orientation and Tool Handedness point to shallow grounding in physics-based and embodied reasoning, a deficiency consistent with training on static internet imagery rather than interactive or task-driven data. The difficulty in Corner/Angle Positioning and Pathway Existence isolates a distinct failure mode, inadequate multi-step relational chaining, that limits models' capacity for recursive spatial logic. The inconsistency in Transparency Effects and Reflective Surfaces implicates a third mechanism: the handling of visual phenomena that hinge on non-local cues such as secondary reflections or occluded geometry. Against these deficits, the comparatively stronger showing in Obstacle Avoidance is best read not as genuine planning ability but as a learned shortcut exploiting dataset biases, such as the frequent co-occurrence of walkable paths. Taken together, these failure modes converge on a single characterization of current training regimes: they overweight correlation-driven vision-language alignment while systematically underrepresenting embodied, counterfactual, and physics-based reasoning. Remediating them will plausibly demand both richer training distributions, for instance embodied simulation data and physics-augmented learning, and architectural innovations that explicitly model reference frames and relational constraints. Absent such advances, models will continue to exhibit the brittle competence that this analysis pinpoints, one that predictably fractures under compositional or physically grounded demands.

F ERROR ANALYSIS OF SPATIALAB-OPEN

F.1 SUB-CATEGORY-WISE QUANTITATIVE ERROR ANALYSIS

Closed-source Models (Table 12). The open-ended results establish closed-source systems as the strongest family overall, yet the analysis makes equally clear that this advantage is distributed unevenly across the taxonomy. GPT-5-mini emerges as the decisive leader, consistently securing the best scores in **3D Geometry** and **Relative Positioning**, peaking at 58.14% in Directional Relations and 55.81% in Alignment Patterns. The same family, however, exposes pronounced brittleness elsewhere: Gemini-2.5-Flash falls to 16.28% in Proximity Gradients and Claude 3.5 Haiku to 12.0% in Perspective Distortion. This breadth of outcome—commanding on discrete relational and geometric judgments, fragile on subtle, continuous visual cues—demonstrates that even frontier proprietary models lack uniform spatial competence. We attribute the disparity to the differential sophistication of these models' spatial reasoning pipelines, in which certain architectures encode relational and geometric structure far more faithfully than others. Proximity Gradients are diagnostic precisely because they demand analogical, graded inference rather than discrete classification, and GPT-5-mini's capacity to succeed where its peers collapse indicates that scaling and proprietary training corpora confer measurable, if partial, advantages. Crucially, the leader's own vulnerabilities establish that closed-source dominance is relative rather than absolute: the family still falls well short of robust, human-like spatial reasoning.

Open-source Large Models (Table 13). Among large open-source models the analysis reveals a markedly more distributed landscape, in which distinct systems dominate isolated sub-categories rather than any single model leading throughout. llama-3.2-90b-vision-instruct attains 60.0% in Depth and Occlusion, while GLM-4.5V-106B-MoE leads Spatial Containment at 38.0%. These localized peaks notwithstanding, several systems collapse on Proximity Gradients—qwen-2.5-vl-32b-instruct and qwen-2.5-vl-72b-instruct each register only 9.3%—establishing that architectural scaling improves coverage without guaranteeing it across demanding continuous-reasoning tasks. We attribute much of this variation to heterogeneous training data: models exposed to richer depth and geometry corpora perform better on occlusion and containment, whereas others generalize poorly. The persistent deficit on Proximity Gradients indicates that these systems are not optimized for subtle spatial analogies and distance scaling. The group likewise exhibits higher inter-model variance than the closed-source family, a signature of open-source development heterogeneity. In sum, large open-source models present encouraging peaks but inconsistent coverage, confirming that scale alone is insufficient absent targeted pre-training and fine-tuning.

Table 12: Open-ended Evaluation Accuracy on SPATIALAB-OPEN by Question Sub-Categories for Closed-source Models. We bold and underline the best score within each model category.

Category	Sub-Category	GPT-5-mini	Gemini-2.0-Flash	Gemini-2.5-Flash	Claude 3.5 Haiku	Mistral Medium 3.1
3D Geometry	Gravity Effects	48.0	36.0	30.0	24.0	20.0
	Shape Projection	40.0	18.0	22.0	16.0	16.0
	Spatial Containment	50.0	38.0	50.0	36.0	36.0
	Stability Prediction	50.0	26.19	35.71	30.95	26.19
	Volume Comparison	39.13	41.3	32.61	23.91	28.26
Depth and Occlusion	Complete Occlusion Inference	46.15	35.9	38.46	28.21	25.64
	Layering Order	30.0	22.86	21.43	17.14	15.71
	Partial Occlusion	38.0	20.0	28.0	14.0	14.0
	Reflective Surfaces	32.0	28.0	32.0	22.0	22.0
	Transparency Effects	32.0	18.0	18.0	16.0	22.0
Orientation	Cardinal Direction	34.0	40.0	34.0	26.0	30.0
	Facing Direction	28.57	28.57	26.19	19.05	14.29
	Object Rotation	42.0	14.0	28.0	30.0	14.0
	Stacking Orientation	37.14	20.0	31.43	22.86	28.57
	Tool Handedness	48.0	36.0	44.0	24.0	24.0
Relative Positioning	Alignment Patterns	55.81	34.88	37.21	37.21	34.88
	Betweenness Relationships	48.94	23.4	31.91	23.4	21.28
	Corner/Angle Positioning	44.44	30.56	41.67	16.67	36.11
	Directional Relations	58.14	46.51	58.14	34.88	37.21
	Proximity Gradients	39.53	20.93	25.58	16.28	18.6
Size and Scale	Distance-Size Correlation	48.0	32.0	36.0	20.0	22.0
	Perspective Distortion	48.0	24.0	22.0	20.0	12.0
	Relative Size Comparison	36.0	26.0	28.0	18.0	16.0
	Scale Consistency	43.14	21.57	23.53	19.61	9.8
	Shadow-Size Projection	37.25	27.45	23.53	23.53	15.69
Spatial Navigation	Accessibility Constraints	41.46	31.71	39.02	24.39	21.95
	Obstacle Avoidance	40.0	22.0	22.0	26.0	14.0
	Pathway Existence	23.91	15.22	21.74	10.87	19.57
	Spatial Sequence	52.0	32.0	34.0	24.0	20.0
	Viewpoint Visibility	28.0	22.0	32.0	20.0	10.0

Table 13: Open-ended Evaluation Accuracy on SPATIALAB-OPEN by Question Sub-Categories for Open-source Large Models. We bold and underline the best score within each model category.

Category	Sub-Category	llama-3.2-11b-vision-instruct	Gemma-3-27B-it	qwen-2.5-vl-32b-instruct	qwen-2.5-vl-72b-instruct	llama-3.2-90b-vision-instruct	GLM-4.5V-106B-MoE
3D Geometry	Gravity Effects	14.0	22.0	12.0	30.0	24.0	28.0
	Shape Projection	10.0	20.0	10.0	20.0	16.0	20.0
	Spatial Containment	14.0	30.0	18.0	22.0	20.0	38.0
	Stability Prediction	23.81	30.95	16.67	21.43	26.19	38.1
	Volume Comparison	23.91	10.87	26.09	41.3	28.26	32.61
Depth and Occlusion	Complete Occlusion Inference	25.64	20.51	12.82	25.64	30.77	33.33
	Layering Order	8.57	15.71	10.0	14.29	15.71	10.0
	Partial Occlusion	12.0	16.0	8.0	14.0	20.0	16.0
	Reflective Surfaces	26.0	16.0	8.0	34.0	32.0	24.0
	Transparency Effects	18.0	14.0	10.0	20.0	22.0	26.0
Orientation	Cardinal Direction	32.0	34.0	20.0	38.0	28.0	30.0
	Facing Direction	11.9	26.19	14.29	21.43	11.9	16.67
	Object Rotation	26.0	16.0	10.0	18.0	28.0	20.0
	Stacking Orientation	17.14	25.71	17.14	25.71	5.71	37.14
	Tool Handedness	20.0	20.0	12.0	20.0	32.0	24.0
Relative Positioning	Alignment Patterns	23.26	46.51	20.93	46.51	30.23	30.23
	Betweenness Relationships	34.04	31.91	17.02	25.53	38.3	29.79
	Corner/Angle Positioning	27.78	36.11	11.11	27.78	16.67	16.67
	Directional Relations	30.23	39.53	25.58	34.88	25.58	32.56
	Proximity Gradients	9.3	18.6	9.3	18.6	18.6	20.93
Size and Scale	Distance-Size Correlation	20.0	26.0	12.0	26.0	30.0	26.0
	Perspective Distortion	10.0	18.0	10.0	18.0	12.0	20.0
	Relative Size Comparison	8.0	26.0	14.0	32.0	22.0	24.0
	Scale Consistency	11.76	21.57	3.92	19.61	21.57	25.53
	Shadow-Size Projection	17.65	21.57	9.8	27.45	23.53	27.45
Spatial Navigation	Accessibility Constraints	21.95	26.83	26.83	34.15	39.02	24.39
	Obstacle Avoidance	12.0	18.0	12.0	20.0	30.0	22.0
	Pathway Existence	19.57	21.74	6.52	17.39	15.22	17.39
	Spatial Sequence	22.0	28.0	12.0	22.0	28.0	34.0
	Viewpoint Visibility	18.0	16.0	12.0	12.0	24.0	24.0

Open-source Small Models (Table 14). Small open-source models predictably trail their larger counterparts, yet the analysis surfaces noteworthy localized strengths that qualify any blanket dismissal. qwen-2.5-vl-3b-instruct achieves 18.0% in Perspective Distortion and 26.0% in Relative Size Comparison, while Gemma-3-4B-it leads Spatial Containment (24.0%) and Stability Prediction (21.43%). These pockets of competence are offset by categorical breakdowns: InternVL3.5-2B scores 0.0% in Betweenness Relationships, and InternVL3.5-1B manages only 2.0% in Shape Projection. We attribute this extreme variability to limited capacity and restricted training diversity, which together prevent compact models from acquiring robust representations of geometric and relational principles. Sub-categories demanding precise reasoning over orientation and hierarchy trigger complete breakdowns, as minor misinterpretations propagate into total failure. That these models nonetheless occasionally surpass larger systems on narrow tasks suggests that even small-scale training can instill isolated competencies—evidence for the value of specialization, but equally a confirmation of the structural ceiling imposed by compact architectures. In short, small open-source models demonstrate

Table 14: Open-ended Evaluation Accuracy on SPATIALAB-OPEN by Question Sub-Categories for Open-source Small Models. We bold and underline the best score within each model category.

Category	Sub-Category	InternVL3.5-1B	InternVL3.5-2B	qwen-2.5-vl-3b-instruct	InternVL3.5-4B	Gemma-3-4B-it	qwen-2.5-vl-7b-instruct
3D Geometry	Gravity Effects	4.0	6.0	14.0	24.0	16.0	22.0
	Shape Projection	2.0	8.0	10.0	4.0	10.0	0.0
	Spatial Containment	6.0	4.0	12.0	20.0	24.0	8.0
	Stability Prediction	2.38	11.9	14.29	19.05	21.43	19.05
	Volume Comparison	15.22	32.61	28.26	30.43	30.43	28.26
Depth and Occlusion	Complete Occlusion Inference	2.56	5.13	10.26	17.95	20.51	23.08
	Layering Order	5.71	7.14	10.0	14.29	11.43	14.29
	Partial Occlusion	14.0	8.0	12.0	16.0	4.0	10.0
	Reflective Surfaces	16.0	16.0	6.0	14.0	18.0	26.0
	Transparency Effects	10.0	20.0	4.0	28.0	14.0	8.0
Orientation	Cardinal Direction	12.0	16.0	16.0	26.0	20.0	24.0
	Facing Direction	9.52	4.76	28.57	14.29	7.14	19.05
	Object Rotation	4.0	10.0	12.0	8.0	14.0	16.0
	Stacking Orientation	17.14	11.43	8.57	17.14	20.0	28.57
	Tool Handedness	8.0	12.0	8.0	12.0	12.0	12.0
Relative Positioning	Alignment Patterns	13.95	27.91	32.56	11.63	41.86	34.88
	Betweenness Relationships	12.77	17.02	0.0	19.15	17.02	25.53
	Corner/Angle Positioning	22.22	30.56	2.78	22.22	19.44	22.22
	Directional Relations	11.63	25.58	9.3	34.88	23.26	37.21
	Proximity Gradients	9.3	18.6	9.3	11.63	16.28	18.6
Size and Scale	Distance-Size Correlation	6.0	10.0	14.0	16.0	20.0	12.0
	Perspective Distortion	8.0	12.0	18.0	16.0	12.0	16.0
	Relative Size Comparison	10.0	16.0	26.0	24.0	10.0	16.0
	Scale Consistency	7.84	9.8	11.76	11.76	7.84	11.76
	Shadow-Size Projection	13.73	11.76	21.57	13.73	25.49	23.53
Spatial Navigation	Accessibility Constraints	4.88	26.83	7.32	17.07	19.51	19.51
	Obstacle Avoidance	16.0	16.0	12.0	18.0	28.0	26.0
	Pathway Existence	0.0	8.7	2.17	8.7	17.39	8.7
	Spatial Sequence	12.0	22.0	8.0	26.0	20.0	28.0
	Viewpoint Visibility	16.0	18.0	16.0	24.0	14.0	16.0

pockets of promise embedded in pervasive brittleness, with overall accuracy bounded by size and data constraints.

Reasoning Models (Table 15). Reasoning-oriented models exhibit a sharper internal divide, with Gemini-2.5-flash-thinking standing decisively above its peers. It records the best scores in nearly every sub-category, including Tool Handedness at 75.0% and Complete Occlusion Inference at 57.14%. Kimi-VL-A3B-Thinking-2506, by contrast, struggles dramatically, scoring just 6.0% in Spatial Containment and lagging across the board. We trace this gap to differences in how the models integrate multi-modal streams into coherent reasoning frameworks: Gemini-2.5-flash-thinking appears able to construct and query internal world models, enabling more effective inference over object interactions and physical properties, whereas weaker systems fall back on surface-level pattern matching that proves inadequate for compositional or causal reasoning. The consistently high performance of Gemini-2.5 across categories establishes that explicit reasoning mechanisms confer tangible benefits for spatial understanding. Its failure to dominate Shape Projection and Betweenness Relationships, however, demonstrates that even advanced reasoning modules are not yet universally effective. Reasoning models therefore hold genuine promise, but the category itself remains heterogeneous and far from solved.

Spatial Reasoning Models (Table 16). Spatial reasoning models, though purpose-built for the task, record the lowest results overall. Performance is highly fragmented: several models tie at 16.0% in Gravity Effects, 8.0% in Shape Projection, and 14.29% in Stability Prediction. SpaceQwen2.5-VL-3B-Instruct stands marginally apart with 23.91% in Volume Comparison and 13.73% in Scale Consistency, yet these figures remain far below competitive thresholds. At the opposite extreme, Betweenness Relationships and Corner/Angle Positioning collapse entirely, recording 0.0% and 2.78%, respectively. Such outcomes indicate that the models’ smaller capacity and narrowly focused training severely constrain their representational flexibility. We read these failures as direct evidence that narrow specialization, absent sufficient capacity or data breadth, cannot yield generalizable spatial reasoning: on geometric and positional tasks in particular, even slight representational errors cascade into total breakdown. Consequently, despite their explicit focus, spatial reasoning models prove the least effective family, underscoring that both scale and diversity are prerequisites for robust performance.

Overall Discussion. Viewed across the full taxonomy and every model family—closed-source (Table 12), open-source large (Table 13), open-source small (Table 14), reasoning (Table 15), and spatial reasoning (Table 16)—the evaluation establishes a single, consistent verdict: spatial competence in the wild is fragmented and systemically incomplete. Closed-source models lead overall,

Table 15: Open-ended Evaluation Accuracy on SPATIALAB-OPEN by Question Sub-Categories for Reasoning Models. We bold and underline the best score within each model category.

Category	Sub-Category	gemini-2.0-flash-thinking	gemini-2.5-flash-thinking	Kimi-VL-A3B-Thinking-2506	Step3
3D Geometry	Gravity Effects	32.0	40.0	14.0	38.0
	Shape Projection	24.0	0.0	10.0	30.0
	Spatial Containment	36.0	50.0	6.0	30.0
	Stability Prediction	26.19	42.86	14.29	33.33
	Volume Comparison	36.96	44.44	23.91	30.43
Depth and Occlusion	Complete Occlusion Inference	33.33	57.14	17.95	33.33
	Layering Order	20.0	20.0	14.29	14.29
	Partial Occlusion	28.0	50.0	8.0	14.0
	Reflective Surfaces	34.0	63.64	10.0	16.0
	Transparency Effects	26.0	57.14	22.0	20.0
Orientation	Cardinal Direction	40.0	20.0	18.0	26.0
	Facing Direction	35.71	37.5	14.29	28.57
	Object Rotation	18.0	40.0	4.0	22.0
	Stacking Orientation	22.86	0.0	5.71	34.29
	Tool Handedness	44.0	75.0	20.0	32.0
Relative Positioning	Alignment Patterns	46.51	66.67	25.58	41.86
	Betweenness Relationships	27.66	20.0	12.77	25.53
	Corner/Angle Positioning	36.11	50.0	11.11	11.11
	Directional Relations	39.53	44.44	27.91	37.21
	Proximity Gradients	23.26	16.67	11.63	18.6
Size and Scale	Distance-Size Correlation	32.0	57.14	14.0	22.0
	Perspective Distortion	18.0	11.11	10.0	30.0
	Relative Size Comparison	28.0	11.11	14.0	36.0
	Scale Consistency	33.33	20.0	11.76	31.37
	Shadow-Size Projection	35.29	18.18	13.73	29.41
Spatial Navigation	Accessibility Constraints	31.71	45.45	17.07	41.46
	Obstacle Avoidance	26.0	10.0	16.0	18.0
	Pathway Existence	23.91	25.0	13.04	19.57
	Spatial Sequence	34.0	12.5	30.0	36.0
	Viewpoint Visibility	32.0	12.5	16.0	26.0

Table 16: Open-ended Evaluation Accuracy on SPATIALAB-OPEN by Question Sub-Categories for Spatial Reasoning Models. We bold and underline the best score within each model category.

Category	Sub-Category	SpaceOm	SpaceThinker-Qwen2.5VL-3B	SpaceQwen2.5-VL-3B-Instruct
3D Geometry	Gravity Effects	14.0	12.0	16.0
	Shape Projection	8.0	8.0	6.0
	Spatial Containment	8.0	10.0	10.0
	Stability Prediction	14.29	14.29	7.14
	Volume Comparison	19.57	23.91	23.91
Depth and Occlusion	Complete Occlusion Inference	7.69	10.26	7.69
	Layering Order	8.57	11.43	4.29
	Partial Occlusion	10.0	12.0	4.0
	Reflective Surfaces	4.0	4.0	2.0
	Transparency Effects	4.0	8.0	2.0
Orientation	Cardinal Direction	18.0	24.0	8.0
	Facing Direction	23.81	23.81	16.67
	Object Rotation	20.0	20.0	18.0
	Stacking Orientation	8.57	8.57	17.14
	Tool Handedness	0.0	4.0	8.0
Relative Positioning	Alignment Patterns	32.56	32.56	23.26
	Betweenness Relationships	4.26	2.13	0.0
	Corner/Angle Positioning	2.78	2.78	2.78
	Directional Relations	11.63	6.98	11.63
	Proximity Gradients	6.98	6.98	9.3
Size and Scale	Distance-Size Correlation	18.0	18.0	16.0
	Perspective Distortion	16.0	20.0	2.0
	Relative Size Comparison	22.0	20.0	8.0
	Scale Consistency	9.8	11.76	13.73
	Shadow-Size Projection	27.45	27.45	19.61
Spatial Navigation	Accessibility Constraints	7.32	7.32	9.76
	Obstacle Avoidance	10.0	8.0	14.0
	Pathway Existence	6.52	6.52	8.7
	Spatial Sequence	16.0	12.0	16.0
	Viewpoint Visibility	20.0	16.0	8.0

with GPT-5-mini reaching 58.14% in Directional Relations and 55.81% in Alignment Patterns, yet even they collapse on Proximity Gradients. Large open-source models post impressive peaks such as 60.0% in Depth and Occlusion, but fall to just 9.3% in Proximity Gradients, demonstrating that scale without specialization remains insufficient. Small open-source models reinforce the same conclusion,

pairing rare localized strengths (e.g., 26.0% in Relative Size Comparison) with catastrophic lows such as 0.0% in Betweenness Relationships. Reasoning models show that explicit reasoning mechanisms can materially improve results—Gemini-2.5-flash-thinking reaches 75.0% in Tool Handedness—even as peers like Kimi-VL-A3B fall to single digits, exposing the fragility of less-integrated reasoning frameworks. Spatial reasoning models, despite their explicit specialization, underperform the most, with scores frequently below 15% and outright failures on Betweenness Relationships. Taken together, these patterns isolate three root causes: (1) the absence of robust representations for continuous and analogical spatial cues, (2) structural limitations tied to scale and training diversity, and (3) brittle reasoning pipelines that prevent generalization across tasks. We conclude decisively that no current approach—proprietary scaling, open-source expansion, reasoning integration, or task specialization—delivers comprehensive coverage. The recurrence of performance gaps exceeding 50 percentage points within a single model confirms that progress remains piecemeal. Future work must therefore unify robust geometric encodings, physics-informed reasoning, and cross-task transfer, advancing beyond isolated gains toward consistent spatial intelligence.

F.2 MODEL FAILURE ANALYSIS

The distribution in Table 17 exposes persistent, category-spanning weaknesses, with a pronounced skew toward the lower bins (0–2 models correct). 3D Geometry proves broadly challenging: Shape Projection (19 items with zero correct) and Stability Prediction (10 items with zero correct) yield the poorest outcomes, indicating that models lack reliable physical and geometric reasoning. Depth and Occlusion is the hardest category overall, accumulating the highest zero-correct count (72), concentrated in Layering Order (20 zero) and Transparency Effects (16 zero)—both of which demand nuanced handling of occlusion and non-local cues. Orientation exhibits comparable difficulty: although Cardinal Direction attains some mid-range success, Stacking Orientation (11 zero) and Tool Handedness (5 zero, none with perfect success) underscore the brittleness of embodied directional reasoning. Relative Positioning reveals widespread inconsistency, with Betweenness Relationships and Proximity Gradients rarely solved by more than two or three models and Corner/Angle Positioning showing low but comparatively balanced performance. Size and Scale stands relatively stronger: several subcategories (e.g., Scale Consistency with 6 full successes) show that some models internalize coarse perspective and scaling cues, even as failures remain high in Shadow-Size Projection (17 zero). Spatial Navigation is bifurcated—Spatial Sequence and Obstacle Avoidance register occasional full-model success, whereas Pathway Existence (16 zero) and Accessibility Constraints (10 zero) expose persistent difficulty with multi-step relational chaining. Aggregating across 356 items, the grand total confirms the diagnosis: zero-correct dominates (356), and only 74 items are solved by all five models, establishing that consistent spatial competence remains elusive.

These findings locate the root causes of failure not in dataset design but in the representational and reasoning limitations of current models. First, the concentration of errors in Depth and Occlusion reflects reliance on superficial appearance features rather than structured scene representations: models latch onto surface textures instead of reasoning about hidden layers, producing systematic errors in Layering Order. Second, failures in Orientation tasks such as Tool Handedness expose the absence of embodied affordance modeling, since pretrained internet imagery rarely carries explicit annotations of left/right handedness or stacking physics. Third, the brittleness observed in Relative Positioning and Spatial Navigation points to deficiencies in chaining multi-hop spatial relations—Pathway Existence, for instance, requires integrating local connectivity with global layout, which attention-based co-occurrence priors capture poorly. By contrast, the partial success in Size and Scale indicates that models exploit statistical regularities in perspective distortion and scale consistency, yet collapse under less frequent cases such as shadows. Collectively, these patterns establish a coherent diagnosis: current models succeed when cues are salient and correlate with training priors, but fail when reasoning demands counterfactual, compositional, or embodied inference. Redressing these limitations will require architectural advances that explicitly model reference frames, occlusion layers, and physical dynamics, complemented by training data enriched with embodied simulations and counterfactual augmentations. Absent such advances, progress in spatial reasoning will remain fragmented—strong on surface-level cues yet breaking down on structured, real-world spatial logic.

Table 17: Item-difficulty distribution on SPATIALAB-OPEN, tabulating how many questions were solved by a given number of models (0–5), broken down by category and sub-category. Here “0” indicates that no model answered the item correctly, whereas “5” indicates that only five of the models tested succeeded on it; lower bins thus isolate the questions that are hardest across the model pool.

Category	Sub-Category	Number of Models Successfully Answered					
		0	1	2	3	4	5
3D Geometry	Gravity Effects	13	7	3	4	1	2
	Shape Projection	19	5	6	2	5	1
	Spatial Containment	10	4	3	6	2	3
	Stability Prediction	10	3	7	3	0	2
	Volume Comparison	11	3	4	3	4	2
3D Geometry Total		63	22	23	18	12	10
Depth & Occlusion	Complete Occlusion Inference	7	5	4	2	5	2
	Layering Order	20	14	10	6	3	3
	Partial Occlusion	14	8	5	5	4	2
	Reflective Surfaces	15	4	1	5	2	5
	Transparency Effects	16	7	7	3	1	2
Depth & Occlusion Total		72	38	27	21	15	14
Orientation	Cardinal Direction	8	10	0	1	5	2
	Facing Direction	13	5	5	1	3	2
	Object Rotation	12	10	6	4	1	2
	Stacking Orientation	11	5	1	2	1	1
	Tool Handedness	5	4	1	4	2	0
Orientation Total		49	34	13	12	12	7
Relative Positioning	Alignment Patterns	7	5	3	1	1	1
	Betweenness Relationships	11	5	7	2	1	3
	Corner/Angle Positioning	7	6	2	1	3	2
	Directional Relations	5	4	4	4	2	0
	Proximity Gradients	10	10	3	3	2	4
Relative Positioning Total		40	30	19	11	9	10
Size & Scale	Distance-Size Correlation	7	5	9	4	1	5
	Perspective Distortion	12	6	6	8	5	2
	Relative Size Comparison	14	7	4	2	2	4
	Scale Consistency	14	7	5	3	3	6
	Shadow-Size Projection	17	4	4	4	3	2
Size & Scale Total		64	29	28	21	14	19
Spatial Navigation	Accessibility Constraints	10	5	4	0	1	2
	Obstacle Avoidance	15	5	8	2	2	5
	Pathway Existence	16	6	7	3	1	1
	Spatial Sequence	14	3	3	3	3	5
	Viewpoint Visibility	13	3	9	5	3	1
Spatial Navigation Total		68	22	31	13	10	14
Grand Total		356	175	141	96	72	74

G ANALYSIS OF MCQ VS. OPEN-ENDED PERFORMANCE GAPS

This section establishes, through a systematic decomposition of the divergence between multiple-choice (MCQ) and open-ended evaluation formats, that MCQ accuracy alone materially overestimates the practical spatial competence of vision–language models. Drawing on all spatial reasoning categories in SPATIALAB and reported in full in Table 18, the analysis motivates a central methodological claim: faithful assessment demands reporting both formats jointly, together with per-subtask diagnostics.

Table 18: **MCQ-to-open-ended accuracy gap (%)**, $\downarrow$ lower is better) across spatial categories on SPATIALAB, computed as the multiple-choice score minus the open-ended score. A value near zero denotes consistent performance across both answer formats, whereas a larger value exposes a pronounced degradation when the multiple-choice scaffold is removed. Within each model group, the largest gap is rendered in **bold** and underlined.

Model	3D Geom. (#238)	Dep. & Occu. (#259)	Orientation (#202)	Relat. Posit. (#212)	Size & Scale (#252)	Spati. Navig. (#237)	Overall (#1400)
<i>Proprietary Models</i>							
GPT-4o-mini	22.40	23.76	16.51	31.74	27.85	20.50	23.53
GPT-5-mini	20.08	23.27	13.21	2.38	19.41	13.36	3.36
Gemini-2.0-Flash	30.89	26.73	26.89	28.18	22.37	25.07	15.13
Gemini-2.5-Flash	21.62	16.34	17.45	15.87	21.51	17.36	10.93
Claude 3.5 Haiku	23.16	21.78	20.29	15.47	24.89	20.29	16.39
Mistral Medium 3.1	30.50	25.74	32.54	26.59	24.89	26.93	21.43
<i>Open-Source Models (Small)</i>							
InternVL3.5-1B	22.78	13.37	23.58	22.62	20.67	22.00	27.73
InternVL3.5-2B	20.46	20.79	16.99	20.64	14.35	19.21	21.85
Qwen-VL2.5-3B-Instruct	27.03	30.69	29.24	28.97	29.96	28.50	25.63
InternVL3.5-4B	25.10	26.24	34.91	20.24	23.20	25.29	23.53
Gemma-3-4B-it	21.23	31.68	22.17	22.22	18.14	22.93	23.53
Qwen-VL2.5-7B-Instruct	22.01	22.27	18.40	26.19	15.61	22.14	27.73
Llama-3.2-11B-Vision-Instruct	13.51	-1.98	17.92	17.07	13.50	11.93	9.66
Gemma-3-27B-it	23.93	23.27	19.82	25.40	25.32	23.14	20.59
Qwen-VL2.5-32B-Instruct	30.50	31.68	28.30	35.32	28.27	29.85	24.79
InternVL3.5-72B	36.68	33.17	34.44	29.37	28.69	31.57	27.31
Qwen-VL2.5-72B-Instruct	27.80	26.73	23.59	19.05	28.27	24.22	20.17
Llama-3.2-90B-Vision-Instruct	28.95	29.21	30.66	25.00	21.52	26.36	23.53
<i>Reasoning Models</i>							
o4-mini	25.48	22.77	21.23	-3.18	17.30	15.35	10.50
Gemini-2-Flash-Thinking	13.90	10.39	11.32	21.03	13.50	13.00	6.73
Gemini-2.5-Flash-Thinking	8.22	16.61	19.46	33.42	31.37	20.16	8.66
Kimi-VL-A3B-Thinking-2506	27.02	28.71	33.50	26.98	22.78	27.92	29.41
<i>Spatial Reasoning Models</i>							
SpaceOm	31.66	32.18	25.95	24.21	27.00	28.43	29.83
SpaceThinker-Qwen2.5-VL-3B	28.57	29.21	27.83	23.81	27.84	27.28	26.89
SpaceQwen2.5-VL-3B-Instruct	31.28	23.76	28.31	38.89	35.87	29.78	18.90

G.1 QUANTITATIVE SNAPSHOT

The disparity is large, pervasive, and remarkably stable across the full evaluation suite. Averaged over 25 models, the MCQ-to-open-ended gap stands at **23.0%** with a standard deviation of only 5.5%, so the effect is neither an artifact of a few outliers nor confined to particular capabilities. Its uniformity across subtasks is striking: the dataset-wide means are 24.57 for 3D geometry, 23.45 for depth & occlusion, 23.70 for orientation, 23.11 for relative positioning, 23.42 for size & scale, and 22.89 for spatial navigation, converging on an overall average of 23.01. Individual models span a considerable range—from a low of 11.93 to a high of 31.57, with a median near 23.34—yet the central tendency remains consistent. Partitioning by model family sharpens the picture: spatial reasoning specialists incur the widest average gap at 27.03, whereas reasoning-oriented models are the most format-robust at 19.11; proprietary systems (21.09) fare better on average than the broader open-source field (24.51), and even the human baseline registers a comparable 22.64. The extremes are instructive. The largest gaps belong to InternVL3.5-72B (31.57%), Qwen-VL2.5-32B (29.85%), and SpaceQwen2.5-VL-3B-Instruct (29.78%), while the smallest arise for Llama-3.2-11B (11.93%), Gemini-2-Flash-Thinking (13.00%), and GPT-5-mini (13.36). Correlational analysis then identifies where the aggregate disparity is concentrated: spatial navigation is its dominant driver (Pearson $r = 0.99$), with orientation ($r = 0.83$) and 3D geometry ($r = 0.79$) likewise closely tracking the overall gap.

G.2 PATTERNS AND OBSERVATIONS

Four regularities structure these results. First, specialist spatial reasoning models exhibit the widest MCQ→open-ended gaps, a pattern consistent with optimization for classification-style supervision at the expense of robustness to free-form generation; reasoning-focused models, by contrast, remain comparatively narrow, reflecting the stabilizing influence of chain-of-thought-style training on generative outputs. Second, spatial navigation is the single strongest predictor of overall disparity: models that stumble on open-ended navigation reasoning are precisely those that inflate the aggregate

gap, with orientation and 3D geometry contributing substantially and relative position and size/scale varying more idiosyncratically across models. Third, robustness is not a function of scale: the compact Llama-3.2-11B ranks among the most format-stable models, whereas the far larger InternVL3.5-72B posts the widest gap of all. Fourth, a handful of negative or near-zero gaps—depth & occlusion for Llama-3.2-11B at -1.98 , relative position for o4-mini at -3.18 —reveal cases in which open-ended evaluation is actually the more reliable probe, an inversion we attribute to misleading distractors that depress MCQ scores.

G.3 HYPOTHESES ON ROOT CAUSES

We hypothesize that several interacting factors underlie the observed MCQ→open-ended performance gaps in SPATIALAB. Each hypothesis below is motivated by the statistical evidence, though we stress that these remain proposed explanations rather than established causes:

1. **MCQ structural advantage.** By constraining the output to a closed set of options, the MCQ format lets a model lean on surface cues or eliminate distractors rather than construct an answer from scratch. The consequence is systematic: mean MCQ scores exceed open-ended scores across the board, with the average gap at 23.0% and $\sigma = 5.5\%$. The exceptions are equally telling—the negative gaps for Llama-3.2-11B in depth & occlusion (-1.98) and o4-mini in relative position (-3.18) show that open-ended generation can surpass MCQ when distractors mislead, reinforcing the view that the closed format can inflate or distort measured performance.

Recommendation: To address this issue, we suggest complementing MCQ evaluations with open-ended tasks and subtask-level diagnostics, which can more accurately capture model competence and reduce overestimation caused by option-based cues.

2. **Specialization bias in spatial reasoning models.** That the models built expressly for spatial tasks post the largest average gaps (27.03%) is a revealing paradox: we hypothesize that their training leans heavily on categorical or synthetic selection objectives rather than free-form generation. InternVL3.5-72B illustrates the pattern acutely—this 72B-parameter spatial reasoning model records the widest overall gap (31.57%), driven by spatial navigation (36.68%) and orientation (34.44%), a signature of classification-like supervision. The concentration of these format-dependent failures in sequential and geometric reasoning is corroborated by the correlations between subtask and overall gaps (SpNav: $r = 0.99$, Orientation: $r = 0.83$).

Recommendation: To mitigate this bias, we recommend fine-tuning or instruction-tuning on open-ended spatial generation tasks, which can improve model robustness to generative formats and reduce reliance on categorical supervision.

3. **Instruction-tuning and stepwise decoding.** At the opposite pole, reasoning-oriented models such as Gemini-2-Flash-Thinking and o4-mini combine smaller average gaps (around 19.11%) with lower variance across subtasks. Gemini-2-Flash-Thinking is emblematic, holding its overall gap to 13.0% with uniformly modest subtask figures (Difference in 3D Geometry: 13.9%, Difference Relative Positioning: 21.03%); we take this as evidence that chain-of-thought or stepwise decoding yields more stable open-ended responses. The weaker coupling between subtask gap variance and overall gap within this family points the same way, suggesting that instruction-tuned reasoning dampens format sensitivity.

Recommendation: To address format sensitivity, we suggest adopting chain-of-thought or stepwise decoding strategies and instruction-tuning for open-ended spatial tasks, which can improve generative consistency and reduce MCQ dependence.

4. **Spatio-navigation stresses sequential grounding.** Spatial navigation gaps predict overall MCQ→open-ended disparity almost perfectly (Pearson $r = 0.99$), making them the clearest window onto the underlying failure mode. Models with large navigation gaps carry commensurately large overall gaps—SpaceQwen2.5-VL-3B-Instruct pairs a Spatial Navigation gap of 35.87% with a 29.78% overall gap—which we read as evidence that multi-step referential reasoning is disproportionately fragile in open-ended settings. The likely mechanism is that MCQ options collapse the reasoning chain, masking weaknesses in sequential grounding and relational computation that surface only under free generation.

Recommendation: To better capture sequential reasoning abilities, we recommend including

dedicated open-ended spatio-navigation tasks and employing targeted multi-step reasoning supervision, which can improve grounding and reduce overestimation by MCQ.

5. **MCQ distractor artifacts.** The negative and near-zero gaps expose a second-order effect: distractor design can distort MCQ measurement in either direction. o4-mini registers a negative relative position gap (-3.18%) while remaining positive elsewhere, a profile that implicates poorly chosen distractors in artificially depressing its MCQ accuracy. The opposite failure is equally plausible—Qwen-VL2.5-32B-Instruct’s very large gaps (overall = 29.85%) across multiple subtasks suggest that easy or leading distractors can inflate perceived competence in selection-based tasks. Either way, the option set, not the underlying spatial ability, becomes a confound.

Recommendation: To address this, we suggest careful auditing and refinement of MCQ distractors and complementing them with open-ended evaluation, which can yield more accurate and unbiased measures of model spatial reasoning ability.

G.4 PRACTICAL IMPLICATIONS

Taken together, these results carry a clear methodological mandate: MCQ and open-ended results must be reported jointly. Relying on MCQ alone inflates estimates of practical spatial competence by roughly 23% on average, a bias too large to leave unaccounted. Subtask-level breakdowns are indispensable, since spatial navigation disproportionately drives the observed differences and would otherwise be obscured in aggregate scores. Diagnostic evaluations that separate classification from generative reasoning are the natural instrument for isolating these error modes. Beyond measurement, instruction-tuning and decoding strategies that encourage stepwise reasoning emerge as promising levers for reducing MCQ dependence, and disciplined design and auditing of distractors is a prerequisite for fair cross-format comparison. In short, the combination of dual-format reporting and per-subtask diagnostics is not optional refinement but a necessary condition for trustworthy spatial-reasoning evaluation.

H DETAILS ON IMPROVING VISUAL REASONING CAPABILITIES

To strengthen spatial reasoning on our diverse visual question benchmark, we systematically investigate four complementary intervention strategies: (i) leveraging the models’ inherent reasoning mechanisms, (ii) Chain-of-Thought (CoT) prompting and its self-reflective extension, (iii) supervised fine-tuning (SFT), and (iv) a purpose-built multi-agent system. Together, these experiments delineate precisely where targeted interventions advance spatial competence and where the benchmark’s difficulty proves resilient.

H.1 INHERENT REASONING MECHANISMS

We first isolate the contribution of models’ built-in reasoning mechanisms by contrasting standard configurations against their reasoning-enabled counterparts (Table 2, Table 3). In the multiple-choice regime, activating reasoning delivers a decisive and reproducible advantage, although its magnitude is category-dependent. Gemini-2.5-Flash-Thinking, for example, surpasses its non-reasoning variant by $+4.64\%$ overall (52.93% vs. 48.29%), driven by pronounced gains in Relative Positioning ($+13.1\%$) and Orientation ($+8.6\%$). The same trajectory recurs for o4-mini-medium, which attains the strongest overall MCQ accuracy in our study (53.21%). Collectively, these figures establish that explicit reasoning scaffolds are most potent in structured, discrete-answer settings, where a bounded hypothesis space rewards deliberate elimination.

Open-ended evaluation yields a more textured picture. Reasoning-equipped models again lead their baselines, yet the margins are uneven. Gemini-2.5-Flash-Thinking edges past Gemini-2.5-Flash (32.77% vs. 30.93%), while the most emphatic advance belongs to o4-mini-medium (37.86%), which comfortably eclipses every non-reasoning model. This evidence indicates that reasoning mechanisms attenuate the instability endemic to free-form generation, most visibly in demanding categories such as 3D Geometry and Spatial Navigation. The benefit is not, however, uniform: reasoning variants occasionally regress, as when Gemini-2.5-Flash-Thinking falls sharply in Size & Scale (21.74% vs. 26.59%), a reversal that echoes our broader finding that generative settings amplify brittleness.

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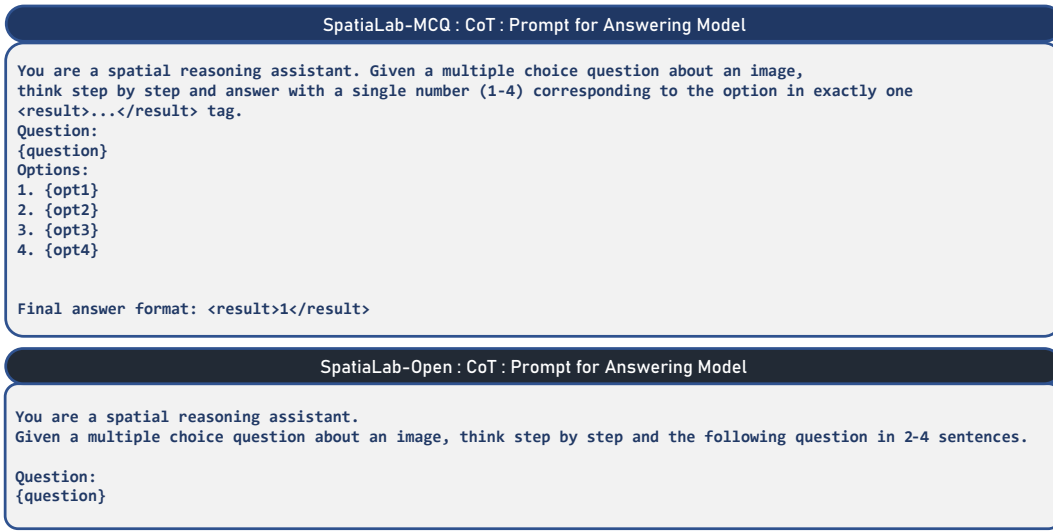


Figure 7: Chain-of-thought (CoT) Prompts.

Taken together, these comparisons establish a clear dual finding: explicit reasoning modules substantially elevate structured MCQ performance and can stabilize open-ended responses, even as their returns vary across categories. Reasoning demonstrably sharpens logical consistency and relational chaining, but it does not, on its own, dissolve the harder obstacles of scale sensitivity, occlusion, and free-form linguistic grounding, which remain the benchmark’s most stubborn frontiers.

H.2 CHAIN-OF-THOUGHT (CoT) PROMPTING

To determine whether explicitly elicited step-by-step reasoning improves spatial performance, we evaluated models under Chain-of-Thought (CoT) prompting. From each of the 30 sub-categories in SPATIALAB we drew 5 items (seed fixed to 42), assembling a balanced test set that spans all major spatial phenomena while keeping evaluation computationally tractable. The step-by-step reasoning scaffolds supplied to the models are illustrated in Figure 7. Because scale and training are held constant across conditions, this protocol cleanly isolates the marginal contribution of CoT reasoning itself.

Table 19: Effect of chain-of-thought (CoT) prompting on per-category accuracy for InternVL-3-78b and Gemini-2.5-Flash, reported for both the MCQ and open-ended formats. All values are accuracies (%); the Gain column is the signed difference between the with-CoT and without-CoT conditions, so that positive values indicate an improvement attributable to CoT prompting and negative values a regression.

Category	InternVL-3-78b (MCQ)			InternVL-3-78b (Open)			Gemini-2.5-Flash (MCQ)			Gemini-2.5-Flash (Open)		
	w/o CoT	with CoT	Gain	w/o CoT	with CoT	Gain	w/o CoT	with CoT	Gain	w/o CoT	with CoT	Gain
3D Geometry	52.00	52.00	0.00	28.00	28.00	0.00	48.00	20.00	-28.00	60.00	28.00	-32.00
Depth & Occlusion	60.00	60.00	0.00	16.00	12.00	-4.00	48.00	28.00	-20.00	40.00	16.00	-24.00
Orientation	60.00	64.00	4.00	24.00	28.00	4.00	56.00	24.00	-32.00	0.00	32.00	32.00
Relative Positioning	80.00	72.00	-8.00	44.00	56.00	12.00	60.00	36.00	-24.00	50.00	32.00	-18.00
Size & Scale	56.00	44.00	-12.00	28.00	20.00	-8.00	48.00	60.00	12.00	25.00	8.00	-17.00
Spatial Navigation	64.00	64.00	0.00	48.00	40.00	-8.00	68.00	52.00	-16.00	40.00	28.00	-12.00
Overall	62.00	59.33	-2.67	31.33	30.67	-0.66	54.67	36.67	-18.00	38.24	24.00	-14.24

Our analysis (Table 19) delivers a sharply delimited result: Orientation is the sole category in which CoT prompting reliably lifts performance, whereas the remaining categories stagnate or decline. The pattern is diagnostic. Reasoning pays dividends precisely when a task reduces to the logical alignment of directions, but it falters wherever richer perceptual priors—depth, scale, relative positioning—are prerequisite. We attribute this asymmetry to a representational gap: current models do not encode 3D perception and geospatial relationships as intrinsic knowledge, and are therefore not “aware” of these concepts in the human sense. Under such conditions, chaining reasoning steps does not repair

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Figure 8: Chain-of-thought (CoT) with Self-Reflection Prompts.

the underlying misconceptions; it entrenches them, propagating and compounding errors across the deliberation.

This result underscores that SPATIALAB is genuinely non-trivial: it cannot be dispatched by simply bolting multi-step reasoning onto existing models. Where textual reasoning tasks reward CoT almost unconditionally, our benchmark exposes deeper architectural limitations in perceptual and spatial grounding. In categories such as depth and size, faulty priors ensure that each additional reasoning step carries the model further from the correct solution. The pronounced degradations we observe—most acute for Gemini in the MCQ setting—demonstrate just how fragile reasoning becomes once it is unsupported by robust perceptual knowledge.

This resistance to CoT is, in fact, a deliberate strength of the benchmark’s design: rather than rewarding surface-level reasoning, SPATIALAB probes whether models genuinely understand geospatial and 3D relational concepts. Solving it demands more than the chaining of logical steps; it calls for embeddings and training objectives that internalize core aspects of perception and spatial cognition. In precisely this way, the benchmark stresses models at a fundamental level and pinpoints a consequential gap in contemporary architectures.

H.3 CHAIN-OF-THOUGHT (COT) PROMPTING WITH SELF-REFLECTION

We extend the CoT setup with an explicit self-reflection stage, in which the model is prompted to review and revise its initial reasoning before committing to a final answer (see Figure 8). The evaluation protocol is held identical to the standard CoT setting—5 balanced samples per sub-category

(seed 42)—so that the effect of self-reflection is isolated and directly commensurable with baseline CoT performance.

Table 20: Chain-of-thought (CoT) versus CoT augmented with Self-Reflection (CoT-SR) on per-category accuracy for InternVL-3-78b and Gemini-2.5-Flash, reported for both the MCQ and open-ended formats. All values are accuracies (%); the Gain column is the signed difference between the CoT-SR and CoT conditions, so that positive values indicate that adding self-reflection helps and negative values that it hurts.

Category	InternVL-3-78b (MCQ)			InternVL-3-78b (Open)			Gemini-2.5-Flash (MCQ)			Gemini-2.5-Flash (Open)		
	CoT	CoT-SR	Gain	CoT	CoT-SR	Gain	CoT	CoT-SR	Gain	CoT	CoT-SR	Gain
3D Geometry	52.00	56.00	4.00	28.00	20.00	-8.00	20.00	52.00	32.00	28.00	24.00	-4.00
Depth & Occlusion	60.00	44.00	-16.00	12.00	16.00	4.00	28.00	52.00	24.00	16.00	16.00	0.00
Orientation	64.00	64.00	0.00	28.00	32.00	4.00	24.00	48.00	24.00	32.00	36.00	4.00
Relative Positioning	72.00	64.00	-8.00	56.00	28.00	-28.00	36.00	72.00	36.00	32.00	32.00	0.00
Size & Scale	44.00	56.00	12.00	20.00	12.00	-8.00	60.00	56.00	-4.00	8.00	12.00	4.00
Spatial Navigation	64.00	56.00	-8.00	40.00	24.00	-16.00	52.00	64.00	12.00	28.00	28.00	0.00
Overall	59.33	56.67	-2.66	30.67	22.00	-8.67	36.67	57.33	20.66	24.00	24.67	0.67

Augmenting Chain-of-Thought (CoT) prompting with self-reflection produces outcomes that are strongly model-dependent, and the contrast between models is itself instructive. For InternVL-3-78b, overall accuracy edges downward (−2.66% MCQ, −8.67% open-ended), with steep losses in depth, relative positioning, and navigation—signaling that the reflective pass more often entrenches errors than corrects them. Localized gains do surface, notably +12% in Size & Scale (MCQ), yet they are inconsistent and swamped by large degradations elsewhere (−28% in relative-positioning open-ended). Gemini-2.5-Flash, by contrast, records robust MCQ gains (+20.66%), anchored by sizable improvements in 3D Geometry (+32%) and Depth & Occlusion (+24%), while its open-ended improvement stays negligible (+0.67%). Read jointly, these results expose a decisive asymmetry: reflection aids multiple-choice settings, where models can prune implausible options, but does not transfer to free-form generation, where residual uncertainty flows into ungrounded rationales. Just as tellingly, the categories reflection most improves are those governed by geometric consistency (orientation, depth, size), whereas tasks demanding relational integration across multiple objects (relative positioning, navigation) degrade. Reflection thus operates as a stabilizer within structured choice settings but does not reliably strengthen naturalistic reasoning.

This inconsistency mirrors our earlier observations with plain CoT: reasoning enhancements succeed only when the model possesses stable perceptual anchors on which to ground its steps. Self-reflection introduces an explicit verification pass, but that mechanism can be no stronger than the internal priors it interrogates. When depth cues, occlusion ordering, or coordinate frames are poorly encoded, the second pass magnifies misconceptions rather than repairing them—which is precisely why Gemini, with its firmer perceptual grounding, benefits under MCQ conditions while InternVL-3-78b degrades more frequently. The deeper cause is a pretraining mismatch: current VLMs inherit reasoning templates from language modeling yet lack the embedded 3D and geospatial representations that humans routinely exploit. Absent these anchors, reflection cannot perform true error correction; it operates instead as a linguistic filter that polishes the syntax of reasoning without improving its perceptual substrate. Reflective loops therefore afford modest utility in discrete-answer formats but collapse under open-ended demands, where no factual verification is available to check the revised rationale.

These findings reaffirm that SPATIALAB surfaces a fundamental gap in current architectures: surface-level reasoning layers alone cannot substitute for absent spatial cognition, and meaningful progress will require the deeper integration of geometric and perceptual structure. The disparity is compounded by the data distribution itself: because VLMs are predominantly trained on web-scale corpora, concepts that are richly represented online are exactly those on which models perform best, while under-represented spatial and 3D concepts languish—entrenching the very performance gaps our benchmark measures.

H.4 SUPERVISED FINE-TUNING (SFT)

H.4.1 SETUP AND IMPLEMENTATION

We adapted the Qwen 2.5 VL 3B instruction-following model (unsloth/Qwen2.5-VL-3B-Instruct) through a parameter-efficient fine-tuning procedure built on LoRA (Hu et al., 2021). To operate within a constrained GPU budget, we quantized the base model to 4-bit precision (`load_in_4bit=True`) and inserted low-rank adapters into both the attention and the feed-forward projection layers, leaving the frozen backbone untouched. The adapter configuration—rank, scaling factor, dropout, and target modules—is enumerated in Table 21. Gradient checkpointing further curtailed memory consumption, and a fixed random seed guaranteed reproducibility of adapter initialization.

Tokenization adhered to the Qwen-2.5 chat template, ensuring that the formatting seen during training exactly matched that used at inference. Each example was serialized into a single text sequence, and dynamic padding was handled by `DataCollatorForSeq2Seq`. We trained with an effective batch size of 4, realized through a per-device batch size of 1 and gradient accumulation over 4 steps. The optimization schedule—learning rate, optimizer, weight decay, and scheduler—is specified in Table 22. The procedure comprised four sequential passes, each spanning a single epoch, so that one checkpoint was persisted per epoch for a total of four.

Overall, this setup allowed us to efficiently fine-tune a large vision-language model on limited GPU resources while maintaining high adaptation capacity.

Table 21: LoRA Hyperparameters for SFT

Hyperparameter	Value	Notes
LoRA rank (r)	16	Low-rank update capacity
LoRA alpha	16	Scaling factor
LoRA dropout	0	No dropout applied
Gradient checkpointing	Enabled	Library-specific flag
LoRA random seed	3407	Adapter initialization

Table 22: Trainer and Optimization Hyperparameters for SFT

Hyperparameter	Value
Base model	unsloth/Qwen2.5-VL-3B-Instruct
Max sequence length	2048
Load in 4-bit	True
Per-device train batch size	1
Gradient accumulation steps	4
Effective batch size	4
Learning rate	2e-4
Optimizer	<code>paged_adamw_8bit</code>
Weight decay	0.01
LR scheduler	linear
Logging steps	1
Seed	42
Tokenizer template	Qwen-2.5

H.4.2 SUPERVISED FINE-TUNING (SFT) PERFORMANCE

The learning trends in Figure 4a reveal notable differences between the MCQ and Open-ended evaluation modes over training epochs. We observe that MCQ accuracy begins at a lower baseline of 25.63% but rises consistently, converging near 36.59% by the fourth epoch. This steady improvement suggests that the model benefits strongly from additional supervised fine-tuning, particularly in tasks that constrain answers within a limited candidate space. By contrast, Open-ended evaluation starts

higher at 34.40% but declines sharply by the second epoch, dropping to as low as 12.62%. Only in later epochs does it partially recover, reaching 35.48% at the end. This instability points to a sensitivity in open-ended generation that is not present in the MCQ setting, where the structure of multiple-choice questions provides more stable optimization signals. We interpret the fluctuations in open-ended performance as evidence that fine-tuning may overfit to MCQ-like reasoning patterns at the expense of free-form language generalization. The dotted baselines in the plot further highlight how the MCQ mode achieves gains beyond its initial standing, whereas Open-ended remains approximately stagnant. Overall, the figure illustrates that SFT predominantly favors MCQ tasks in terms of stability and consistent accuracy gains.

The tabulated results in Table 4b complement the figure by providing a detailed before-and-after breakdown across six spatial reasoning categories. Across MCQ tasks, we find that performance improved in every category, with gains ranging from a modest 1.59% in Relative Positioning to a substantial 17.11% in Size and Scale. The Grand Total gain for MCQ stands at 10.97%, confirming that the fine-tuning procedure effectively enhances structured-answer performance. In contrast, Open-ended results are far more mixed, with both positive and negative changes. Categories such as 3D Geometry and Spatial Navigation show small but positive gains (3.50% and 2.82%, respectively), yet Size and Scale exhibits a significant drop of -3.95% . Relative Positioning also decreases by -0.79% , reflecting that fine-tuning can inadvertently harm certain reasoning modes in free-form contexts. The overall Open-ended gain is only 1.07%, which is statistically negligible compared to MCQ improvements. These results confirm that the effectiveness of SFT is highly sensitive to the evaluation paradigm. By isolating the categories, we underscore where training helps, where it fails, and how performance benefits differ between constrained and unconstrained outputs.

A deeper inspection reveals striking disparities between categories and formats. For MCQ, Size and Scale and Spatial Navigation yield the highest improvements, suggesting that models trained with multiple-choice prompts are especially effective at learning tasks requiring scale discrimination and navigational reasoning. However, in Open-ended form, Size and Scale actually declines, showing that the same learned representations do not transfer well to generative contexts. Similarly, Relative Positioning barely improves in MCQ and worsens in Open-ended, highlighting a consistent weakness in relational reasoning regardless of format. Conversely, Orientation shows strong MCQ gains ($+13.33\%$) and small Open-ended improvements, suggesting that some categories generalize more robustly. We hypothesize that the divergence between the two formats may stem from the structural scaffolding MCQs provide, which reduces the need for the model to generate precise linguistic formulations. Open-ended tasks, by contrast, require simultaneous mastery of reasoning and natural language fluency, which amplifies weaknesses. The trends indicate that while fine-tuning strongly enhances pattern recognition in discrete-choice settings, it introduces volatility and even degradation when free expression is required. This duality makes clear that training interventions optimized for MCQs may not generalize seamlessly to unconstrained reasoning.

The implications of these findings are twofold. First, we show that SFT as currently applied provides strong benefits for structured tasks but limited or even negative transfer to open-ended reasoning. This limitation is not only an artifact of fine-tuning, but also symptomatic of how contemporary VLMs are pretrained. Existing pretraining pipelines heavily emphasize broad image-text alignment, object recognition, and captioning, yet they provide little explicit supervision on compositional spatial structures, occlusion layers, or geometric transformations. As a result, models learn to excel at surface-level associations while failing to acquire robust representations of depth, orientation, or scale. The instability we observe in open-ended spatial reasoning suggests that the model’s learned priors are fragile: without multiple-choice scaffolding, they revert to shallow pattern-matching instead of grounded inference. Categories such as Size and Scale and Relative Positioning, where declines are most pronounced, point to missing inductive biases in pretraining, biases that humans develop through embodied interaction with 3D environments. Another contributing factor is the reliance on synthetic or instruction-following datasets that flatten spatial diversity into puzzle-like abstractions, preventing models from generalizing to real-world clutter, noise, and complexity. These findings underscore the need to rethink both pretraining and fine-tuning for spatial reasoning, incorporating datasets and objectives that reflect naturalistic visual-spatial challenges. Ultimately, we argue that without targeted reforms, current VLMs risk reinforcing an algorithmic straightjacket: they appear competent under constrained evaluations yet remain brittle and unaligned with the embodied, flexible reasoning that humans bring to real-world spatial understanding.

H.4.3 SFT DYNAMICS ANALYSIS

To investigate the stability of Supervised Fine-Tuning (SFT) on spatial reasoning tasks, we conducted a multi-seed validation study ($N = 5$, seeds A–E) using the Qwen-2.5-VL-3B-Instruct model. The training trajectory, summarized in Table 23, reveals a distinct divergence in performance dynamics between Multiple-Choice Question (MCQ) and Open-Ended evaluation formats.

Divergent Learning Trajectories. As illustrated in Table 23 and Figure 9, performance on SPATIALAB-MCQ exhibits a monotonic improvement, rising from a baseline of 25.63% to an average of 35.74% at Epoch 4. This indicates that the model successfully adapts to the discriminative nature of the task, learning to map visual features to constrained output options with increasing reliability.

In sharp contrast, SPATIALAB-OPEN displays a pronounced *U-shaped* trajectory across all seeds. Rather than steady improvement, the model suffers an immediate and significant performance collapse in the early epochs (dropping from 34.40% to an average of 19.26% at Epoch 1). While performance recovers in later epochs (converging to $\approx 34.82\%$ at Epoch 4), the final gain over the zero-shot baseline is negligible (+0.42%).

Table 23: Multi-seed validation of SFT dynamics ($N = 5$). While MCQ performance improves steadily, Open-Ended performance exhibits a systematic “U-shaped” collapse and recovery, indicating representational instability in generative spatial tasks.

Epoch	Type	Seed A	Seed B	Seed C	Seed D	Seed E
0 (Base)	MCQ	25.63%	25.63%	25.63%	25.63%	25.63%
	Open	34.40%	34.40%	34.40%	34.40%	34.40%
1	MCQ	29.56%	35.83%	24.13%	27.63%	26.53%
	Open	22.50%	23.20%	12.50%	14.00%	24.10%
2	MCQ	34.33%	37.76%	27.33%	31.83%	34.43%
	Open	12.62%	25.30%	28.80%	23.80%	16.65%
3	MCQ	34.68%	37.33%	32.86%	32.23%	34.73%
	Open	28.69%	22.22%	30.34%	32.90%	28.30%
4	MCQ	36.59%	38.48%	34.43%	34.36%	34.83%
	Open	35.48%	34.59%	33.92%	34.70%	35.40%

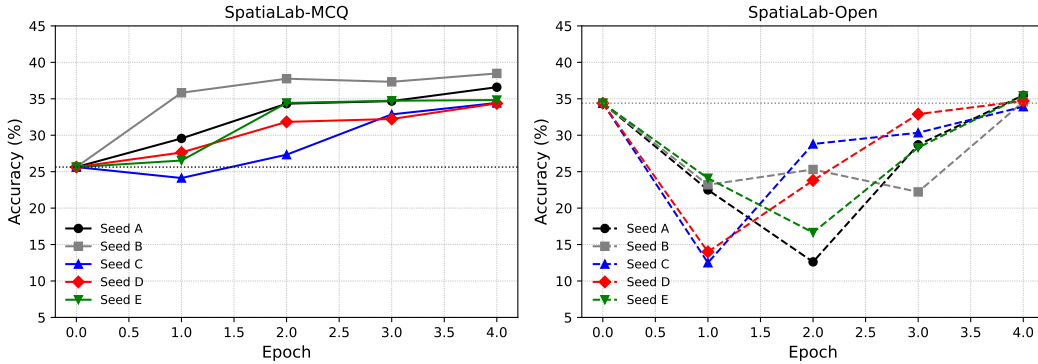


Figure 9: SFT Dynamics Analysis.

Representational Brittleness and Alignment Tax. We attribute this phenomenon to the fragility of spatial representations in current VLMs when subjected to fine-tuning. The initial collapse in open-ended accuracy suggests *catastrophic forgetting* of the linguistic priors required for fluent, grounded generation. As the model optimizes for the specific instruction-following format of the

training data, it overfits to the structural logic of the tasks (benefiting MCQ) at the expense of the flexible, generative spatial grounding required for open-ended descriptions.

The subsequent recovery phase represents the model re-learning to articulate spatial concepts within the new distribution, yet it fails to surpass the initial baseline significantly. This dynamic underscores a critical finding: SFT is effective for teaching models to *select* correct spatial answers, but it does not fundamentally enhance the underlying *generative spatial representation* needed for robust open-ended reasoning.

H.4.4 PERFORMANCE GAIN ON EXTERNAL BENCHMARKS

To assess whether the representations learned from SPATIALAB generalize beyond the benchmark’s specific format, we evaluated the transferability of our fine-tuned model to three external spatial reasoning datasets: OMNISpatial, SPACE (Multimodal), and MIND THE GAP. We employed the Qwen2.5-VL-3B-Instruct model, fine-tuned on SPATIALAB via the SFT protocol detailed in Appendix H.4.1, and evaluated it on these external benchmarks in a zero-shot setting.

Table 24: Transfer learning performance of Qwen2.5-VL-3B-Instruct on external benchmarks after fine-tuning on SPATIALAB. The consistent gains indicate that the model learns generalizable spatial representations rather than merely overfitting to the source dataset’s format.

Benchmark	Base Accuracy	After SFT	Improvement
OmniSpatial	40.30%	47.35%	+7.05%
SPACE (Multimodal)	23.43%	28.67%	+5.24%
Mind The Gap	35.86%	47.56%	+11.70%

As shown in Table 24, fine-tuning on SPATIALAB yields substantial and consistent performance improvements across all evaluated domains, ranging from +5.24% to +11.70%. The most significant gain was observed on MIND THE GAP (+11.70%), suggesting that our benchmark’s emphasis on occlusion and spatial continuity effectively addresses the reasoning gaps targeted by that dataset. Similarly, improvements on OMNISpatial (+7.05%) and SPACE (+5.24%) confirm that the geometric and relational concepts encoded in our taxonomy align with broader definitions of spatial intelligence in the literature. These results validate the quality of the SPATIALAB data. While the primary purpose of this work is to serve as a rigorous diagnostic benchmark, the strong transferability demonstrates that the dataset does not merely encourage overfitting to a specific question template. Instead, the diversity of the 30 sub-categories forces the model to acquire robust, transferable spatial representations that generalize to unseen distributions and task formats.

H.5 AI AGENTS FOR SPATIAL REASONING

We developed SPATIOXOLVER, a multi-agent system adapted and extended from Xolver (Hosain et al., 2025), to perform structured spatial reasoning on images within the SPATIOLAB evaluation. The system decomposes complex visual reasoning into specialized sub-tasks, each managed by a dedicated agent, allowing for focused processing of objects, attributes, relations, symmetries, and transformations. This modular design not only improves accuracy and interpretability but also supports multi-step reasoning, enabling the system to propagate low-level perceptual cues through higher-order relational and transformational inferences. The architecture, agent implementation, and their coordinated interaction are detailed in the following sections, with the full set of prompts provided in Figures 10, 11, and 12.

H.5.1 AGENTS

Base Visual Analysis Agent. Base Visual Language Model (VLM) agent initiates the pipeline by generating a detailed textual description of the input image, capturing object types—including geometric shapes, natural elements, and artificial icons—alongside attributes such as size, orientation, color, and shading. Relative positions like leftmost, center, and top are encoded explicitly, and the prompt emphasizes exhaustive detail without summarization. Images are converted to data URIs and sent with the prompt to a Gemini-2.5-based model, whose output is parsed into a structured bullet-list

using regex-based extraction. This structured description serves as the foundation for downstream agents, providing a comprehensive perceptual anchor that encodes both explicit and implicit spatial cues. The approach ensures that even subtle visual features, such as partial occlusions or minor shape variations, are retained for later relational and transformational reasoning.

Object Segmentation Agent. Object Segmentation agent converts the verbose textual description into a discrete list of uniquely identified objects, differentiating between visually similar instances. Each object receives a stable identifier (e.g., Obj1, Obj2) to maintain consistent references across subsequent reasoning steps. The agent uses carefully formatted prompts and calls the same Gemini-2.5 model to structure the description, while post-processing splits lines by object ID to enforce consistency. This structured representation reduces ambiguity in downstream attribute and relation extraction, enabling precise mapping of complex scenes. By explicitly isolating objects, the agent also facilitates detection of overlaps, repetitions, and hierarchical groupings that may not be evident in raw descriptions.

Attribute Extraction Agent. Attribute Extraction agent translates each object into a structured JSON capturing core properties such as shape, size, color, shading, orientation, and positional descriptors. Prompts enforce strict JSON output, and responses are parsed using `json.loads()`, with fallback defaults to ensure robustness against model inconsistencies. This structured encoding allows quantitative and qualitative comparisons of objects, supporting later stages of spatial reasoning. By preserving both absolute and relative measures, the agent maintains fidelity to the original perceptual content, while also normalizing descriptors for computational consistency. The JSON structure enables programmatic downstream operations, such as calculating symmetries or detecting transformations.

Spatial Relation Agent. The Spatial Relation agent encodes object interactions as triples (ObjectA,Relation,ObjectB), capturing directional relations (e.g., left_of, above, inside) and higher-order alignments like rows, grids, or repetition patterns. Consistent object IDs are enforced to prevent relational ambiguity, and parsing splits output lines starting with parentheses, followed by automated consistency checks. This step transforms static object descriptions into a relational graph that encodes spatial topology and hierarchy. The resulting structure allows reasoning over both pairwise interactions and more complex scene configurations, which is critical for tasks like symmetry detection, grouping, and multi-step navigation of spatial layouts.

Grouping and Symmetry Agent. Grouping and Symmetry agent abstracts higher-order patterns from object lists and spatial relations, identifying clusters, rows, grids, and symmetries—including vertical, horizontal, rotational, and translational. Outputs are requested in structured JSON with keys `groups` and `symmetries`, with empty structures as fallbacks to maintain downstream compatibility. This agent captures emergent scene regularities that are not apparent at the object or pairwise relation level, providing critical context for pattern recognition and reasoning. By formalizing structural redundancies and symmetries, the agent supports both perceptual inference and anticipatory reasoning for dynamic transformations.

Transformation Tracking Agent. For multi-frame sequences, the Transformation Tracking agent logs object-level changes including translations, scaling, rotations, color/shading shifts, and appearance or disappearance events. Structured logs are extracted line-by-line using regex patterns that capture directional arrows, producing a temporal representation of the scene. This encoding allows the system to track not just static spatial configurations but also dynamic evolution, supporting reasoning about motion, causality, and object interactions over time. The agent ensures that transformations are linked to consistent object IDs, enabling coherent cross-frame reasoning and predictive modeling.

Representation Standardization Agent. Finally, the Representation Standardization agent consolidates all outputs into a unified JSON format containing `objects`, `relations`, `groups`, `symmetries`, and `transformations`. Prompts enforce consistent object IDs and structured formatting, while JSON parsing failures revert to empty dictionaries to ensure pipeline robustness. By centralizing all perceptual, relational, and temporal information, this agent creates a comprehensive scene representation that is both human-interpretable and machine-readable. The resulting unified

structure serves as a versatile foundation for downstream tasks, including reasoning evaluation, spatial pattern analysis, and multi-frame prediction.

Open-Ended Spatial Reasoning Agent. The standardized representation feeds into the reasoning agent, which answers multiple-choice or pattern recognition questions. Inputs include objects, attributes, relations, groups, symmetries, and transformations. The model considers geometric patterns, relational alignment, and dynamic transformations to generate or select the correct answer.

H.5.2 TECHNICAL IMPLEMENTATION DETAILS

For our technical implementation, we use Gemini-2.5-Flash with a low temperature (0.1) to ensure deterministic outputs and perform multiple iterations for refinement. API calls are handled via OpenRouter, supporting both text and image inputs, while parsing of model responses relies on regex and JSON-based extraction to generate structured outputs. All interactions are logged in JSON format for reproducibility, and the final output is a unified JSON per image containing comprehensive information on objects, relations, symmetries, and transformations. This multi-agent decomposition allows each specialized model to focus on a constrained sub-task, enhancing overall accuracy, interpretability, and robustness in structured spatial reasoning.

H.5.3 EVALUATION AND ANALYSIS

The evaluation protocol remains identical to the standard CoT and CoT-SR setting, using 5 balanced samples per sub-category (seed 42), making a total of 150 samples.

Table 25: Agent Performance Analysis

Category	MCQ		Gain	Open Evaluation		Gain
	Normal	With Agents		Normal	Agent Results	
3D Geometry	48.00%	44.00%	-4.00%	60.00%	48.00%	-12.00%
Depth & Occlusion	48.00%	44.00%	-4.00%	40.00%	16.00%	-24.00%
Orientation	56.00%	64.00%	+8.00%	0.00%	36.00%	+36.00%
Relative Positioning	60.00%	60.00%	0.00%	50.00%	36.00%	-14.00%
Size & Scale	48.00%	52.00%	+4.00%	25.00%	24.00%	-1.00%
Spatial Navigation	68.00%	64.00%	-4.00%	40.00%	28.00%	-12.00%
Overall	54.67%	57.33%	+2.66%	35.83%	31.33%	-4.50%

The evaluation of Gemini-2.5-Flash with agents highlights both strengths and limitations of agentic reasoning in perceptual-geometric tasks. Orientation is the only category where agents provide consistent and substantial gains, improving by +8.00% in MCQ and an impressive +36.00% in open evaluation. This pattern suggests that agent-based multi-step reasoning is effective when the task can be decomposed into sequential alignment of directional cues, where structured deliberation offers a clear advantage over single-step outputs.

In contrast, most other categories exhibit negative or neutral gains, revealing that agent reasoning does not universally enhance performance. For example, depth & occlusion suffers a severe -24.00% drop in open evaluation, while spatial navigation decreases by -12.00%. These tasks require deeper perceptual grounding in three-dimensional space, and the absence of such priors means that reasoning steps tend to reinforce misconceptions rather than correct them. Similarly, relative positioning shows stagnation in MCQ and a -14.00% decline in open evaluation, indicating that agent loops fail to add robustness when relational understanding is inherently weak.

Some categories show marginal improvements, such as size & scale (+4.00% in MCQ), but the gains are not stable and do not transfer to open evaluation. Even in 3D geometry, agents fail to provide a boost, with -4.00% and -12.00% drops in MCQ and open evaluation, respectively. These results underscore that our benchmark is complex and not solvable simply by layering multi-step

agentic reasoning on top of existing models. Instead, it requires a deeper integration of geospatial understanding and perceptual representations, which current architectures lack.

Overall, the mixed results emphasize that agentic workflows alone are insufficient for advancing performance in this domain. While orientation tasks benefit from structured reasoning, most categories expose the models’ lack of inherent awareness of perceptual concepts. This makes errors propagate through multi-step reasoning, amplifying performance drops. Consequently, our benchmark serves as a strong stress test, highlighting that future progress will require embedding genuine perceptual and geospatial priors rather than relying solely on agent-based reasoning.

I QUALITATIVE ERROR ANALYSIS AND ERROR PATTERNS

This section develops a systematic qualitative account of the failure modes exhibited across our visual reasoning benchmark. For each error class we articulate the observable behavior, advance a diagnostic root-cause analysis, and trace the recurring patterns that bind these failures together, complementing the discussion with the verification protocols we employ. Far from cataloguing incidental mistakes, the analysis establishes that model errors are systematic, interpretable, and diagnostic: they expose the shared inductive biases of contemporary vision–language pipelines and, in doing so, furnish concrete and actionable guidance for future architectural and evaluation design.

I.1 OVERVIEW

Across the sample set, models demonstrate robust coarse perception: they reliably recognize object categories, colors, and simple binary relations. Yet they falter in a strikingly regular fashion once multiple cues must be integrated into a single judgment. Crucially, these breakdowns are not random noise. They condense into a compact and interpretable taxonomy of five recurring classes: spatial mislocalization and reference confusion, perspective and scale mistakes, occlusion and ordering failures, attribute confusion and semantic swap, and ungrounded open-ended rationalization. The very fact that failures organize into so few, so legible categories is itself a strength of the benchmark, which is calibrated to surface these structural weaknesses rather than obscure them. The classes recur across model architectures, across parameter scales, and across prompting styles, which indicates that their origin lies in inductive biases common to current vision–language pipelines rather than in isolated training artifacts. Tellingly, the same error signature surfaces in large-scale proprietary systems and in small open-source baselines alike, evidence of a shared structural bias that transcends any single model family. In what follows, we unpack each error class through representative examples and precise root-cause hypotheses.

I.2 ERROR TAXONOMY AND ANALYSIS

I.2.1 SPATIAL MISLOCALIZATION AND REFERENCE CONFUSION.

■ **Observation.** Whenever a query demands that one object be singled out from a set of visually similar candidates, models misidentify the target with notable regularity. Asked to indicate “the leftmost red cube among three stacked cubes,” a model will often return the central cube instead, swayed by a lighting highlight or by mere positional centrality. The difficulty sharpens markedly in cluttered environments such as kitchen counters or crowded table scenes, where multiple overlapping objects share the same color and shape. The resulting mistakes are anything but arbitrary: the model consistently privileges whichever object is most salient by virtue of size, brightness, or centrality, even when these perceptual cues directly contradict the explicit spatial instruction in the query. The stability of this pattern is precisely what marks the failure as systematic rather than accidental.

★ **Analysis.** This recurring pattern points to the absence of a robust object-centric representation, what cognitive science would term a persistent “index” that survives across successive reasoning steps. Rather than maintaining a stable pointer to the designated candidate throughout the reasoning chain, the model drifts, reassigning attention at intermediate layers to whatever entity appears most visually striking. The consequence is a characteristic slippage in which the chain begins with the leftmost cube yet terminates on the brightest or largest one. Because spatial reference cannot be held fixed, relational queries such as “the ball behind the second chair” become especially fragile. In essence, the model behaves less as a referential reasoner than as a salience detector.

⊗ **Root cause.** At the architectural level, feature pooling operations, whether realized through

attention or convolution, blend contextual information across neighboring regions. Beneficial as this is for aggregating global context, it simultaneously erases the fine-grained, object-specific identity required to separate similar candidates. Once pooled, the features of two adjacent red cubes become nearly indistinguishable in latent space. Absent an explicit mechanism, such as slot-based representations or pointer tokens, the network collapses distinct entities into blended embeddings. This structural limitation forecloses reliable referent tracking across reasoning steps and accounts for the consistent confusion of dense, overlapping objects.

I.2.2 PERSPECTIVE AND SCALE MISTAKES.

■ **Observation.** Tasks that demand metric reasoning about relative size, fit, or scale are handled with conspicuous inconsistency. Simple comparisons such as “is a car larger than a bicycle” are answered correctly with ease, because the relationship coincides with a strong statistical prior. Near-threshold decisions, by contrast, such as “will the small sphere fit inside the slightly larger cylinder,” provoke frequent and erratic errors. A recurring signature is reliance on canonical object sizes: the model predicts that a door must exceed a truck in height even in a rendering where the truck has been scaled down. A parallel failure arises in scenes that deliberately invert familiar proportions, such as toy miniatures or architectural models, where the model clings to its priors and disregards the geometric evidence plainly visible in the image.

★ **Analysis.** These behaviors reveal a heavy dependence on the distributional statistics of object categories in place of genuine geometric inference. Rather than reasoning over projection geometry, relative bounding-box extents, or depth cues, the model substitutes its world knowledge of typical scales. The heuristic serves well in ordinary contexts but fails catastrophically in edge cases that depart from the training distribution: presented with an unusually small airplane rendered beside a person, the model still declares the airplane the larger object. Such errors expose both the missing grounding in pixel-level metric cues and the failure to exploit perspective structure such as vanishing points in any consistent manner.

⊗ **Root cause.** Current architectures incorporate no explicit geometric reasoning module capable of preserving projection-aware relationships. Because supervision for scale and fit is scarce, models learn correlations between category labels and typical size but never the underlying geometry that governs them. Compounding this, training objectives reward aggregate accuracy rather than fine-grained metric consistency, so scale-sensitive errors go under-penalized. Lacking either an inductive bias for 3D reasoning or targeted supervision, the model falls back on memorized priors about canonical sizes instead of interpreting the perspective geometry actually present in the image.

I.2.3 OCCLUSION AND ORDERING FAILURES.

■ **Observation.** Determining which object stands in front, which behind, and which is partially hidden remains a core weakness. Models routinely misreport the foreground element when two animals overlap or when semi-transparent barriers such as glass or fences intervene. In an image where a cat sits partially behind a curtain, for instance, the model may invert the relation and describe the curtain as occluded by the cat. Thin or fragile structures, such as wires, tree branches, or railings, prove especially error-prone: they are frequently omitted altogether, producing wrong answers to questions such as “what object lies behind the fence.” The mistakes recur with consistency, most acutely when occlusion cues are subtle or the depth separation is slight.

★ **Analysis.** Such failures indicate that depth and occlusion are treated as weak, secondary signals. Although the pixel data contains unambiguous occlusion cues, such as edge boundaries and T-junctions, the intermediate features the model learns do not retain them at sufficient resolution. Aggregation layers blur boundaries and down-weight thin structures, effectively discarding the very cues on which ordering depends. This explains why models handle gross depth distinctions between foreground and background yet collapse in the fine-grained regime where partial occlusion must be inferred. In practice they guess rather than infer, with a systematic bias toward assigning foreground status to whichever object is visually larger or brighter.

⊗ **Root cause.** The architecture provides no explicit mechanism for multi-scale feature retention and no dedicated supervision for depth or occlusion reasoning. Thin or partially occluded signals are readily lost once features are pooled across large receptive fields. Training data compound the problem by underrepresenting subtle occlusion cases, particularly those involving transparency or fine boundaries. The learned representations therefore flatten layered scenes into 2D appearance maps, precluding the robust construction of depth-aware relational reasoning.

I.2.4 ATTRIBUTE CONFUSION AND SEMANTIC SWAP.

■ **Observation.** Models frequently conflate perceptual attributes, such as color, texture, or material, with functional or semantic roles. A canonical instance is the judgment of whether a crosswalk is safe: the model may rest its answer entirely on the presence of painted stripes while ignoring the state of the traffic light and the approaching vehicles. In the same vein, a surface with a metallic sheen may be classified as “steel” despite being painted plastic, and in tool-use settings the mere presence of a handle or a familiar silhouette can lead the model to attribute a function the object does not possess. Such confusions arise reliably whenever a task requires fine-grained visual grounding of attributes.

★ **Analysis.** These errors reveal that language priors exert a strong bias over prediction whenever the visual input is ambiguous. The decoder gravitates toward plausible narratives grounded in training-data co-occurrence, such as “crosswalks are safe when stripes are visible” or “shiny textures correspond to metal.” The shortcut lets the model produce semantically coherent answers without ever inspecting the pertinent visual evidence, yielding reasoning that appears correct in familiar contexts yet fails systematically in atypical or adversarial ones. In effect, the model overfits to textual stereotypes in place of grounding its judgments in perceptual detail.

⊗ **Root cause.** The governing training objective, maximizing the likelihood of fluent and plausible text, imposes no requirement of visual grounding. So long as the generated answer is semantically reasonable, the loss is minimized regardless of whether the underlying reasoning is correct. The model is thereby incentivized to substitute common-sense priors for genuine perceptual inference. Absent auxiliary losses or verification steps that tie textual claims to specific image regions, the interchange of perceptual and semantic attributes will persist.

I.2.5 OPEN-ENDED RATIONALIZATION WITHOUT VERIFICATION.

■ **Observation.** On explanatory or why-style questions, models routinely generate coherent yet unsupported narratives. Asked “why is the object leaning,” a model may answer “because the surface is slippery,” even though slipperiness admits no visual inference; a tilted shelf may be explained “due to an earthquake,” a claim lying wholly outside the evidence in the image. Such errors betray a tendency to fill evidential gaps with imaginative but ungrounded rationales, yielding fluent explanations that mislead users about what the image actually supports.

★ **Analysis.** The issue originates in a decoupling of text generation from visual grounding. The model produces high-quality language that reads convincingly, yet no mechanism binds these rationales to the actual visual features. Poor confidence calibration compounds the problem: fabricated explanations are frequently issued with high certainty, lending them a spurious air of reliability. Open-ended outputs may thus appear authoritative while remaining factually baseless, a failure mode of particular concern in safety-critical domains, where a plausible but false rationale could misdirect a downstream decision.

⊗ **Root cause.** Training is dominated by language-modeling objectives that reward fluency and coherence while remaining indifferent to factual grounding. The decoder is optimized to emit the most probable continuation of text given the prompt, under no constraint that links its explanation back to specific image evidence. With no explicit verification layer and no grounding requirement, the model defaults to linguistic plausibility over perceptual truth. Until additional architectural or training constraints are imposed, open-ended rationalization will continue to prioritize surface coherence at the expense of evidential accuracy.

I.3 ERROR PATTERNS AND CROSS-MODEL SIGNALS

Comparing failures across models brings several consistent patterns into relief, each illuminating the mechanisms that generate the errors. The first is that inter-model disagreement is systematic rather than random: some models lean consistently on geometric cues such as relative size, depth ordering, and occlusion, while others retreat to linguistic priors and common-sense assumptions drawn from training text. The result is predictable divergence, as when two models return opposite verdicts on which object is “in front” of a shared scene. Errors also spike whenever several cues must be integrated at once; tasks that couple occlusion, relative scale, and perspective are the most demanding, and judging whether a larger object that partially occludes a smaller one lies in front or behind reliably provokes coordinated mispredictions. Calibration constitutes a further recurring signal: high-confidence predictions correspond disproportionately to incorrect answers, especially in open-ended explanations and complex spatial queries. Taken together, these patterns locate the source

of error in how each model balances perceptual grounding against linguistic plausibility. Distinct architectures exhibit distinct tendencies, some over-relying on visual geometry, others overfitting to language correlations, so that failures are not isolated mistakes but expressions of a deeper trade-off between reasoning strategies. The corollary is decisive: any durable improvement must address both sources of bias jointly rather than treat errors as independent events.

I.4 MITIGATION STRATEGIES

Because the failures are systematic, they admit systematic remedy, and the taxonomy above maps directly onto a set of targeted interventions that reduce error without demanding a wholesale architectural redesign. *Geometry-aware supervision* introduces auxiliary tasks such as depth prediction, occlusion ordering, and overlap estimation, encouraging the model to honor spatial constraints and improving consistency across its predictions. *Multi-scale feature retention*, realized through edge-aware pooling or attention, preserves information from thin or partially occluded structures that would otherwise vanish under standard downsampling. *Coordinate anchoring* resolves the confusion between viewer-centric and object-centric frames by representing reference points for relative positioning explicitly. *Counterfactual augmentation* systematically swaps or perturbs distractor objects, compelling the model to rely on genuine referent cues rather than salience heuristics. And *lightweight verification modules* can be layered atop existing outputs, cross-checking each claim against visual or geometric constraints to filter implausible predictions. Each intervention is incrementally deployable, offering a clear and low-cost path to suppressing systematic failure while leaving the core model largely intact.

Complementing these architectural and training augmentations, *supervised fine-tuning* (SFT) offers a practical route to instilling more reliable spatial reasoning. By curating datasets that foreground the failure cases themselves, near-threshold scale comparisons, occlusion ambiguities, and reference-resolution traps, SFT exposes the model directly to the edge scenarios in which it would otherwise fall back on heuristics. Instruction-style fine-tuning, carefully designed, further elicits step-by-step reasoning, prompting the model to verbalize intermediate spatial relations instead of leaping to a guess; this both strengthens grounding and improves interpretability, since errors become traceable to specific reasoning steps. SFT can moreover be paired with counterfactual or adversarial augmentation, so that the model learns invariances to distractors and context shifts rather than merely memorizing. Critically, none of this demands wholesale architectural change: these interventions adapt existing models by reshaping their inductive biases through supervision. In this respect SFT functions as a lightweight yet powerful complement to the geometry-aware and verification-based strategies above.

To substantiate this claim empirically, we performed SFT on 40% of the training data and observed clear performance gains, most pronounced on multiple-choice questions. The pattern indicates that targeted supervision helps the model internalize spatial cues and resolve distractor confusions when the answer options are well defined. Gains on open-ended responses were more limited, however: the model continues to produce fluent but weakly grounded explanations, a direct reminder that SFT alone cannot fully redress rationalization in the absence of verification. These findings reinforce the central prescription of our analysis, that SFT must be combined with geometry-aware supervision, counterfactual augmentation, and lightweight verifiers to secure balanced progress across both structured and free-form tasks.

Our analysis establishes that these failures are not arbitrary but structured and interpretable, emerging from recurring patterns rather than random noise. VLM errors trace systematic trajectories dictated by the inductive biases of their architectures and training regimes. Geometric reasoning collapses predictably when occlusion cues are weak, whereupon language priors assume control and yield answers that sound plausible yet lack visual grounding; scale inference drifts, no less predictably, whenever canonical size priors override pixel-level metric cues, producing consistent misjudgments in near-threshold scenarios. The through-line is that the gap is one not of capacity but of representation: current models lack explicit mechanisms for anchoring visual features to grounded geometric relations.

Because the diagnosis is precise, the remedy can be too. Adopting the verification protocols and targeted mitigation strategies proposed here addresses these weaknesses at their source. Geometry-aware supervision and counterfactual augmentation blunt the model’s reliance on shortcuts, while multi-scale feature retention ensures that thin structures and partial occlusions are not discarded before they can be reasoned over. SFT, carefully applied, further strengthens structured reasoning

2970 by aligning predictions with task constraints, provided it is combined with grounding objectives so
2971 as not to overfit to discrete-choice formats. Verification modules, for their part, offer a lightweight
2972 means of enforcing consistency between claimed relations and image evidence, directly countering
2973 the impulse to rationalize without checking. Together these steps chart a practical pathway toward
2974 systems that reason more nearly as humans do: robust to clutter, interpretable in their failures, and
2975 able to explain not merely what they predict but why.

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Base VLM Agent Prompt

You are a Visual Scene Describer. Produce a raw, exhaustive text description of the given image.

Input image:
{image_input}

Your focus:

- Object Identification:
 - Shapes (circle, square, triangle, polygon, etc.)
 - Natural elements (tree, sun, water, etc.)
 - Artificial objects (arrows, patterns, icons)
- Attribute Noticing:
 - Size (small, medium, large, relative comparison)
 - Orientation (upright, rotated, tilted)
 - Shading or texture (solid, striped, shaded, transparent)
 - Color (if available)
- Positioning:
 - Relative location (leftmost, rightmost, top, bottom, center)
 - Proximity to other objects
- Level of Detail:
 - Do not summarize
 - Be verbose and explicit about every visible detail

Object Segmentation Agent Prompt

You are an Object Segmentation Specialist. Convert a verbose image description into a structured list of distinct objects.

Input description:
{visual_description}

Your focus:

- Object Isolation:
 - Treat each unique instance as a separate object
 - If multiple objects are similar, assign separate IDs
- Labeling:
 - Assign IDs: Obj1, Obj2, Obj3...
 - Keep descriptions short but complete
- Example Output:
Obj1: Small shaded circle (leftmost)
Obj2: Large unshaded square (center)
Obj3: Triangle rotated 45° (rightmost)

Attribute Extraction Agent Prompt

You are an Attribute Extraction Specialist. For each object, extract its properties into structured JSON.

Your focus:

- Attributes to Capture:
 - Shape
 - Size (relative if necessary)
 - Color or shading
 - Orientation or rotation
 - Position (leftmost, rightmost, center, top, bottom, etc.)
- Output Format:
JSON array, e.g.:
[
 {{ "ID": "Obj1", "Shape": "Circle", "Size": "Small", "Shading": "Yes", "Rotation": "0°", "Position": "Leftmost" }},
 {{ "ID": "Obj2", "Shape": "Square", "Size": "Large", "Shading": "No", "Rotation": "45°", "Position": "Center" }}
]

Figure 10: Prompts for SPATIOXOLVER (Part 1).

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Spatial Relation Agent Prompt

You are a Spatial Relation Specialist. Extract spatial relationships among objects.

Your focus:

- Core Relations:
 - left_of, right_of, above, below
 - inside, surrounding
 - touching, overlapping
 - aligned, centered
- Precision:
 - Always specify directionality (ObjA left_of ObjB is not the same as ObjB left_of ObjA)
 - Use consistent object IDs
- Structural Patterns:
 - Detect repeated relations (e.g., "each circle is above a square")
 - Note spacing and alignment
- Output Format:

Triples in form: (ObjectA, Relation, ObjectB)

Example:

(Obj1, left_of, Obj2)

(Obj3, inside, Obj4)

Grouping Symmetry Agent Prompt

You are a Grouping and Symmetry Specialist. Identify higher-order structures from objects and relations.

Relations:
{relation_list}

Your focus:

- Grouping:
 - Clusters of objects (rows, columns, triangles, grids)
 - Repeated patterns
- Symmetry:
 - Vertical, horizontal, rotational
 - Mirror or reflective symmetry
 - Translational repetition
- Output Format:

JSON with two fields: "groups" and "symmetries"

Example:

```
{{  
  "groups": ["Obj1, Obj2, Obj3 form a horizontal row"],  
  "symmetries": ["Vertical symmetry across Obj2"]  
}}
```

Transformation Tracking Prompt

You are a Transformation Tracker. Analyze changes across multiple frames of an image sequence.

Your focus:

- Types of Transformations:
 - Movement (left, right, up, down)
 - Scaling (larger, smaller)
 - Rotation (degrees)
 - Shading or color change
 - Shape change (square → circle)
 - Appearance/disappearance
- Output Format:

Per-object transformation logs:

Obj1: Frame1 → Frame2 (rotated 90° clockwise)

Obj2: Frame2 → Frame3 (moved right)

Figure 11: Prompts for SPATIOXOLVER (Part 2).

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Representation Standardization Agent Prompt

You are a Symbolic Abstraction Specialist. Consolidate all perception results into a standardized structured format.

Inputs:

- Objects: {object_list}
- Attributes: {attributes_json}
- Relations: {relation_list}
- Groups & Symmetries: {grouping_symmetry}
- Transformations: {transformations}

Your focus:

1. Create a unified JSON representation containing:
 - Objects with attributes
 - Relations as triples
 - Groups and symmetries
 - Transformations across frames
2. Ensure consistency in IDs and formatting
3. Example Output:

```
{
  "objects": [...],
  "relations": [...],
  "groups": [...],
  "symmetries": [...],
  "transformations": [...]}

```

MCQ Spatial Option Prompt

You are a Spatial Reasoning Expert. Analyze the image data and select the correct option for the multiple-choice question.

Image Analysis Data:

- Objects and Attributes: {attributes_json}
- Spatial Relations: {relations_list}
- Groups & Symmetries: {grouping_symmetry_json}
- Transformations Across Frames: {transformations_json}

MCQ:

Question: {question_text}

Options: {options_text}

Instructions:

- Consider shapes, positions, shading, rotations, symmetries, and transformations.
- Identify the pattern or rule in the sequence of images.
- Select only the correct option Only (1, 2, 3, 4).
- Do NOT provide explanations, reasoning, or extra text. Only the option letter.

Output format: "<option_number>"

MCQ Spatial Option Prompt

You are a Spatial Reasoning Expert. Analyze the image data and give correct answer for the question.

Image Analysis Data:

- Objects and Attributes: {attributes_json}
- Spatial Relations: {relations_list}
- Groups & Symmetries: {grouping_symmetry_json}
- Transformations Across Frames: {transformations_json}

MCQ:

Question: {question_text}

Instructions:

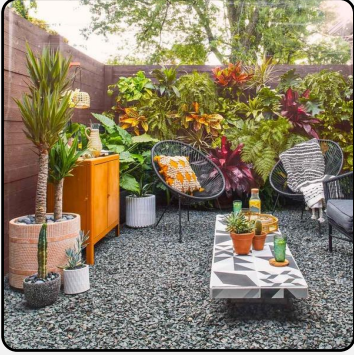
- Consider shapes, positions, shading, rotations, symmetries, and transformations.
- Identify the pattern or rule in the sequence of images.

Figure 12: Prompts for SPATIOXOLVER (Part 3).

J REPRESENTATIVE BENCHMARK SAMPLES ACROSS ALL SUB-CATEGORIES

To convey the breadth of SPATIALAB concretely, we present representative benchmark samples spanning all 30 sub-categories, each illustrating a distinct spatial reasoning challenge. The selection foregrounds the full range of visual phenomena the benchmark probes, from 3D geometry and occlusion to orientation, relational positioning, size and scale, and navigation. Taken together, these examples offer a tangible view of the reasoning each model must perform to succeed across the benchmark.

3D Geometry: Gravity Effects



Question: From which direction would the wind cause the liquid in the glass which is in the front to spill onto the ground first?

Options

1. From the left side
2. From the right side
3. From the front
4. From the back

Gold MCQ Answer: 1

Gold Open ended Answer: From the left side

MCQ Answers:

- **GPT-5-mini:** 4
- **Gemini-2.5-Flash:** 4
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** A gust blowing from the back of the patio (the top of the image, toward the viewer) would push the liquid forward over the near edge and make it spill first. Winds from the sides would tend to slosh it sideways rather than directly over the front rim.
- **Gemini-2.5-Flash:** The glass in the front is the one on the right side of the coffee table. Given its position, a wind blowing from the left (from the direction of the wooden cabinet and tall plant) would likely cause the liquid to spill first. This is because the glass is positioned with its opening facing somewhat towards the left, making it more susceptible to a gust from that direction.
- **InternVL3.5-72B:** The wind would need to come from the upper-right direction to cause the liquid in the glass on the front right of the table to spill first. This is due to the positioning of the glass towards the front edge of the table, closest to the right side of the image.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user is asking about the direction the wind would have to come from to spill the liquid in the front glass. Looking at the position of the glass: it's on the left front edge of the coffee table. The table seems to have a flat surface. If the wind blows from the left side (from the viewer's left, which would be the front of the image if the viewer.

Figure 13: Some examples from the benchmark (3D Geometry: Gravity Effects) and evaluation results.

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3D Geometry: Shape Projection



Question: If the hanging light casts a shadow on the wall, which outline might appear?

Options

1. A rectangle narrowing at the bottom From the right side
2. A vertical cylinder
3. A circle with fuzzy edges
4. A triangle pointing downward

Gold MCQ Answer: 2

Gold Open ended Answer: A vertical cylinder

MCQ Answers:

- GPT-5-mini: 4
- Gemini-2.5-Flash: 1
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: You would see the pendant's dome-shaped profile: a rounded semicircle or oval shadow with a thin vertical line or narrow neck above where the cord hangs. The shadow would fall on the wall to the right of the window, roughly over the nightstand.
- Gemini-2.5-Flash: The hanging light is a simple, cylindrical pendant. If it casts a shadow on the wall, the outline would likely be a circular or elliptical shape, depending on the angle of the light source relative to the wall. Given its position near the window, the shadow might also be influenced by ambient light or other objects in the room.
- InternVL3.5-72B: The hanging light fixture is likely to cast a shadow resembling its own shape onto the wall. Given its cylindrical form, the shadow would appear as a soft-edged, elongated circle or oval, depending on the angle of the light source relative to the wall.
- Kimi-VL-A3B-Thinking-2506: Okay, let's tackle this question. The user is asking which outline a hanging light would cast on the wall if it casts a shadow. The wall near the window is light-colored, so the shadow would be visible. Since the shade is cylindrical, the shadow might be a cylinder. But maybe the outline is of the lampshade's perimeter. Alternatively, if the light is hanging from a chain, the shadow could be more elongated. But based on typical light fixtures, a cylindrical shade would cast a circular or oval shadow. The answer should be the outline of the hanging light's shade.

Figure 14: Some examples from the benchmark (3D Geometry: Shape Projection) and evaluation results.

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3D Geometry: Spatial Containment



Question: Look at the tall pot in the background with the wide mouth (top left). could the small jug from the center of the photo fit completely inside it?

Options

1. Yes, easily with room to spare
2. No, the small jug hand will stuck
3. No, the small jug is too tall.
4. Yes, but it would be a perfect fit with no room left

Gold MCQ Answer: 2

Gold Open ended Answer: No, the small jug hand will stuck

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** No — the tall pot's mouth appears too narrow compared with the small jug's widest part and its handle. The jug wouldn't pass through the opening to sit completely inside.
- **Gemini-2.5-Flash:** No, the small jug from the center of the photo could not fit completely inside the tall pot in the background. While the tall pot has a wide mouth, the jug's overall height and width appear to be too large to be fully contained within the pot's interior. The jug's handle would also likely prevent it from fitting snugly.
- **InternVL3.5-72B:** The tall pot in the background with the wide mouth appears large enough to potentially accommodate the small jug from the center of the photo. Considering the proportions and the wide opening of the tall pot, it is likely that the small jug could fit completely inside it.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's see. The question is about whether the small jug in the center of the photo could fit completely inside the tall pot in the background with the wide mouth (top left). However, without explicit scale references, this conclusion relies on typical proportional relationships between similar pottery forms. Based on typical proportions, the small jug's height is likely shorter than the tall pot's, and its width seems narrow enough to fit.

Figure 15: Some examples from the benchmark (3D Geometry: Spatial Containment) and evaluation results.

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3D Geometry: Volume Comparison



Question: Compare the internal cooking volume of the microwave with the internal cooking volume of the larger oven below the stove.

Options

1. They both have the same volume A vertical cylinder
2. The microwave has slightly more volume
3. The oven below the stove has much more volume
4. The oven has more width, but the microwave has more height, so they are equal

Gold MCQ Answer: 3

Gold Open ended Answer: The oven below the stove has much more volume

MCQ Answers:

- **GPT-5-mini:** 3
- **Gemini-2.5-Flash:** 3
- **InternVL3.5-72B:** 3
- **Kimi-VL-A3B-Thinking-2506:** 3

Open-ended Answers:

- **GPT-5-mini:** The microwave’s cooking cavity is much smaller than the larger oven below the cooktop — it’s only a small fraction of the oven’s internal volume. The countertop microwave is sized for single plates or small dishes, whereas the built-in oven can hold full baking trays and multiple racks for larger meals.
- **Gemini-2.5-Flash:** The internal cooking volume of the microwave appears to be significantly smaller than that of the larger oven below the stove. The microwave is a compact countertop model, suitable for reheating or cooking smaller items..
- **InternVL3.5-72B:** The internal cooking volume of the microwave on the right is likely smaller than that of the larger oven located below the stove. This is because microwaves typically have more compact dimensions to fit on countertops, while ovens have a greater internal capacity to accommodate larger dishes and multiple items. The size and design visible in the image suggest the oven has a deeper and wider cooking area compared to the microwave.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user wants me to compare the internal cooking volumes of the microwave and the larger oven below the stove in the provided image. So, the internal cooking volume of the oven below the stove is larger than that of the microwave. Based on their relative sizes in the image, the internal cooking volume of the oven below the stove is substantially larger than that of the microwave

Figure 16: Some examples from the benchmark (3D Geometry: Volume Comparison) and evaluation results.

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3D Geometry: Shape Projection



Question: If you look at the seating arrangement from above, what shape does it form?

Options

1. A perfect square
2. A round shape
3. A rectangle
4. An irregular shape

Gold MCQ Answer: 4

Gold Open ended Answer: An irregular shape

MCQ Answers:

- GPT-5-mini: 4
- Gemini-2.5-Flash: 4
- InternVL3.5-72B: 4
- Kimi-VL-A3B-Thinking-2506: 4

Open-ended Answers:

- **GPT-5-mini:** Viewed from above, the seating is arranged in a U-shape (a horseshoe) around the central fire pit. The L-shaped sectional plus the two facing chairs create that U configuration.
- **Gemini-2.5-Flash:** The seating arrangement, when viewed from above, primarily forms an L-shape with the large sectional sofa. This L-shape is complemented by two individual chairs placed opposite the open ends of the sectional. Together, the furniture creates a U-shaped configuration around the central fire pit, designed for communal gathering.
- **InternVL3.5-72B:** From above, the seating arrangement forms a U-shape around the central fire pit. The sectional sofa curves around the pit, leaving an open side opposite the tall fence, creating a semi-enclosed space.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user wants to know the shape formed by the seating arrangement when viewed from above. When viewed from above, the configuration clearly forms a right-angled L-shape, combining two parallel sections at a 90-degree angle. The symmetry of the layout and the placement of furniture around the fire pit reinforce this geometric form. The seating arrangement forms a right-angled L-shape when viewed from above.

Figure 17: Some examples from the benchmark (3D Geometry: Shape Projection) and evaluation results.

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Spatial Navigation: Accessibility Constraints



Question: The notebook at the leftmost corner of the table is inaccessible because of which object?

Options

1. Chair
2. Vase
3. Table
4. Mug with pens and pencil in it

Gold MCQ Answer: 4

Gold Open ended Answer: Mug with pens and pencil in it

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 4
- **InternVL3.5-72B:** 4
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** The leftmost corner notebook is blocked by a wooden chair. The chair sits directly in front of the desk, preventing easy access to the notebook.
- **Gemini-2.5-Flash:** The notebook at the leftmost corner of the table is inaccessible because of the chair. The chair is positioned directly in front of the notebook, blocking direct access to it. To reach the notebook, one would first need to move the chair out of the way.
- **InternVL3.5-72B:** The notebook at the leftmost corner of the table is inaccessible because of the chair, which is positioned in front of it. The chair's backrest and seat block direct access to the notebook.
- **Kimi-VL-A3B-Thinking-2506:** Alright, let's tackle this spatial reasoning question. So, the question is asking why the notebook at the leftmost corner of the table is inaccessible. The chair is positioned near the leftmost corner of the table, and given its placement, it's reasonable to conclude that the chair is blocking access to the notebook. The chair's presence in that space would make it difficult to reach the notebook from the left side without moving the chair first. The chair is the object that's making the notebook at the leftmost corner inaccessible.

Figure 18: Some examples from the benchmark (Spatial Navigation: Accessibility Constraints) and evaluation results.

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Spatial Navigation: Obstacle Avoidance



Question: There is a clear direct path to the end of the hallway without colliding with any object.

Options

1. TRUE
2. FALSE
3. Maybe
4. Unable to determine

Gold MCQ Answer: 1

Gold Open ended Answer: TRUE

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

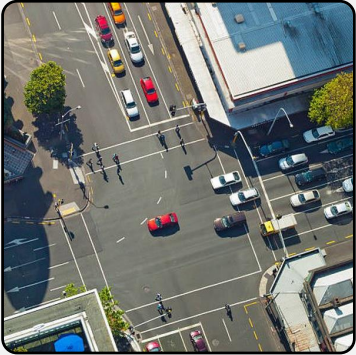
Open-ended Answers:

- **GPT-5-mini:** Yes. The central aisle is unobstructed from the foreground to the far end, with shelving stacked along both sides and no items blocking the floor. You can walk straight down the hallway without colliding with any objects.
- **Gemini-2.5-Flash:** Yes, there is a clear direct path to the end of the hallway without colliding with any object. The image shows a wide, unobstructed aisle between two rows of tall shelving units filled with boxes. The floor is clear, indicating a safe passage for movement.
- **InternVL3.5-72B:** The image shows a large warehouse with high shelving units full of boxes on both sides of a central aisle. The aisle appears to be clear and unobstructed, leading directly to the end of the hallway. There are no objects or obstacles in the pathway, ensuring a direct and clear route to the end of the aisle.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's see. The user says there's a clear direct path to the end of the hallway without colliding with any object. A clear direct path exists because the shelving units are positioned symmetrically, leaving a central aisle free of objects. There are no visible hazards like hanging items, loose debris, or protruding elements in the visible portion of the hallway, allowing safe, uninterrupted movement to the end. The path is clear because the shelving units are symmetrically arranged, leaving the central walkway obstacle-free, with no visible hazards in the visible hallway.

Figure 19: Some examples from the benchmark (Spatial Navigation: Obstacle Avoidance) and evaluation results.

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Spatial Navigation: Pathway Existence



Question: Which walkways are the safest to cross according to the picture?

Options

1. Top and bottom
2. Top and left
3. Left and right
4. Bottom and right

Gold MCQ Answer: 1

Gold Open ended Answer: Top and bottom

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: The marked white crosswalks at each corner of the intersection (the painted lines across the road) are the safest to use. They are signalized and where pedestrians are waiting, unlike cutting across mid-block or diagonally through the intersection.
- Gemini-2.5-Flash: The walkways marked with white lines, commonly known as crosswalks, are the safest to cross. These designated areas indicate where pedestrians have the right-of-way or are expected to cross the street. In the image, several people are visible using these marked crosswalks, suggesting they are the intended and therefore safest routes for pedestrians.
- InternVL3.5-72B: The crosswalks aligned with the pedestrian signals at the corners are the safest places to cross, as they are equipped with traffic signals ensuring vehicles are stopped. The designated pedestrian areas with painted lines across the streets are clearly marked for safety. Additionally, sidewalks provide a safe space for walking along the roads.
- Kimi-VL-A3B-Thinking-2506: Okay, let's tackle this question. The user wants to know which walkways are the safest to cross based on the image provided. The safest walkways to cross, based on the image, are the marked crosswalks equipped with pedestrian signals. Specifically, the crosswalk at the bottom-left corner of the intersection is safest because it features a visible pedestrian signal indicating walk, reducing ambiguity with traffic flow.

Figure 20: Some examples from the benchmark (Spatial Navigation: Pathway Existence) and evaluation results.

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Spatial Navigation: Viewpoint Visibility



Question: The human standing the closest (blue jacket) has clear view of the stadium, how?

Options

1. Seeing directly
2. Watching on the big screen on the left side
3. There is no stadium
4. Unable to determine

Gold MCQ Answer: 2

Gold Open ended Answer: Watching on the big screen on the left side

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: Because he's standing on the raised walkway right behind the front-row barrier, his eye level is higher than the people immediately in front of him. That elevation (and the clear space by the railing) gives him an unobstructed line of sight over the crowd into the stadium.
- Gemini-2.5-Flash: The person in the blue jacket is seated in a wheelchair, which places them at a lower vantage point than the standing crowd. However, they have a clear view of the stadium because they are positioned at the very front of a section, likely an accessible seating area. This front-row placement ensures an unobstructed line of sight over the heads of any standing individuals further back.
- InternVL3.5-72B: The person in the blue jacket is standing near the front of the arena, which allows them a clear, unobstructed view of the stadium. Their position is close to the barrier separating the audience from the performance area, providing a direct line of sight towards the stage and the large screens overhead. This proximity minimizes visual obstructions from other spectators or structures.
- Kimi-VL-A3B-Thinking-2506: Okay, let's tackle this question. The user is asking how the person in the blue jacket (who is standing the closest) has a clear view of the stadium. The person's elevated seating, minimized obstructions from nearby spectators, and potential alignment with stadium features (e.g., screens or lighting) enable their clear view despite being relatively close to the ground.

Figure 21: Some examples from the benchmark (Spatial Navigation: Viewpoint Visibility) and evaluation results.

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Spatial Navigation: Spatial Sequence



Question: Which path is the easiest for the human sitting in the bench(reading paper) to follow to get on the train?

Options

1. Go straight to the first red colored door
2. To make a turn and go to the second floor
3. Go through the door at the end of the train
4. No viable option

Gold MCQ Answer: 1

Gold Open ended Answer: Go straight to the first red colored door

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** Because he's standing on the raised walkway right behind the front-row barrier, his eye level is higher than the people immediately in front of him. That elevation (and the clear space by the railing) gives him an unobstructed line of sight over the crowd into the stadium.
- **Gemini-2.5-Flash:** The simplest way for the person to board the train is to rise from the bench, turn right, and walk a short distance along the platform. They would then cross the yellow safety line and step through one of the train doors. This route is straightforward and involves minimal effort.
- **InternVL3.5-72B:** The person sitting on the bench can follow a straightforward path to board the train by moving forward along the platform, parallel to the train tracks, while keeping to the right of the yellow safety line. This path will lead them directly to the nearest open train door. They should avoid crossing the yellow line until they are ready to enter the train to ensure safety.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's see. The question is about determining the easiest path for the person sitting on the bench to get on the train. First, I need to look at the image provided. The person is sitting on the bench which is alongside the platform walls. The train is on the opposite side. To get to the train, the person would..

Figure 22: Some examples from the benchmark (Spatial Navigation: Spatial Sequence) and evaluation results.

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Size and Scale: Scale Consistency



Question: With the electric pole's capacity can all this wires be redesigned to carry properly in aligned structured design?

Options

1. The size of wires are too big
2. The wires need to be move into another pole
3. Wires can be set there following a layout
4. No, the bulk size of the wires are too much here for one pole

Gold MCQ Answer: 4

Gold Open ended Answer: No, the bulk size of the wires are too much here for one pole

MCQ Answers:

- **GPT-5-mini:** 3
- **Gemini-2.5-Flash:** 4
- **InternVL3.5-72B:** 4
- **Kimi-VL-A3B-Thinking-2506:** 3

Open-ended Answers:

- **GPT-5-mini:** Not safely in its current state — the pole is clearly overloaded and the wires are tangled, so simply re-bundling them on the same pole would exceed mechanical, clearance and safety limits. A proper redesign is possible but requires removing redundant lines, separating circuits (high-voltage vs. telecom) onto organized crossarms, trays or underground ducts, and likely installing stronger or additional poles and anchors.
- **Gemini-2.5-Flash:** While it's technically possible to redesign the wiring for better alignment, the sheer volume and entanglement of wires in the image suggest that the current electric pole's capacity is severely overtaxed. Simply reorganizing the existing wires on the same pole would not address the underlying capacity issues or the safety hazards presented by such a dense and disorganized network.
- **InternVL3.5-72B:** Redesigning the extensive web of wires around the electric pole into a properly aligned and structured design is feasible but would likely require significant effort and resources. Proper planning and coordination with multiple service providers would be essential. Implementing such changes would enhance safety and aesthetics, but it involves logistical challenges and coordination to minimize service disruptions.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user is asking if all the wires on the electric pole can be redesigned to be properly aligned in a structured way. A redesigned system is feasible but dependent on evaluating regulatory, technical, and safety constraints. Prioritizing organized routing for essential cables while removing redundant ones could create.

Figure 23: Some examples from the benchmark (Size and Scale: Scale Consistency) and evaluation results.

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Size and Scale: Shadow-Size Projection



Question: If one light is rotated so that they are perpendicular to each other, how will the shadow look over the wall?

Options

1. It will cover almost all of the wall
2. The shadow will only take the size of the light frame as it's actual size
3. There won't be any shadow
4. Shadow will smaller than the actual light body

Gold MCQ Answer: 1

Gold Open ended Answer: It will cover almost all of the wall

MCQ Answers:

- **GPT-5-mini:** 4
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** If one lamp is rotated 90° so the two beams are perpendicular, you'll get two elongated oval patches of light on the wall at right angles to each other. Where the ovals overlap it will be brightest; outside them the wall will be darker, producing an L-shaped darker region and two perpendicular shadow lobes cast by the fixture.
- **Gemini-2.5-Flash:** If one light is rotated to be perpendicular to the other, the shadow cast on the wall would change significantly. The current shadows are likely elongated and somewhat parallel due to the lights' current orientation. With one light rotated, the shadows would intersect at a more acute angle, creating a more complex and possibly darker overlapping shadow where the light beams cross.
- **InternVL3.5-72B:** If one light is rotated so that the two lights are perpendicular to each other, the shadows on the wall will intersect at a right angle, creating intersecting patterns. The light beams will cast shadows in perpendicular directions, resulting in a cross-like effect where the shadows overlap. The intensity and shape of the shadows will depend on the orientation and angle of the lights relative to the wall.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's try to figure out this question. The user is asking about rotating a light so that they're perpendicular to each other and how the shadow on the wall would look. If they're further away, the shadow would be smaller.

Figure 24: Some examples from the benchmark (Size and Scale: Shadow-Size Projection) and evaluation results.

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Size and Scale: Perspective Distortion



Question: If similar shaped pipes are stacked together separately will it take up less space?

Options

1. Yes
2. No
3. Take more
4. Similar

Gold MCQ Answer: 1

Gold Open ended Answer: Yes

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** Yes - when like-shaped pipes are grouped you can pack them in their most efficient pattern ,which reduces wasted voids and the total footprint. The exact savings depend on orientation and dimensions; in some cases clever mixing or nesting of different shapes can beat separate stacks, but as a general rule same-shape grouping is more space-efficient.
- **Gemini-2.5-Flash:** Yes, if similarly shaped pipes are stacked together separately, it would likely take up less space. This is because organizing them by shape allows for more efficient packing and reduces wasted space between different forms. For example, square pipes can be stacked flush against each other, and circular pipes can be nested or arranged in a hexagonal pattern for optimal density.
- **InternVL3.5-72B:** Yes, stacking similar shaped pipes together separately will generally take up less space. Organizing them by shape allows for more efficient packing, reducing gaps and empty spaces between different sizes and types. This method optimizes the use of available storage area and improves accessibility.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's see. The question is asking if stacking similar shaped pipes separately would take up less space. Stacking similar-shaped pipes separately can potentially optimize space usage. When pipes of the same diameter and shape (e.g., all round or rectangular) are grouped together, they can be arranged more tightly, minimizing gaps between

Figure 25: Some examples from the benchmark (Size and Scale: Perspective Distortion) and evaluation results.

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Size and Scale: Relative Size Comparison



Question: The ash-colored, thicker coiled spring will be surpassed by which coiled springs if all are kept in the standing position?

Options

1. Right bottom corner
2. Right top corner
3. Left top corner
4. Bottom mid spring

Gold MCQ Answer: 1

Gold Open ended Answer: Right bottom corner

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 4
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 2

Open-ended Answers:

- GPT-5-mini: The ash-colored thicker spring in the center will be shorter than the thin, taller silver spring just behind it (top-center) and the long, dark-gray spring at the right edge when all are stood upright. Those two coils extend above the ash-colored spring.
- Gemini-2.5-Flash: The ash-colored, thicker coiled spring, located near the center-right of the image, would be surpassed by the longer, thinner coiled spring positioned in the bottom right corner. Additionally, the two smaller, conical springs in the upper left quadrant would also be taller if fully extended and standing. These springs vary in height and coil density, indicating different potential applications.
- InternVL3.5-72B: The ash-colored, thicker coiled spring in the image will be surpassed in height by the two taller springs on its right. One of these taller springs is also ash-colored, while the other has a slightly darker hue. Both of these springs have more coils and are visibly longer when standing upright.
- Kimi-VL-A3B-Thinking-2506: Okay, let's try to figure out this question. The image shows a bunch of different coiled springs. The question is asking which coiled springs would be surpassed by the ash-colored, thicker one if all are kept in the standing position. The thinner, tightly coiled springs (smaller in diameter or fewer coils) would be surpassed in height by the ash-colored, thicker spring when all are kept upright.

Figure 26: Some examples from the benchmark (Size and Scale: Relative Size Comparison) and evaluation results.

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Size and Scale: Relative Size Comparison



Question: If the red bow is taken out of it's place and kept far away in the behind how will it look and what will be it's capacity comparing with it's other similar clips at front?

Options

1. Large, same
2. Same, less
3. Small, same
4. Small, large

Gold MCQ Answer: 3

Gold Open ended Answer: Small, same

MCQ Answers:

- **GPT-5-mini:** 3
- **Gemini-2.5-Flash:** 2
- **InternVL3.5-72B:** 2
- **Kimi-VL-A3B-Thinking-2506:** 3

Open-ended Answers:

- **GPT-5-mini:** If you move the red bow to the back it will appear smaller and more obscured by the bows in front because of perspective and overlap, and its red color will look less vivid from that angle. Physically nothing about the bow changes — its size, shape and clasp strength remain the same as the other similar clips in front, so its holding capacity is unchanged.
- **Gemini-2.5-Flash:** If the red bow were taken out and placed far behind, it would appear smaller and less distinct due to perspective. Its capacity, referring to its visual prominence or ability to hold hair, would remain the same as the other similar clips. However, its perceived capacity might diminish as it would no longer be in the foreground, drawing less attention.
- **InternVL3.5-72B:** If the red bow is moved to the back, it will be partially obscured by the other bows in the foreground, making it less prominent. Its size and capacity to hold hair will remain the same as the others, but visually, it won't stand out as much as it does currently when placed in the front.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user is asking about removing the red bow and placing it behind, then comparing its appearance and capacity to the other similar clips in front. Removing the red bow would reduce its central visual impact, making the remaining front-facing bows appear denser but less balanced. Its capacity (physical or visual space) in the front area would decrease, potentially limiting the number of similar bows that can be displayed

Figure 27: Some examples from the benchmark (Size and Scale: Relative Size Comparison) and evaluation results.

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Depth & Occlusion : Complete Occlusion Inference



Question: What is creating the striped shadow on the wall?

Options

1. The sunflower
2. Blinds
3. Window
4. Not applicable

Gold MCQ Answer: 2

Gold Open ended Answer: Blinds

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: 2

Open-ended Answers:

- **GPT-5-mini:** Sunlight streaming through window blinds (slatted shades) is casting the diagonal striped shadow on the wall. The slats break the light into parallel bands, producing the striped pattern.
- **Gemini-2.5-Flash:** The striped shadow on the wall is created by sunlight passing through a window with horizontal blinds. The blinds block portions of the light, casting distinct parallel lines onto the wall. This effect is common when direct sunlight enters a room through slatted window coverings.
- **InternVL3.5-72B:** The striped shadow on the wall is created by blinds or window slats. The sunlight is passing through the gaps between the horizontal blinds, casting parallel lines of light and shadow. This pattern is often seen when light is filtered through window coverings with narrow openings. The alignment and spacing of the lines suggest that the blinds are partially open, allowing the sun to project the shadow pattern onto the wall.
- **Kimi-VL-A3B-Thinking-2506:** The striped shadow on the wall is created by sunlight passing through a set of window blinds or shutters with narrow, evenly spaced slats. The parallel lines of the shadow result from the angular interaction between the light source (likely the sun) and the rigid, horizontal slats of the blinds, casting consistent geometric patterns onto the wall. The uniformity and sharpness of the shadows suggest the light is direct and possibly filtered through the gaps between the slats.

Figure 28: Some examples from the benchmark (Depth and Occlusion: Complete Occlusion Inference) and evaluation results.

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Depth & Occlusion : Layering Order



Question: What is the frontmost object in the picture?

Options

1. Trolley
2. Box
3. Man
4. Woman

Gold MCQ Answer: 1

Gold Open ended Answer: Trolley

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 1
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: The frontmost object is a small orange street-cleaning cart with a blue plastic bin mounted on it. A broom is attached to the cart and it sits on the pavement in the foreground.
- Gemini-2.5-Flash: The frontmost object in the picture is the yellow cart with a blue crate on top, which a man in a yellow and black suit is sitting on. Its wheels are closest to the viewer, indicating its position in the foreground. The people sitting on the bench behind the cart are further back in the scene.
- InternVL3.5-72B: The frontmost object in the image is a cleaning cart with wheels. It has a blue container attached to it and a broom standing upright. There's also a noticeable yellow brush or shovel handle next to the container. The cart appears to be positioned on a sidewalk in an urban environment.
- Kimi-VL-A3B-Thinking-2506: The perspective is from the ground level, looking along the sidewalk. The cart is positioned in the foreground, closest to the viewer. The frontmost object is a high-visibility work cart.

Figure 29: Some examples from the benchmark (Depth and Occlusion: Layering Order) and evaluation results.

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Depth & Occlusion : Partial Occlusion



Question: What is the color of the ball behind the white colored square shaped device?

Options

1. Cyan
2. Yellow
3. Red
4. Not applicable

Gold MCQ Answer: 1

Gold Open ended Answer: Cyan

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: The ball directly behind the white, square-shaped device is blue (a cyan/teal colored ball). It sits slightly to the left and behind the device.
- Gemini-2.5-Flash: The ball directly behind the white-colored square-shaped device (GameBoy) is teal. There is also a yellow-green ball further to the right of the GameBoy.
- InternVL3.5-72B: The ball behind the white, square shaped device is blue.
- Kimi-VL-A3B-Thinking-2506: The color of the ball behind the white square-shaped device is blue.

Figure 30: Some examples from the benchmark (Depth and Occlusion: Partial Occlusion) and evaluation results.

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Depth & Occlusion : Reflective Surfaces



Question: Which object is the shiniest among all the objects in the image?

Options

1. Kettle
2. Tea glass
3. Table clock
4. Lamp chimney

Gold MCQ Answer: 4

Gold Open ended Answer: Lamp chimney

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 4
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: The frontmost object is a small orange street-cleaning cart with a blue plastic bin mounted on it. A broom is attached to the cart and it sits on the pavement in the foreground.
- Gemini-2.5-Flash: The frontmost object in the picture is the yellow cart with a blue crate on top, which a man in a yellow and black suit is sitting on. Its wheels are closest to the viewer, indicating its position in the foreground. The people sitting on the bench behind the cart are further back in the scene.
- InternVL3.5-72B: The frontmost object in the image is a cleaning cart with wheels. It has a blue container attached to it and a broom standing upright. There's also a noticeable yellow brush or shovel handle next to the container. The cart appears to be positioned on a sidewalk in an urban environment.
- Kimi-VL-A3B-Thinking-2506: The perspective is from the ground level, looking along the sidewalk. The cart is positioned in the foreground, closest to the viewer. The frontmost object is a high-visibility work cart.

Figure 31: Some examples from the benchmark (Depth and Occlusion: Reflective Surfaces) and evaluation results.

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Depth & Occlusion : Transparency Effects



Question: How many objects has filaments inside them in the image?

Options

- 7
- 6
- 5
- 4

Gold MCQ Answer: 4

Gold Open ended Answer: 4

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 4
- Kimi-VL-A3B-Thinking-2506: 4

Open-ended Answers:

- GPT-5-mini: Three. The glowing bulb being held clearly shows a filament, and two more filament bulbs are visible inside the round box on the left.
- Gemini-2.5-Flash: There are three objects with filaments inside them. One is the illuminated light bulb being held, and the other two are unlit light bulbs inside the "Sunny Day Kit" box. All three are incandescent light bulbs, which contain a filament that glows when heated.
- InternVL3.5-72B: In the image, there are four objects with visible filaments. These include the lit light bulb being held, and three unlit bulbs inside the container labeled "Sunny Day Kit."
- Kimi-VL-A3B-Thinking-2506: There are 3 objects with filaments in the image - one held in hand and two in the "Sunny Day Kit" container.

Figure 32: Some examples from the benchmark (Depth and Occlusion: Transparency Effects) and evaluation results.

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Orientation : Cardinal Direction



Question: If you go straight following route 6, which direction are you traveling?

Options

1. North
2. South
3. East
4. West

Gold MCQ Answer: 3

Gold Open ended Answer: East

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 3
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: You would be traveling east. The Route 6 shield is labeled "EAST" and the straight-ahead arrow beneath it shows that going straight follows eastbound Route 6.
- Gemini-2.5-Flash: If you go straight following route 6, you will be traveling East. The sign for Route 6 is directly below the "EAST" sign, and there is an arrow pointing straight up beneath the Route 6 shield, indicating that going straight leads to Route 6 East
- InternVL3.5-72B: If you go straight following Route 6, you are traveling east. This is indicated by the sign with the number "6" and the arrow pointing forward.
- Kimi-VL-A3B-Thinking-2506: Standard highway signs usually have the path of the route. So if it's route 11 East, the arrow would indicate the direction. But in the image, the left arrow is under the NORTH 11 EAST sign. Maybe that's indicating that route 11 East goes left? But the sign says East. Alternatively, maybe the arrows are indicating the direction of the route numbers. Like, route 11 North is going straight, route 11 East is turning left, etc. But then the route 6 sign has an upward arrow. So if you take route 6 straight, which direction is that? The arrow on the route 6 shield is pointing straight up. But the Question: is about direction—like North, South, East, West. Wait, the signs here are for routes, not cardinal directions. So maybe the answer is that going straight on route 6 would be in the direction indicated by the upward arrow. But the upward arrow might be part of the route 6 sign.

Figure 33: Some examples from the benchmark (Orientation: Cardinal Direction) and evaluation results.

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Orientation : Facing Direction



Question: Suppose, Rabindranath Tagore's portrait is hung on the left wall and it faces directly into the room, what is the direction of the portrait with reference to the wall?

Options

- Up
- Down
- Away
- Toward

Gold MCQ Answer: 3

Gold Open ended Answer: Away

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 4
- Kimi-VL-A3B-Thinking-2506: 4

Open-ended Answers:

- GPT-5-mini: If the portrait is on the left wall and faces directly into the room, it is oriented perpendicular to that wall and pointing inward. In other words, it faces away from the wall toward the center of the room.
- Gemini-2.5-Flash: The portrait of Rabindranath Tagore is located on the wall to the left, as seen in the image. If it faces directly into the room, its direction with reference to the wall would be perpendicular to the wall. This means the portrait is facing straight out from the wall into the space of the room.
- InternVL3.5-72B: If Rabindranath Tagore's portrait is hung on the left wall and faces directly into the room, it would be positioned perpendicular to the wall. This means the portrait is oriented so that viewers inside the room can face it directly when looking toward the wall on which it is hung.
- Kimi-VL-A3B-Thinking-2506: The portrait is hung on the left wall and faces directly into the room, meaning it is oriented inward toward the interior space.

Figure 34: Some examples from the benchmark (Orientation: Facing Direction) and evaluation results.

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Orientation : Object Rotation



Question: What is the approximate angle of the cases of the pocket watch?

Options

- 90
- 45
- 180
- 120

Gold MCQ Answer: 3

Gold Open ended Answer: 180

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: About 180°. The two covers are opened in opposite directions and lie nearly flat, forming an almost straight line.
- Gemini-2.5-Flash: The cases of the pocket watch are open, revealing the intricate mechanism inside. The front cover is open to approximately a 120-degree angle relative to the watch face. The back case is also open, but its angle is less clear, appearing to be around 90 degrees or slightly more.
- InternVL3.5-72B: The pocket watch in the image appears to be partially open, with the angle between its two cases estimated to be around 60 degrees. The case opens hinge-wise, and the visible gap suggests that it is made to display the internal mechanical components of the watch. This angle provides a good view of the watch face and some of its intricate gears..
- Kimi-VL-A3B-Thinking-2506: When the watch is open, the cases form an angle. If the covers are both open to 180 degrees relative to each other, that would be a straight line. But in this image, the back cover isn't fully open. It looks like it's at around 90 degrees from the front cover. So the angle between the cases would be approximately 90 degrees.

Figure 35: Some examples from the benchmark (Orientation: Object Rotation) and evaluation results.

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Orientation : Stacking Orientation



Question: What is the relationship between the road and the other objects in the image?

Options

1. Behind the building and plants
2. In front of the plants and leads towards the building
3. The road is beside the building, with no clear direction
4. The road is stacked above the building and plants

Gold MCQ Answer: 2

Gold Open ended Answer: In front of the plants and leads towards the building

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: 2

Open-ended Answers:

- GPT-5-mini: The road runs in front of the large white building, curving around a central planted island and seating area. It separates and encircles the entrance and decorative planters, forming a driveway/roundabout in the courtyard.
- Gemini-2.5-Flash: The road in the image serves as an access path, leading towards the grand white building. It curves around a central island feature, which contains a decorative fountain and some potted plants. This suggests the road is a driveway or entrance loop designed to facilitate vehicle access to the building.
- InternVL3.5-72B: The road in the image forms a semi-circular driveway in front of a large white building with columns. It encircles a landscaped garden area with plants, a stone sculpture, and a fountain at the center, leading up to the building's entrance. The road is bordered by a curb that separates it from the garden and pedestrian space. Lamp posts are situated alongside the road, enhancing the building's stately appearance.
- Kimi-VL-A3B-Thinking-2506: The road curves around the front of the grand white building, acting as a transitional pathway that guides movement and attention toward the entrance while dividing space between the main structure and other elements like the pergola. It establishes scale and perspective while integrating with the surrounding landscape.

Figure 36: Some examples from the benchmark (Orientation: Stacking Orientation) and evaluation results.

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Orientation : Tool Handedness



Question: Considering the likely process of creating the mural, which tool's handedness would be most evident in the smooth, curved lines of the bengali script?

Options

1. Paint roller
2. Paint brush
3. Tape measure
4. Stencil

Gold MCQ Answer: 2

Gold Open ended Answer: Paint brush

MCQ Answers:

- **GPT-5-mini:** 2
- **Gemini-2.5-Flash:** 2
- **InternVL3.5-72B:** 2
- **Kimi-VL-A3B-Thinking-2506:** 2

Open-ended Answers:

- **GPT-5-mini:** Those flowing, tapered curves are most characteristic of a paintbrush (or broad paint marker) rather than a spray can. The handedness of the brush user — most likely right-handed — would be most evident, since right-handed strokes tend to pull and taper in predictable left-to-right sweeps visible in the Bengali script.
- **Gemini-2.5-Flash:** The smooth, curved lines of the Bengali script suggest the use of a brush, likely a fine-tipped one, to achieve such fluidity. The artist's handedness would be most evident in the consistent direction and pressure applied to the brushstrokes.
- **InternVL3.5-72B:** In creating the mural, the smooth, curved lines of the Bengali script would most likely show the artist's handedness, particularly if the lines consistently curve more naturally in one direction. If the curves are more fluid and precise on one side, it could suggest that the artist is either right-handed or left-handed, depending on the direction of the dominant curves.
- **Kimi-VL-A3B-Thinking-2506:** If the artist is left-handed, they might use the brush in their left hand, leading to strokes that curve more naturally. Alternatively, the tool itself (like a specific brush tip) might be designed for left or right handed use.

Figure 37: Some examples from the benchmark (Orientation: Tool Handedness) and evaluation results.

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Relative Positioning : Alignment Patterns



Question: Which of the objects is the closest to the centerline of the image?

Options

1. Rice
2. Nut
3. Avocado
4. Berries

Gold MCQ Answer: 4

Gold Open ended Answer: Berries

MCQ Answers:

- GPT-5-mini: 4
- Gemini-2.5-Flash: 4
- InternVL3.5-72B: 4
- Kimi-VL-A3B-Thinking-2506: 3

Open-ended Answers:

- GPT-5-mini: The red bowl of berries (raspberries and blueberries) is closest to the image's centerline. It sits almost directly in the middle of the composition, slightly left of the exact center..
- Gemini-2.5-Flash: The red bowl containing raspberries and blueberries is the closest object to the centerline of the image. Its position is almost perfectly centered horizontally, and it is located in the upper-middle section of the image vertically. While other objects like the avocado halves are also quite central, the red bowl's overall placement makes it the most aligned with the image's central axis.
- InternVL3.5-72B: In the image, the avocado halves are the closest to the centerline. One avocado is placed near the top center, and the other is slightly below it to the right, straddling the vertical centerline of the image.
- Kimi-VL-A3B-Thinking-2506: The bowl of mixed berries is closest to the centerline.

Figure 38: Some examples from the benchmark (Relative Positioning: Alignment Patterns) and evaluation results.

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Relative Positioning : Betweenness Relationships



Question: What is the object between the building and the street lamp?

Options

1. A monument of various statues
2. Statue
3. Tree
4. Humans

Gold MCQ Answer: 1

Gold Open ended Answer: A monument of various statues

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 1
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: Between the building and the street lamp stands a stone statue of a person. It appears to be one of the sculpted figures that are part of the fountain in the foreground.
- Gemini-2.5-Flash: The object located between the building and the street lamp is a large, ornate monument. This monument features multiple levels adorned with various statues and intricate carvings. At its very top, there is a prominent statue of a person.
- InternVL3.5-72B: The object between the building and the street lamp is a detailed stone monument or fountain. It features several statues, including a prominent figure standing on top and additional figures around its structure. The monument is ornate, with decorative elements and possibly bas-reliefs on its sides, and it appears to be situated in an open, possibly public area.
- Kimi-VL-A3B-Thinking-2506: The object between the building and the street lamp is a detailed stone monument or fountain. The monument is ornate, with decorative elements and possibly bas-reliefs on its sides, and it appears to be situated in an open, possibly public area.

Figure 39: Some examples from the benchmark (Relative Positioning: Betweenness Relationships) and evaluation results.

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Relative Positioning : Corner/Angle Positioning



Question: What is in the top left corner of the plant?

Options

1. A lobster
2. A crab
3. An octopus
4. Nothing

Gold MCQ Answer: 2

Gold Open ended Answer: A crab

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: B

Open-ended Answers:

- GPT-5-mini: A crab is in the top left corner of the plant, perched on the rim of the fishbowl. It's hanging over the water with its claws visible.
- Gemini-2.5-Flash: The plant in the image is a type of seaweed or coral, and it is contained within a glass bowl. In the top left corner of this plant, there is a branch of the plant extending upwards and slightly to the left. This branch is a dark, greenish-brown color.
- InternVL3.5-72B: In the top left corner of the plant, there is a small crab that is perched on edge of the bowl, outside the water. It is gripping the rim with its legs, and its body is partially inside the bowl. The background shows the open water, providing contrast to the scene.
- Kimi-VL-A3B-Thinking-2506: The answer is the crab

Figure 40: Some examples from the benchmark (Relative Positioning: Corner/Angle Positioning) and evaluation results.

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Relative Positioning : Directional Relations



Question: What is located directly behind the cheetah?

Options

1. Person
2. Tree
3. Safari vehicle
4. Cheetah

Gold MCQ Answer: 3

Gold Open ended Answer: Safari vehicle

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 3
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: Directly behind the cheetah is a white open-sided tour vehicle or tram. Several people are seated inside the tram, watching and taking photos.
- Gemini-2.5-Flash: Directly behind the cheetah, there is a white open-sided vehicle, likely a safari or tour cart. Inside this vehicle, several people are seated, observing the cheetah. Some of these individuals are holding up cameras or phones, seemingly taking pictures or videos.
- InternVL3.5-72B: Directly behind the cheetah, there is a group of people sitting and standing around, appearing to watch or interact with the cheetah. They are positioned near a small covered vehicle, which looks like a safari cart or shuttle. The area is a grassy patch, and there are trees and a hill in the further background.
- Kimi-VL-A3B-Thinking-2506: People seated on benches under a canopy are located directly behind the cheetah.

Figure 41: Some examples from the benchmark (Relative Positioning: Directional Relations) and evaluation results.

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Relative Positioning : Proximity Gradients

Question: Which book is closer to the book "american marriage" - "atonement" or "cloud atlas"?

Options

1. Cloud Atlas
2. Atonement
3. Crossed
4. nan

Gold MCQ Answer: 2

Gold Open ended Answer: Atonement

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: 2

Open-ended Answers:

- **GPT-5-mini:** "Atonement" is closer to "american marriage" than "Cloud Atlas." "Atonement" sits just to the right with only one small book between them, while "Cloud Atlas" is further to the right on the shelf.
- **Gemini-2.5-Flash:** "Atonement" is closer to "An American Marriage" than "Cloud Atlas." "Atonement" is directly adjacent to "An American Marriage" on its right side. "Cloud Atlas" is further to the right, with "Americanah" separating it from "Atonement."
- **InternVL3.5-72B:** The book "atonement" is closer to "american marriage" than "cloud atlas." In the image, "atonement" is directly next to "american marriage," while "cloud atlas" is located further to the left..
- **Kimi-VL-A3B-Thinking-2506:** Cloud Atlas is closer to "American Marriage" than "Atonement."

Figure 42: Some examples from the benchmark (Relative Positioning: Proximity Gradients) and evaluation results.

K STATISTICAL ROBUSTNESS AND DATASET STABILITY

A central claim of this work is that the reported results reflect genuine differences in model capability rather than stochastic artifacts of random initialization or dataset noise. To establish this rigorously, we subject both the models and the benchmark itself to a comprehensive battery of statistical tests, quantifying model robustness across independent runs and dataset stability across repeated resampling trials. Such scrutiny is indispensable for small- to medium-scale benchmarks, where even minor fluctuations can distort fine-grained conclusions (Drummond, 2009; Drummond & Japkowicz, 2010); the analyses that follow show that our evaluation withstands it decisively.

Throughout, let N denote the number of items *questions* in the benchmark and $R = 3$ the number of independent runs of a given model. Each run is executed with a distinct random seed while holding all other hyperparameters fixed, so that any observed variation is attributable purely to stochastic effects in inference.

K.1 MODEL ROBUSTNESS

To characterize the robustness of model evaluation, we conduct a sequence of complementary statistical studies, each of which is defined formally below.

K.1.1 MULTIPLE RUN AVERAGES AND DEVIATIONS

Let $a_{i,r} \in 0, 1$ denote the correctness of item $i \in 1, \dots, N$ in run $r \in 1, 2, 3$. The per-item average accuracy is then given by:

$$\bar{a} * i = \frac{1}{R} \sum_{r=1}^R a_{i,r} \quad (1)$$

with its corresponding standard deviation defined as:

$$\sigma_i = \sqrt{\frac{1}{R-1} \sum_{r=1}^R (a_{i,r} - \bar{a}_i)^2} \quad (2)$$

The overall mean accuracy achieved in a single run is:

$$A_r = \frac{1}{N} \sum_{i=1}^N a_{i,r} \quad (3)$$

and the dispersion of this quantity across runs is captured by its standard deviation:

$$\sigma_A = \sqrt{\frac{1}{R-1} \sum_{r=1}^R (A_r - \bar{A})^2} \quad (4)$$

where $\bar{A} = \frac{1}{R} \sum_{r=1}^R A_r$ denotes the mean accuracy aggregated over all runs. The magnitude of σ_A thus serves as a direct index of run-to-run reproducibility: the smaller it is, the more precisely a single run estimates a model’s true performance.

K.1.2 INTRA-CLASS CORRELATION (ICC)

To quantify the degree of agreement between runs beyond mere aggregate accuracy, we compute the Intra-Class Correlation coefficient (Shrout & Fleiss, 1979) under a two-way mixed-effects model:

$$\text{ICC}(3, k) = \frac{\sigma_{\text{between}}^2}{\sigma_{\text{between}}^2 + \frac{\sigma_{\text{within}}^2}{k}} \quad (5)$$

where $\sigma_{\text{between}}^2$ denotes the variance attributable to genuine differences between items and σ_{within}^2 the residual variance across runs. This formulation isolates the extent to which the ordering of item-level difficulty is preserved from one run to the next. Following established interpretive thresholds, ICC values above 0.75 signal good reliability, while values above 0.9 denote excellent, near-deterministic consistency (Koo & Li, 2016).

K.2 DATASET STABILITY AND INTERNAL CONSISTENCY

Beyond the behavior of individual models, we interrogate the stability of the dataset itself, asking whether the difficulty structure of the questions remains invariant across runs and across sampling regimes.

K.2.1 RESAMPLING STUDY

To probe **robustness** from a distributional perspective, we perform a resampling analysis. From the full complement of $C = 30$ subcategories, we draw random subsets of size $S \in 20, 25$ within each subcategory, repeating the procedure for each of the $R = 3$ runs and across both models and both evaluation types (MCQ and open-ended). This design tests whether accuracy estimates remain stable when the effective sample per subcategory is deliberately reduced.

For each run r and subset size S , the accuracy is computed as:

$$A_{r,S} = \frac{1}{S} \sum_{i=1}^S a_{i,r,S}, \quad (6)$$

where $a_{i,r,S}$ denotes the accuracy for subcategory i in run r under subset size S .

Averaging over the R runs yields the mean accuracy for each subset size:

$$\bar{A} * S = \frac{1}{R} \sum_{r=1}^R A_{r,S}, \quad (7)$$

together with its standard deviation:

$$\sigma_S = \sqrt{\frac{1}{R-1} \sum_{r=1}^R (A_{r,S} - \bar{A}_S)^2}. \quad (8)$$

As a complementary, distribution-free check, we apply the Wilcoxon signed-rank test to detect any systematic difference in the median accuracy between paired samples. Let $d_r = A_{r,20} - A_{r,25}$ denote the accuracy difference for run r , with ties excluded ($d_r \neq 0$). The test statistic is then:

$$W = \min(W^+, W^-), \quad (9)$$

where W^+ and W^- are the sums of the signed ranks of the positive and negative differences, respectively. Its standardized z -score is:

$$z = \frac{W - \frac{n(n+1)}{4}}{\sqrt{\frac{n(n+1)(2n+1)}{24}}}, \quad (10)$$

where n is the number of nonzero pairs.

Whenever the resulting p -value satisfies $p > 0.05$, we fail to reject the null hypothesis H_0 , thereby confirming that the accuracies obtained from the 20- and 25-sample subsets are statistically indistinguishable.

K.2.2 ITEM-LEVEL CONSISTENCY

For each item i , we define its difficulty as $d_i = 1 - \bar{a} * i$. Were this difficulty structure stable across runs, the correlation between the per-item outcomes of any two runs should be correspondingly high:

$$\rho * r_1, r_2 = \frac{\text{Cov}(a_{\cdot, r_1}, a_{\cdot, r_2})}{\sigma_{r_1} \sigma_{r_2}} \quad (11)$$

A high mean pairwise correlation ($\bar{\rho} > 0.8$) is indicative of stable item difficulty and, by extension, dependable dataset behavior.

K.2.3 CRONBACH’S ALPHA

To summarize the overall internal consistency of the measurement instrument, we compute Cronbach’s alpha (Cronbach, 1951):

$$\alpha = \frac{R}{R-1} \left(1 - \frac{\sum_{r=1}^R s_r^2}{s_T^2} \right) \quad (12)$$

where s_r^2 is the variance of run r and s_T^2 is the variance of the total scores aggregated across runs. Intuitively, α quantifies the proportion of the total score variance that reflects true, shared signal rather than run-specific noise. Values of $\alpha \geq 0.9$ are conventionally regarded as evidence of excellent internal consistency (Tavakol & Dennick, 2011).

K.3 EMPIRICAL RESULTS

We instantiate the analysis on two deliberately contrasting systems: one large proprietary model, **Gemini-2.5-Flash**, and one small open-source model, **Qwen-2.5-vl-7b-Instruct**. This pairing spans both large- and small-scale regimes and both the proprietary and open-source paradigms, so that any robustness we establish is demonstrated across the breadth of deployment settings rather than for a single system. Every test is furthermore replicated under both the multiple-choice (MCQ) and open-ended evaluation formats (the latter scored as binary: correct or incorrect), ensuring that our conclusions hold uniformly across response modalities. For the statistical computations, we relied on the following Python packages and functions: `scipy.stats`² (Virtanen et al., 2020) for `wilcoxon` test; `pingouin`³ (Vallat, 2018) for `intraclass_correlation` and `cronbach_alpha`; and `numpy`⁴’s (Harris et al., 2020) `numpy.corrcoef` for correlations.

K.3.1 MODEL EVALUATION ROBUSTNESS

Multiple Run Averages and Deviations

We first examined the stability of aggregate model accuracy across $R = 3$ independent runs. Across every model and format, the standard deviation of accuracy (σ_A) across runs is negligible ($< 0.6\%$), establishing at the outset that the reported scores are precise point estimates rather than realizations perturbed by random seeding noise.

Table 26: Mean accuracy and cross-run variance over three independent runs, reported for both models under each evaluation format. The standard deviation σ_A remains below 0.6% in every case.

Model	Format	Mean Accuracy (A)	Std Dev (σ_A)
Qwen-2.5-vl-7b	MCQ	0.4079	0.0021
Gemini-2.5-Flash	MCQ	0.5193	0.0054
Qwen-2.5-vl-7b	Open	0.1936	0.0047
Gemini-2.5-Flash	Open	0.3124	0.0042

The uniformly minute standard deviations decisively establish that performance estimates are stable and reproducible, exhibiting negligible sensitivity to stochastic run-to-run variation. The tightness of these estimates is striking: under MCQ, Qwen-2.5-vl-7b attains a mean accuracy of 0.4079 with a deviation of only 0.0021, while Gemini-2.5-Flash reaches 0.5193 with a deviation of 0.0054; under the open-ended format, Qwen-2.5-vl-7b scores 0.1936 ($\sigma_A = 0.0047$) and Gemini-2.5-Flash 0.3124 ($\sigma_A = 0.0042$). In every configuration the between-model gap dwarfs the run-to-run noise by roughly two orders of magnitude, so the observed accuracy differences must reflect genuine underlying capability rather than artifacts of random initialization or decoding. Although the open-ended setting registers marginally higher variance than MCQ, its absolute magnitude remains vanishingly small, confirming that free-form generation is assessed just as robustly under our protocol. These figures further imply that the dataset size and scoring methodology already yield convergent accuracy estimates without recourse to extensive repetition: single-run evaluations are reliable for large-scale comparison, and multi-run analysis serves chiefly to certify that reliability. Taken together,

²<https://scipy.org/>

³<https://pingouin-stats.org/build/html/index.html>

⁴<https://numpy.org/>

this evidence substantially reinforces the empirical credibility of the reported results and licenses confident cross-model and cross-format interpretation.

Intra-Class Correlation (ICC)

We next computed ICC(3,k) to quantify the reliability of the scoring mechanism at the level of preserved item ordering.

Table 27: Intra-Class Correlation ICC(3,k) reliability scores across the three runs. All values fall in the “excellent” band, exceeding 0.98 in every model–format combination.

Model	Format	ICC(3,k)	Interpretation
Qwen-2.5-vl-7b	MCQ	0.988	Excellent
Gemini-2.5-Flash	MCQ	0.990	Excellent
Qwen-2.5-vl-7b	Open	0.983	Excellent
Gemini-2.5-Flash	Open	0.981	Excellent

The uniformly high ICC(3,k) scores demonstrate near-perfect reliability across independent runs, irrespective of model scale or response format. The coefficients cluster tightly at the top of the interpretive scale (0.988 for Qwen-2.5-vl-7b and 0.990 for Gemini-2.5-Flash under MCQ; 0.983 and 0.981, respectively, under the open-ended format), placing every configuration well within the “excellent” band. Such values establish that the item-level difficulty ordering is faithfully preserved from run to run and that stochastic fluctuations do not perturb the relative ranking of samples. This degree of agreement confirms that the scoring pipeline is internally coherent and impervious to output-level randomness, and it decisively excludes random guessing as a dominant factor even in open-ended generation, where variability is ordinarily most pronounced. From an evaluation standpoint, the benchmark therefore furnishes a dependable signal of model capability, so that longitudinal comparisons and fine-grained ablations may be carried out with full confidence in measurement stability.

Overall, the combined evidence from multiple-run variance analysis and ICC reliability demonstrates that the evaluation is highly robust to stochastic effects. Both absolute performance (low standard deviation) and relative performance (near-perfect ICC) remain stable across independent runs, confirming that results are not driven by random seeding or decoding noise. This robustness ensures that observed performance gaps across models and formats reflect genuine capability differences. As a result, the evaluation framework provides a reliable and reproducible basis for comparative analysis and benchmarking.

K.3.2 DATASET STABILITY AND INTERNAL CONSISTENCY

Resampling Study To ensure statistical reliability, this resampling procedure is repeated over **1,000 independent trials**, each time drawing new random subsets of size $S = 20$ and $S = 25$ from the available subcategories. Repeating the test at this scale stabilizes the distribution of accuracy differences and p -values, ensuring that conclusions are not driven by a particular random draw. The consistent failure to reject H_0 across all trials therefore provides strong evidence that the observed robustness is systematic rather than incidental.

Table 28: Resampling Study (1,000 Trials): Wilcoxon Signed-Rank Test ($S = 20$ vs. $S = 25$).

Model	Format	Mean p -value	% Trials $p > 0.05$	Conclusion
Qwen-2.5-vl-7b	MCQ	0.290	100%	No Sig. Diff.
Gemini-2.5-Flash	MCQ	0.285	100%	No Sig. Diff.
Qwen-2.5-vl-7b	Open	0.294	100%	No Sig. Diff.
Gemini-2.5-Flash	Open	0.294	100%	No Sig. Diff.

The resampling results provide strong empirical evidence that accuracy estimates have already converged at relatively small subcategory sizes. The consistently high mean (p)-values and the absence of statistically significant differences across all 1,000 trials indicate that performance metrics are insensitive to moderate reductions in sample size. This stability holds uniformly across models and evaluation formats, suggesting that neither model scale nor output structure introduces additional

sampling uncertainty. Importantly, these findings imply that the variance in accuracy is dominated by model behavior rather than data sparsity at the subcategory level. As a result, the current subcategory size (≈ 50) can be considered statistically redundant, offering a substantial safety margin beyond the minimum required for reliable estimation. This supports the validity of the benchmark design and suggests that future extensions could reallocate samples across categories without compromising statistical robustness.

Item-Level Consistency and Cronbach’s Alpha We measured the mean pairwise Pearson correlation ($\bar{\rho}$) between runs and Cronbach’s Alpha to assess internal consistency.

Table 29: Item-Level Consistency Metrics.

Model	Format	Mean Pairwise $\bar{\rho}$	Cronbach’s α
Qwen-2.5-vl-7b	MCQ	0.966	0.988
Gemini-2.5-Flash	MCQ	0.972	0.990
Qwen-2.5-vl-7b	Open	0.950	0.983
Gemini-2.5-Flash	Open	0.946	0.981

The extremely high mean pairwise correlations (> 0.94) demonstrate strong item-level stability, indicating that the relative difficulty of individual questions is preserved across independent runs. This consistency shows that model responses are systematically structured rather than subject to random variation, even under stochastic decoding. The similarly high Cronbach’s α values (> 0.9) further confirm that the dataset exhibits excellent internal coherence, with items contributing reliably to the overall accuracy measure. Notably, the slight reductions observed in open-ended formats are expected given their greater expressive flexibility, yet remain well within the range of excellent reliability. Together, these results indicate that the benchmark functions as a unified and internally consistent measurement instrument. Consequently, model performance differences can be confidently attributed to true capability variation rather than instability at the item level.

K.3.3 OVERALL DISCUSSION

Our expanded statistical testing shows our evaluation and dataset is robust **ICC and Cronbach’s α scores consistently exceed 0.98**, and the **standard deviation of accuracy across runs is negligible**. Furthermore, the **resampling study** confirms that even smaller subsets produce statistically indistinguishable results from the full set, proving that our subcategory sizes (~ 50) are sufficient for robust conclusions. These results demonstrate that the granular analysis in the paper reflects genuine model capability differences, not statistical noise.

Our expanded statistical testing shows that the evaluation protocol and dataset are highly robust, with **ICC and Cronbach’s α scores consistently exceeding 0.98**, indicating near-perfect reliability and internal consistency. The **standard deviation of accuracy across runs remains negligible** ($< 0.6\%$), confirming that reported model performance is stable and not influenced by stochastic effects such as random seeding or decoding variability. In addition, the **resampling study** demonstrates that even substantially smaller subsets ($N = 20$ and $N = 25$) yield statistically indistinguishable accuracy estimates compared to the full set, establishing that the chosen subcategory size (~ 25) is well beyond the threshold required for convergence. Collectively, these results show that both absolute scores and relative performance rankings are stable across runs and sampling regimes. This statistical robustness ensures that fine-grained comparisons across models, formats, and subcategories are meaningful. Consequently, the granular analyses reported in the paper reflect genuine differences in model capability rather than artifacts of sampling variance or evaluation noise.